

Admissibility and the Well-Definedness of Yang–Mills Endpoint Constructions

Determinacy, Identity, and Uniqueness on Fixed Domains

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This paper is a standalone extension of the predecessor local-net Yang–Mills mass-gap theorem. The predecessor package constructs the bounded-region local gauge-invariant Haag–Kastler/Wightman theory, its non-trivial vacuum representation, its exact strictly local endpoint, and its vacuum Hamiltonian gap. The present paper recalls that theorem package as a fixed input and proves the completion-control part of the Clay-scope proof chain. It fixes a theorem-scope class of extended-support line and surface data, shows that these data are not bounded-region local observables, organizes their realized sector layer over the fixed local net, and proves that sector data enter a complete-theory Hamiltonian claim only after passing a scope-faithful, kernel-faithful sector-to-completion admissibility filter. The endpoint theorem is a GNS/spectral bridge: every theorem-scope admissible sectoral or ultraviolet completion of the fixed local vacuum theory has no new locally detectable subgap state and no additional subgap copy of the vacuum representation, hence $E_H([0, m^*))\mathcal{H} = \mathbb{C}\Omega$. Consequently the predecessor vacuum gap remains the complete-theory gap for every same-domain admissible completion, and any same-domain Clay-scope realization is locally equivalent on its canonical local vacuum sector to the predecessor local theorem object. The result is a self-contained extension theorem: the earlier paper remains the proof home for local existence and the vacuum-sector gap, while the present paper supplies the sector, admissible-completion, and complete-theory endpoint bridge needed for the Clay-scope formulation.

Mathematical-status lock. AASC is used in this manuscript as a mathematical constraint formalism for fixed-domain construction, claim standing, admissibility, and endpoint-preserving continuation. It is not philosophical commentary, metaphor, optional narration, or an interpretive overlay. A same-mode objection must identify a failed definition, theorem link, dependency claim, hypothesis match, GNS/spectral bridge step, or fixed-domain consequence.

Proof-class lock. The proof class is

fixed-base completion control
+ sector-to-completion admissibility filtering
+ kernel-faithful GNS/spectral bridging.

It is not a fresh local-net construction proof, not a new OS/Wightman reconstruction proof, not a charged-sector Hamiltonian theorem, not a classification of every possible nonlocal sector, not an arbitrary direct-sum or direct-integral enlargement argument, and not a replacement for the predecessor local Yang–Mills proof home. The predecessor package remains the proof home for local existence and the vacuum-sector gap; this manuscript proves the theorem-scope sector, completion-control, and complete-theory endpoint bridge over that fixed local object.

Claim-class lock. The manuscript proves the following theorem-scope claim. Given the predecessor fixed local Yang–Mills theorem object, theorem-scope extended-support data are organized as sector data over the fixed local net. Sector data enter a complete-theory Hamiltonian claim only after passing the ATS/UEAP, scope-faithful, kernel-faithful sector-to-completion admissibility filter. Every theorem-scope admissible sectoral or ultraviolet completion then preserves the predecessor gap as the complete-theory gap. The manuscript does not assert that every sector representation is a same-theory completion, and it does not assert a mass-gap theorem for arbitrary nonvacuum sectors outside the filtered completion class.

Source-use lock. Classical AQFT and QFT sources supply standard background: local nets, GNS uniqueness, sector representations, spectral projections, and the Clay problem statement. The predecessor Yang–Mills package supplies the constructed local theorem object and its vacuum gap. The present manuscript supplies the completion-control bridge over that object. AASC supplies fixed-domain admissibility, standing, and endpoint-continuation constraints. None of these source classes is used to perform the proof work of another source class.

Read this first: fixed-base completion proof spine. The proof works over the fixed predecessor local theorem object

$$\mathfrak{B}(G) = (\mathcal{A}_{\text{loc}}(G), \pi_{\text{vac}}, \mathcal{H}_{\text{vac}}, \Omega_{\text{vac}}, H_{\text{vac}}).$$

The theorem-scope route is

$$\begin{aligned} \mathfrak{B}(G) &\implies \mathfrak{E}(G) \implies \mathfrak{S}_{\text{ext}}(G) \\ &\implies \text{Filter}_{\text{sec} \rightarrow \text{comp}} \implies B_{\text{YM}}^{\text{GNS}}(G; \mathcal{A}, \omega, U) \\ &\implies E_H([0, m^*))\mathcal{H} = \mathbb{C}\Omega. \end{aligned}$$

The first arrow fixes the local theorem object supplied by the predecessor package. The middle arrows organize extended-support data as sector data over the fixed net. The filter arrow blocks unlicensed promotion from sector representation to same-theory completion. The final arrow is the kernel-faithful GNS/spectral bridge proving the complete-theory mass-gap endpoint for theorem-scope admissible completions.

No one-step sector-to-completion promotion. The proof does not use the invalid schema

$$\begin{aligned} \text{sector representation} &\implies \text{same-theory completion} \\ &\implies \text{complete-theory Hamiltonian gap.} \end{aligned}$$

A sector datum first remains representation-theoretic structure over the fixed local net. It enters the complete-theory Hamiltonian claim only after the sector-to-completion admissibility filter supplies carrier, scope, standing, admissibility, dynamics, and kernel fidelity. This is the Yang–Mills no-one-step-promotion lock: lawful sector structure is not completion eligibility by notation alone.

Immediate denial fork. A same-mode refutation must produce one of the following: a failure of the fixed-base handoff, a failure of the local-net boundary, a sector datum that automatically supplies same-theory completion eligibility without the filter, a scope-faithful and kernel-faithful same-domain completion with new locally detectable subgap content, an additional subgap vacuum multiplicity compatible with same-domain kernel fidelity, a failure of the GNS/spectral bridge, or a mismatch between the Lean audit surface and a manuscript claim where Lean support is invoked.

I. INTRODUCTION

A. Problem and scope

A basic distinction in gauge theory is the distinction between *strictly local observables*, attached to bounded spacetime regions, and *extended-support observables*, such as line and surface operators, whose support is not confined to bounded regions. The earlier local theorem package constructed a bounded-region local gauge-invariant net together with its vacuum-sector reconstruction and vacuum mass-gap theorem^{1–4}. The theorem-level endpoint there was deliberately local: the object proved was a strictly local gauge-invariant net in the vacuum sector, not a theorem about line or surface sectors.

The present paper addresses that omitted layer. The governing question is how extended-support structure is organized *over* the fixed local net and which global-form distinctions are thereby recovered. The paper is therefore an extension of a proved base theory. It does not reopen the local construction, and it does not treat nonlocal observables as though they were already elements of the bounded-region local algebras. The predecessor package is the proof home for the local Haag–Kastler/Wightman realization, non-triviality witness, vacuum reconstruction, and vacuum-sector mass gap; the present manuscript is the proof home only for the sectoral, fixed-domain, and admissible-completion layer over that object.

Standalone extension posture. Because this manuscript is read after, and independently of, the local-net paper, it restates the predecessor result only at the theorem-package interface needed here. The reader does not need the predecessor proof in hand to follow the extension argument: Sections II and III recall the exact local theorem object, the vacuum gap, the exact-local endpoint boundary, and the reconstruction-side sector package used as inputs. What is not repeated is the analytic proof of those inputs. Thus the paper is standalone as an extension theorem, while remaining faithful about proof home: the earlier paper proves local existence and the vacuum-sector gap, and the present paper proves the admissible-completion bridge from that fixed local object to the Clay-scope endpoint.

Scope clarification. The present manuscript does not impose external constraints on Yang–Mills theory. It identifies the minimal conditions under which a construction counts as a determinate, well-defined, validity-assessable derivation of a Yang–Mills endpoint object on a fixed domain. Any construction that is claimed to yield such an endpoint is thereby

treated as a validity-assessable derivation of a fixed-domain mathematical object, so the structural conditions developed here are conditions on the construction itself rather than an external interpretive gloss.

Structural sense of derivation. Throughout, the terms derivation, construction step, and validity-assessable are used in a structural sense. They concern the conditions under which a Yang–Mills construction yields a determinate mathematical object whose identity is preserved under admissible transformation. No psychological or merely philosophical reading is involved.

These conditions are not external to the Yang–Mills construction. They are the minimal structural requirements under which the endpoint object itself is well-defined and identity-preserving; any construction that fails to satisfy them does not produce a determinate Yang–Mills object on the fixed domain.

For clarity, the endpoint object refers to the standard Yang–Mills quantum field theoretic object (e.g. the reconstructed Wightman theory or Haag–Kastler net), viewed here as a fixed-domain construction target.

B. Why the problem is nontrivial

The passage from local observables to extended-support observables is not a matter of notation. A bounded-region local net is designed to encode observables localized in bounded regions and to satisfy the corresponding locality, isotony, and covariance conditions. Extended-support observables are different in kind: their support behavior is different, and the information they can carry is different. In particular, global-form-sensitive data that are invisible at the level of strictly local gauge-invariant composites may reappear at the level of line and surface sectors.

That distinction has two immediate consequences. First, if one treats extended-support objects as though they already belonged to the bounded-region local algebras, one changes the theorem-level object while still speaking as though one had preserved it. Second, if one introduces a sectoral layer without specifying how it sits over the fixed local net, one no longer knows whether one has constructed a genuine extension, a change of domain, or merely a reformulation of local data. The present paper is written to avoid exactly those ambiguities.

C. Structural source posture

A further point of proof posture should be made explicit. The structural primitives used in the present paper are not introduced here as stipulative axioms of a chosen formalism. They are used only through previously derived theorem-content at the upstream layer. In particular, the source posture of the present paper treats bivalence, the AMetric boundary, uniqueness of the admissible interior, and standing–reuse-stable-admissibility equivalence as theorem-derived necessity results, not as optional preferences of exposition^{13–16}.

Accordingly, the present manuscript does not rely on the upstream corpus as an authority substitute for the endpoint. It records the restricted kernel fragment needed here in hypothesis/conclusion form, verifies the fixed Yang–Mills theorem object against that fragment, and then states in operator-algebraic language what follows once the predecessor local construction and the theorem-derived structural constraints are both in force. This proof-home posture matters because it determines what counts as a same-theory continuation, what counts as mere bookkeeping, and what counts as an illicit change of theorem object.

Source-neutral dependency discipline. The present manuscript does not use the upstream AASC corpus as an authority source. It uses only theorem-content: stated hypotheses, stated conclusions, and verified transport into the fixed Yang–Mills theorem object. The relevant mathematical question is not authorship, citation distribution, or sociological provenance, but whether the kernel results are correct and whether the fixed local Yang–Mills endpoint satisfies their hypotheses. The structural dependency is therefore exposed in hypothesis/conclusion form below, and the endpoint application is made only after the Yang–Mills theorem object is checked against those hypotheses.

D. Bridge posture for theorem-derived necessity constraints

The endpoint argument uses the following proof posture. The structural kernel is not introduced as a local axiom package. The restricted kernel fragment needed for the endpoint is stated internally in the manuscript, with upstream references serving as expanded provenance. The present paper does not leave its applicability to the Yang–Mills setting at the level of prose. The fixed local theorem object produced by the required existence standard is treated as a nontrivial standing-bearing domain with lawful redescriptions given by unitary

equivalence preserving local observable content, vacuum state, and local dynamics. The endpoint section then proves, inside the manuscript, that this is enough to place the Yang–Mills theorem object on the side of the theorem-derived necessity constraints relevant to same-domain comparison.

E. On the status of the structural kernel

The admissibility/standing kernel used in this manuscript is not introduced as a stipulative axiom system. It is treated as theorem-level necessity content, recorded here in the restricted form needed for the endpoint and developed more fully in the upstream corpus: within any non-degenerate compositional regime supporting stable reference, lawful redescription, and irreversible commitment, the AMetric boundary, bivalence, unique admissible interior, and conservation are forced, and no faithful lower generator or same-domain extension can produce an alternative architecture.

Accordingly, the logical direction in this paper is:

$$\begin{aligned} \text{non-degenerate regime} &\Rightarrow \text{kernel instantiated by necessity} \\ &\Rightarrow \text{constraints on every kernel-faithful Yang–Mills realization.} \end{aligned}$$

The kernel is not a parameter that the Yang–Mills construction may choose to accept or reject. Once a construction is treated as determinate, validity-assessable, identity-preserving, and irreversible on a fixed domain, the kernel is already instantiated as the condition of that non-degenerate status.

Requests to derive this kernel from within a downstream axiomatic regime (for example, OS/Wightman or AQFT) invert the proved dependency order. The closure theorems explicitly rule out any faithful same-domain lower derivation or extension that would generate the kernel from below.

The role of the present manuscript is therefore not to generate the kernel, but to show that once a Yang–Mills theory satisfying the required existence standard is constructed, that theory necessarily lies within the scope of the kernel and inherits its consequences. In particular, admissibility exhausts faithful same-domain mathematical validity on the fixed Yang–Mills domain, eliminating all same-domain alternatives.

F. Status of the Yang–Mills proof home

The predecessor Yang–Mills theorem package does not provide only the bounded-region local net and vacuum-sector gap. At its reconstruction endpoint it also furnishes the standard local QFT realization route, the exact local-net endpoint, the non-triviality witness, and the reconstruction-side sector and realization machinery needed for the present extension analysis, including reconstructed sectors, OS sectors, and realization maps linking reconstruction data to sectoral realization. The role of the present manuscript is not to generate that machinery from first principles, but to fix a theorem-scope extended-support comparison class over an already-constructed Yang–Mills theorem base and to derive the corresponding fixed-domain boundary, sectoral distinction, admissible-completion classification, and complete-theory endpoint consequences.

G. Dependency map and proof-home ledger

For clarity, the proof dependencies used in this manuscript are separated as follows.

Dependency class		Content	Proof home
Standard background	AQFT	C^* -nets, GNS uniqueness, sector representations, local vector states, and spectral projections	Standard AQFT literature
Predecessor Mills package	Yang–Mills	Local net, vacuum representation, OS/Wightman/Haag–Kastler reconstruction route, non-triviality, exact local endpoint, and vacuum mass gap	Predecessor Yang–Mills theorem package
Kernel packet	necessity	Admissibility, standing, determinate reference, irreversibility, no lower generator, no faithful same-domain extension, uniqueness, and conservation	Compressed theorem packet recorded in the present manuscript, with upstream corpus provenance
ATS/UEAP endpoint-use packet		Anchor/tensor/skin role typing, fail-closed role ambiguity, carrier/scope/standing adequacy, and no unlicensed claim promotion in sector-to-completion use	Restricted packet proved in the present manuscript, with upstream ATS/UEAP provenance
Present manuscript		Extended-support class, sector-role realization, local-net boundary, global-form recovery, sector-to-completion filter, kernel-faithful completion classification, and complete-theory spectral bridge	This paper

This ledger is source-neutral: the listed inputs are used through their theorem content and verified hypotheses, not by appeal to authorship. It also fixes the carrier of each claim before the endpoint argument begins.

H. Local-net endpoint and start-point coherence

The handoff from the predecessor local theorem package to the present manuscript is exact. The predecessor endpoint is

$$\boxed{\mathcal{A}_{\text{loc}}(G), (\pi_{\text{vac}}, \mathcal{H}_{\text{vac}}, \Omega_{\text{vac}}, H_{\text{vac}}), \text{Spec}(H_{\text{vac}}) \subset \{0\} \cup [m^*, \infty)}.$$

It is the bounded-region local gauge-invariant sharp-local net generated by flowed curvature composites, together with its OS/Wightman vacuum representation, non-triviality witness, and vacuum-sector mass gap. Extended-support line and surface sectors are excluded from that theorem-level object by localization: they are not elements of the bounded-region local algebra.

The present paper starts from precisely that object. Its base object $\mathfrak{B}(G)$ is not a replacement, strengthening, or reinterpretation of the predecessor endpoint; it is the same local theorem object read as fixed input. The new work begins only after that endpoint has been fixed. It asks which theorem-scope extended-support and completion structures may be organized over the fixed local net without altering the local observable carrier, vacuum representation, local dynamics, or predecessor spectral statement.

Thus the proof-home alignment is one-directional: the predecessor package supplies local existence and the vacuum gap, while the present paper supplies fixed-domain sector and completion control. No theorem proved here back-propagates into the predecessor local net, and no extended-support construction in this paper is allowed to become part of the bounded-region local theorem object by notation alone.

I. Clay-scope endpoint ledger

The official Yang–Mills existence-and-mass-gap formulation asks for a non-trivial quantum Yang–Mills theory on $\mathbb{R}^4$ for compact simple gauge group G , with axiomatic strength at least comparable to the Wightman/Osterwalder–Schrader frameworks, local gauge-invariant field content corresponding to curvature polynomials and covariant derivatives, positive energy, and a strictly positive mass gap⁵. In the present proof architecture those existence-side requirements are proof-home facts of the predecessor Yang–Mills package, not new burdens of this paper.

Clay/local-field component	Proof home	Role in this manuscript
Bounded-region local gauge-invariant net on $\mathbb{R}^4$	Predecessor local theorem package	Fixed input $\mathfrak{B}(G)$
OS/Wightman/Haag–Kastler reconstruction, locality, covariance, and positive energy	Predecessor endpoint packet	Recalled only to fix the vacuum representation and dynamics
Gauge-invariant local curvature-polynomial field correspondence	Predecessor reconstruction and field-correspondence packet	Supplies the local observable carrier whose identity must be preserved
Non-triviality witness and vacuum-sector mass gap	Predecessor Route 1/Lane A/QE3 endpoint	Starting spectral datum m^* for the completion bridge
Extended-support line/surface sectors	Excluded from the predecessor local theorem object	Analyzed here only as a downstream sector/completion problem over the fixed net
Complete-theory gap under same-domain completion	Present manuscript	Kernel-faithful GNS/spectral bridge for theorem-scope admissible completions

This ledger prevents a category error. The current manuscript is not a substitute for the local-existence proof home; it is the theorem-scope completion layer over the local object already supplied there.

J. Main question

The main question may therefore be stated in the following form.

Given a fixed local gauge-invariant observable net, how can extended-support

observables and the corresponding sector data be organized over that net without altering the local theorem-level object, and which global-form distinctions are thereby recovered?

This question has three parts. One must first fix the class of extended-support objects to be studied. One must then determine the correct mathematical home for those objects: not every nonlocal construction belongs in the bounded-region observable algebras, and representation-theoretic sector language is often the natural replacement. Finally, one must identify exactly what information is invisible locally and prove that the sector layer distinguishes it.

K. What is proved and what is not

The paper proves six things.

- (i) it fixes a theorem-scope class of extended-support objects over the local net and proves that these are not elements of bounded-region local algebras;
- (ii) it identifies the theorem-scope extended-support layer over the fixed local net by linking the chosen support-and-label class to the reconstruction-side sector and realization machinery furnished by the predecessor Yang–Mills theorem package;
- (iii) it proves that the corresponding theorem-scope global-form data are invisible to the strictly local net but are distinguished at the sector level over that reconstruction-side realization layer;
- (iv) it proves that the sector construction leaves unchanged the bounded-region local net, its vacuum reconstruction, and the predecessor vacuum-sector mass-gap theorem;
- (v) it proves that sector representations enter the complete-theory spectral bridge only after satisfying the ATS/UEAP sector-to-completion admissibility filter, so sector classification is not conflated with same-theory completion; and
- (vi) it proves that theorem-scope admissible completions preserve the fixed local theorem object and inherit the predecessor vacuum gap as the complete-theory gap.

The paper is equally explicit about its non-claims. It does *not* identify nonlocal observables with bounded-region local observables. It does *not* claim a classification of every possible

nonlocal sector over the local net. And it does *not* infer a charged-sector, defect-sector, or other nonvacuum mass-gap theorem from the existence of the sector structure alone.

L. Fixed-base posture

The technical posture of the paper is conservative. The local net furnished by the earlier local theorem package is treated as a fixed base object. Any later construction is assessed relative to that base. In particular, the paper distinguishes three possibilities whenever extended-support structure is introduced:

- (1) it is a bookkeeping reformulation of data already present in the local theory;
- (2) it is genuinely additional but sits lawfully *over* the fixed local net without changing it;
or
- (3) it changes the underlying theorem object and therefore lies outside the scope of the present paper.

This is a standard mathematical distinction, but it is worth making explicit because informal discussions of line and surface observables often drift between these cases.

Same-domain criterion. Throughout, “same-domain” means preservation of the admissibility-relevant invariant bundle—reference, standing, irreversibility, and admissibility boundary—and hence preservation of the class of licensed constructions up to lawful equivalence.

M. Plan of the paper

Section II states the main theorem, its theorem boundaries, and the proof-home posture. Section III fixes the base object recalled from the earlier local theorem package. Section IV defines the support class and the corresponding support labels and deformation notions. Section V proves the local-net boundary theorem: the chosen extended-support objects are not bounded-region local observables, and the strictly local net is insensitive to the associated global-form data. Section VI constructs the sector layer over the fixed local net. Section VII proves the recovery theorem for the theorem-scope global-form data. Section VIII proves compatibility with the fixed local theorem object and states the spectral boundary.

N. Relation to existing approaches

The paper is adjacent to several familiar lines of work without collapsing into any one of them. It is adjacent to algebraic quantum field theory because its starting point is a fixed observable net and because the enlargement problem is naturally phrased in terms of representations and sectors^{9–12}. It is adjacent to the gauge-theory literature on line and surface observables because those are the prototype extended-support objects that may distinguish global forms. It is also adjacent to superselection analysis more generally, since one natural way to organize nonlocal information over a fixed observable net is through a representation-theoretic extension rather than an enlargement of the local algebras.

What is specific here is the fixed-base discipline. The local theorem is treated as already proved. The question is then what can be added over that theorem object without silently changing it.

O. Formalization posture

The theorem-routing and proof architecture underlying the present manuscript are supported by a dedicated Lean 4 audit archive for the Yang–Mills local-net and Clay-extension endpoint layer²⁷. The archive formalizes the manuscript-facing proof route: fixed-base hand-off, endpoint/seam checking, sector-to-completion audit control, and the theorem-routing layer connecting the local-net construction to the Clay-scope completion endpoint. It reports no live project-level `axiom`, `sorry`, `admit`, or `unsafe` on the audited Yang–Mills surface, while keeping standard Lean/classical foundations and imported mathematical background inside an explicit boundary. The role of the formal layer is corroborative rather than substitutional: it audits the theorem-bearing structure once the relevant objects and theorem boundaries have been fixed; it does not replace the analytic proof homes for the predecessor local construction.

P. Proof synopsis

In synopsis, the paper proves the following. Starting from a fixed bounded-region local gauge-invariant observable net, one can construct a sectoral layer carrying a specified class of extended-support observables; these observables are not elements of the bounded-region local

net itself, the corresponding sector structure distinguishes theorem-scope global-form data invisible to the local net, the original local vacuum-sector theorem remains unchanged, and a kernel-faithful GNS spectral bridge shows that every theorem-scope admissible completion inherits the predecessor vacuum gap as its complete-theory gap.

II. MAIN RESULTS, THEOREM BOUNDARIES, AND PROOF-HOME POSTURE

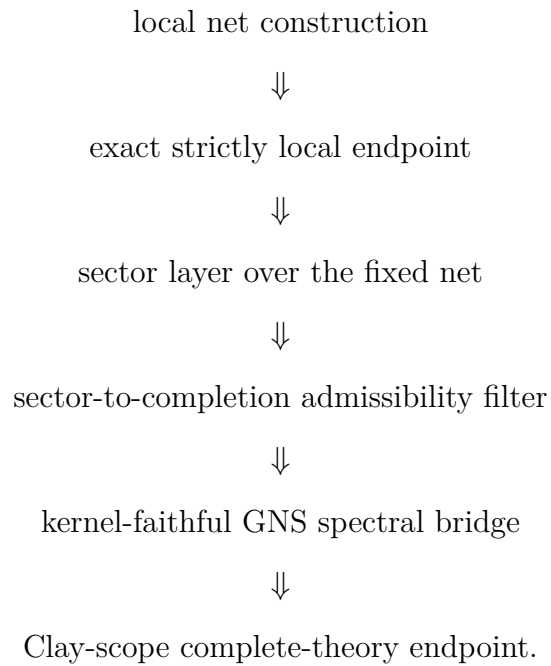
The earlier local theorem package constructed a bounded-region local gauge-invariant Yang–Mills net together with its vacuum-sector reconstruction and vacuum mass-gap theorem. For standalone reading, this section restates exactly the theorem-level part of that package used below, then appends the present extension layer. The present paper does not revisit the analytic local construction. Its task is to identify the theorem-scope sector layer associated with a fixed class of extended-support observables over that local base object. Any construction claimed to yield a determinate Yang–Mills endpoint object in this theorem scope is treated here as a fixed-domain, validity-assessable construction. The theorem boundaries below therefore concern what counts as a lawful continuation of that construction, not an external commentary on it.

Dependency map. For proof-home discipline the load-bearing dependencies of the manuscript are separated as follows.

Dependency class	Content	Proof home
Standard AQFT background	C^* -nets, GNS uniqueness, sector representations, unitary equivalence of covariant systems, and spectral projections	standard observable-net literature
Predecessor Yang–Mills package	local net, vacuum representation, OS/Wightman/Haag–Kastler reconstruction, non-triviality, exact local endpoint, and vacuum mass gap	predecessor Yang–Mills theorem package
Kernel necessity packet	admissibility, standing, reference, irreversibility, no faithful lower generator, no faithful same-domain extension, AMetric/bivalent boundary, uniqueness, and conservation	compressed theorem packet recorded in this manuscript, with upstream AASC theorem provenance
ATS/UEAP endpoint-use packet	anchor/tensor/skin role typing, fail-closed role ambiguity, UEAP carrier/scope/standing adequacy, and no unlicensed claim promotion in sector-to-completion use	restricted packet proved in this manuscript, with upstream ATS/UEAP theorem provenance
Present manuscript	extended-support role layer, sector realization over the fixed local net, sector-to-completion filter, kernel-faithful completion classification, and complete-theory spectral bridge	this paper

A. Single proof chain to the Clay endpoint

Read as a Clay-scope theorem package, the local Yang–Mills proof and the present extension form one proof chain with explicit handoffs:



More explicitly:

Stage	Theorem content	Handoff to the next stage
Local construction	The predecessor package constructs the bounded-region local gauge-invariant Yang–Mills net, its OS/Wightman vacuum representation, non-triviality witness, and vacuum Hamiltonian gap.	Produces the fixed local theorem object $\mathfrak{B}(G)$.
Exact local endpoint	The predecessor endpoint identifies the theorem object as strictly local and excludes extended-support sector data from bounded-region local generators.	Makes extended-support data admissible only as structure over the fixed net, not inside it.
Sector extension	The present paper fixes $\mathfrak{E}(G)$, realizes its theorem-scope data through the predecessor reconstruction-side sector package, and organizes the sector class $\mathfrak{S}_{\text{ext}}(G)$ over $\mathfrak{B}(G)$.	Supplies sector data without changing the local theorem object.
Completion filter	Sector data enter a same-theory completion claim only after passing the ATS/UEAP, scope-faithful, kernel-faithful filter.	Prevents sector representation from being promoted automatically to a complete-theory Hamiltonian claim.
Spectral bridge	Every theorem-scope admissible completion has no new locally detectable subgap state and no additional subgap copy of the vacuum representation.	Forces $E_H([0, m^*))\mathcal{H} = \mathbb{C}\Omega$.
Clay endpoint	Any same-domain Clay-scope realization is identified, up to lawful AQFT equivalence, with the fixed local theorem object and its admissible completions.	The predecessor gap is the complete-theory gap in the theorem scope of this package.

Thus the present paper is not a second proof of the local Yang–Mills construction. It is the completion-control part of the same proof chain: it shows that the exact local theorem object supplied by the predecessor package is not weakened, enlarged, or given new subgap content by theorem-scope sectoral or ultraviolet completions.

B. Main theorem and claim boundaries

Theorem II.1 (Main theorem). *Fix a compact connected simple gauge group G , and let $\mathcal{A}_{\text{loc}}(G)$ be the bounded-region local gauge-invariant Haag–Kastler net furnished by the earlier local theorem package together with its vacuum representation. Let $\mathfrak{E}(G)$ denote the extended-support class fixed in Section IV. Then the following hold.*

- (A) **Strictly local insensitivity.** *The strictly local observable net $\mathcal{A}_{\text{loc}}(G)$ does not detect the theorem-scope global-form data carried by $\mathfrak{E}(G)$.*
- (B) **Exclusion from bounded-region local algebras.** *No nontrivial object in $\mathfrak{E}(G)$ belongs to a bounded-region local algebra $\mathcal{A}_{\text{loc}}(O)$ with O bounded.*
- (C) **Sectoral realization over the fixed net.** *The predecessor Yang–Mills reconstruction package, together with the theorem-scope extended-support class fixed in Section IV, determines a canonically defined sector class $\mathfrak{S}_{\text{ext}}(G)$ over the fixed local net.*
- (D) **Global-form recovery.** *The sector class $\mathfrak{S}_{\text{ext}}(G)$ distinguishes the theorem-scope global-form data that are invisible at the strictly local level through that realized sector layer.*
- (E) **Compatibility with the predecessor local theorem.** *The construction of $\mathfrak{S}_{\text{ext}}(G)$ leaves unchanged the bounded-region local net, its vacuum reconstruction, and the predecessor vacuum-sector mass-gap theorem.*
- (F) **Kernel-faithful GNS spectral bridge and complete-theory endpoint.** *Every theorem-scope admissible sectoral or ultraviolet completion of the fixed local vacuum theory satisfies subgap local-state conservation and subgap local-state rigidity relative to the fixed local algebra. Consequently, the predecessor vacuum gap remains the complete-theory gap on every such completion.*

The theorem-level endpoint. Part (F) is the complete-theory endpoint of the manuscript. It is not a heuristic transfer principle but a kernel-faithful GNS/spectral theorem using standard operator-algebraic machinery: once a completion is shown to be theorem-scope admissible over the fixed local vacuum theory, subgap local-state conservation and subgap rigidity force

$$E_H([0, m^*))\mathcal{H} = \mathbb{C}\Omega,$$

so the predecessor vacuum gap m^* is the complete-theory gap on that completion.

Structural provenance of the closure layer. The closure language used in Part (F) of the main theorem is not a new axiomatic layer local to the present manuscript. The restricted kernel consequences needed here are stated internally below, with upstream AASC references providing expanded theorem provenance: bivalence as a governance-level necessity theorem, the AMetric boundary and non-parameterization package, uniqueness of the admissible interior, and standing–reuse-stable-admissibility equivalence^{13–16}. The present paper uses operator-algebraic translations and fixed-domain consequences of those results over the fixed local Yang–Mills theorem object; it does not treat them as stipulative local axioms.

Bridge claim for the endpoint. The endpoint does not rely merely on the assertion that “any coherent domain” instantiates the kernel. Rather, the manuscript’s claim is narrower and theorem-scoped: once a non-trivial Yang–Mills theory satisfying the required existence standard has been constructed, its fixed local theorem object supplies the exact side conditions needed for the theorem-derived necessity constraints used at the endpoint. The closure argument is therefore attached to the constructed theorem object itself, not to an uninstantiated meta-level slogan.

The contribution is therefore layered. The predecessor Yang–Mills paper remains the proof home for the local observable net, the Haag–Kastler/Wightman realization route, the non-triviality witness, and the vacuum-sector theorem. The present paper adds the extended-support and sector layer over that fixed object, proves the corresponding subgap conservation and rigidity statements in translated operator-algebraic form, and uses them to obtain the kernel-faithful complete-theory spectral bridge in the theorem scope fixed here.

Non-claim. The present paper does *not* claim that the chosen extended-support objects are elements of the bounded-region local observable algebras.

Non-claim. The present paper does *not* claim a fresh lower derivation of the structural

primitives used in the closure layer. The restricted kernel consequences required for the endpoint are stated in this manuscript, with upstream AASC work serving as expanded theorem provenance.

Non-claim. The present paper does *not* claim a classification of every possible nonlocal sector over the fixed local net. It treats only the class $\mathfrak{E}(G)$ fixed in Section IV.

Non-claim. The present paper does *not* claim that an arbitrary extension of the fixed local theory preserves the predecessor gap. The complete-theory endpoint proved here is only for theorem-scope admissible sectoral or ultraviolet completions.

Non-claim. The present paper does *not* claim that every non-vacuum sector representation is a scope-faithful completion of the vacuum theory. Sector representations enter the complete-theory spectral bridge only after satisfying the sector-to-completion admissibility filter of Section VIII E.

Non-claim. The present paper does *not* tie the endpoint object to an auxiliary faithful Wilson regularization. Such choices belong to the proof route of the predecessor package; the theorem object is attached to the compact connected simple gauge group G and its fixed bounded local gauge-invariant net.

C. Fixed base theorem package

The present manuscript uses the predecessor Yang–Mills result only through the following theorem-level package.

Proposition II.2 (Recalled predecessor local theorem package). *For each compact connected simple gauge group G , the predecessor work furnishes a local theorem package with the following properties.*

- (i) *there exists a bounded-region local gauge-invariant Haag–Kastler net $O \mapsto \mathcal{A}_{\text{loc}}(O; G)$ and its quasi-local algebra $\mathcal{A}_{\text{loc}}(G)$;*
- (ii) *there exists a distinguished vacuum representation $(\pi_{\text{vac}}, \mathcal{H}_{\text{vac}}, \Omega_{\text{vac}})$ with vacuum Hamiltonian H_{vac} ;*
- (iii) *the predecessor theorem package establishes the corresponding OS/Wightman/Haag–Kastler reconstruction route and vacuum mass-gap theorem;*

- (iv) *the predecessor theorem package supplies the non-triviality witness and the Minkowski Hamiltonian spectral transfer by which the OS-side gap is inherited by the reconstructed local vacuum representation;*
- (v) *the predecessor theorem package furnishes the reconstruction-side sector and realization data appearing at the endpoint packet level, including reconstructed sectors, realized sector objects, and realization maps linking reconstruction data to sectoral realization;*
- (vi) *the predecessor theorem package establishes the exact local-net endpoint together with exclusion of extended-support sector data from the bounded-region local generators of that endpoint; and*
- (vii) *the theorem-level object is strictly local: bounded-region local observables are the proved object, while extended-support observables are not part of the bounded-region local generators.*

Proposition II.3 (Exact handoff from local-net endpoint to present start point). *The fixed base object $\mathfrak{B}(G)$ used in the present manuscript is precisely the theorem-level local endpoint supplied by the predecessor Yang–Mills package: the bounded-region local gauge-invariant net, its quasi-local algebra, its OS/Wightman vacuum representation, and its vacuum Hamiltonian with predecessor mass gap. The present manuscript neither changes this object nor uses its sector/completion results to strengthen the predecessor local theorem.*

Proof. By Theorem II.2, the predecessor package supplies the local net, vacuum representation, standard-form reconstruction route, non-triviality witness, exact local endpoint, and vacuum-sector mass-gap theorem. By definition, $\mathfrak{B}(G)$ retains exactly the local net, vacuum representation, vacuum vector, and Hamiltonian from that endpoint. Extended-support sector data are excluded from the bounded local generators by item (vi) of Theorem II.2; hence the sectoral and completion structures studied later can only be organized over the fixed object. They cannot redefine the local theorem object without leaving the theorem scope of the present paper. $\square$

Proposition II.4 (Recalled predecessor reconstruction-side sector package). *For each compact connected simple gauge group G , the predecessor Yang–Mills theorem package, in particular at the reconstruction endpoint recorded in Companion III^A, furnishes reconstruction-side sector data consisting of:*

- (i) *reconstructed sectors and their realization at the endpoint level;*
- (ii) *realized sector objects associated to those reconstruction data;*
- (iii) *realization maps linking reconstruction-side sector data to their realized sector representations; and*
- (iv) *a nonempty compatibility package linking the realized sector structure to the Minkowski-space vacuum representation and its spectral evaluation at the endpoint packet level.*

Corollary II.5 (Scope of the predecessor vacuum theorem). *The predecessor vacuum-sector mass-gap theorem is a theorem about the vacuum representation of the bounded-region local net $\mathcal{A}_{\text{loc}}(G)$. It is not already a theorem about charged, defect, or other nonvacuum sectors associated with $\mathfrak{E}(G)$.*

Proof. The earlier local theorem package fixes the local observable net and its vacuum representation. The extended-support sector layer studied here is additional structure over that fixed base and is not part of the original vacuum theorem unless separately proved. $\square$

D. Proof-home posture

The proof-home posture is one-directional.

- (i) The predecessor Yang–Mills project remains the proof home for the local net and its vacuum-sector theorem.
- (ii) The present paper is the proof home for the theorem-scope extended-support comparison class, the bridge from that comparison class to the predecessor reconstruction-side sector package, the corresponding sectoral distinction theorem, and the admissible-completion endpoint consequences.
- (iii) No theorem proved in the present paper is used to strengthen or reinterpret the predecessor local theorem package.

The only background imported here is the standard observable-net and sector language needed to formulate an extension problem over a fixed Haag–Kastler net. No hidden enlargement of the theorem-level object is introduced through notation alone.

Remark II.6 (Why the theorem boundaries are stated first). The main technical risk in a paper of this kind is theorem-object drift. Once the local theorem has already been proved, one must separate that theorem from any additional nonlocal or sectoral structure organized over it. This section states that separation before the construction begins.

III. PRELIMINARIES AND FIXED BASE OBJECT

This section fixes the base object over which the present sector analysis is carried out. The aim is not to reproduce the predecessor Yang–Mills theorem in full detail, but to isolate the part of that theorem package that is used here and to fix notation for the local net, the vacuum representation, and the vacuum Hamiltonian.

A. Notation for the local net and vacuum representation

Fix a compact connected simple gauge group G . For a bounded spacetime region O , write

$$O \longmapsto \mathcal{A}_{\text{loc}}(O; G)$$

for the bounded-region local gauge-invariant observable net furnished by the earlier local theorem package. When the gauge group is fixed, we suppress G and write simply $\mathcal{A}_{\text{loc}}(O)$. The quasi-local C^* -algebra generated by the bounded-region net is denoted by

$$\mathcal{A}_{\text{loc}}(G) := C^*\left(\bigcup_O \mathcal{A}_{\text{loc}}(O; G)\right).$$

The distinguished vacuum representation is written

$$\pi_{\text{vac}} : \mathcal{A}_{\text{loc}}(G) \rightarrow B(\mathcal{H}_{\text{vac}}),$$

with vacuum Hilbert space $\mathcal{H}_{\text{vac}}$, vacuum vector $\Omega_{\text{vac}} \in \mathcal{H}_{\text{vac}}$, and vacuum Hamiltonian H_{vac} .

Definition III.1 (Fixed local observable net). For each compact connected simple gauge group G , the *fixed local observable net* means the bounded-region local gauge-invariant net

$$O \longmapsto \mathcal{A}_{\text{loc}}(O; G)$$

together with its quasi-local algebra $\mathcal{A}_{\text{loc}}(G)$ and the distinguished vacuum representation $(\pi_{\text{vac}}, \mathcal{H}_{\text{vac}}, \Omega_{\text{vac}})$.

B. Recalled earlier local theorem package

Proposition III.2 (Recalled predecessor local theorem package). *For each compact connected simple gauge group G , the earlier local theorem package furnishes a fixed local theorem package with the following properties.*

- (i) *there exists a bounded-region local gauge-invariant Haag–Kastler net $O \mapsto \mathcal{A}_{\text{loc}}(O; G)$ and its quasi-local algebra $\mathcal{A}_{\text{loc}}(G)$;*
- (ii) *there exists a distinguished vacuum representation $(\pi_{\text{vac}}, \mathcal{H}_{\text{vac}}, \Omega_{\text{vac}})$ with vacuum Hamiltonian H_{vac} ;*
- (iii) *that fixed local theorem package establishes the OS/Wightman/Haag–Kastler reconstruction route and the corresponding vacuum-sector mass-gap theorem;*
- (iv) *the Minkowski Hamiltonian in the reconstructed vacuum representation inherits the predecessor spectral gap; and*
- (v) *the theorem-level object constructed there is strictly local.*

Proposition III.3 (Recalled predecessor reconstruction-side sector package). *For each compact connected simple gauge group G , the predecessor Yang–Mills theorem package furnishes, at the reconstruction endpoint, theorem-level reconstruction-side sector and realization data over the fixed local theorem object, including:*

- (i) *a reconstruction-side sector class;*
- (ii) *a realized sector class;*
- (iii) *a realization map from reconstruction-side sector data, together with the corresponding correlation data, to realized sector objects; and*
- (iv) *a nonempty compatibility package linking transport, realization, and Minkowski-gap evaluation at the endpoint packet level.*

In particular, the predecessor theorem package does not stop at the bounded local net and its vacuum-sector gap, but already furnishes the reconstruction-side sector/realization machinery on which the present manuscript builds.

Proof. This is recalled theorem-level content from the predecessor Yang–Mills endpoint packet. The reconstruction-side package supplies reconstructed sectors, realized sector objects, the

corresponding realization map, and the associated transport/compatibility bundle at the endpoint route. The present manuscript uses these data only through their theorem-level consequences over the fixed local theorem object. $\square$

Proposition III.4 (Recalled exact-local-endpoint exclusion principle). *For each compact connected simple gauge group G , the predecessor Yang–Mills theorem package establishes an exact local-net endpoint whose bounded-region local generators exclude extended-support sector data. Equivalently, extended-support sectoral data are not already elements of the bounded-region local observable algebras of the exact local endpoint.*

Proof. This is recalled theorem-level content from the predecessor Yang–Mills endpoint theorem package: the exact local endpoint is strictly local, while extended-support sector data are excluded from its bounded local generators. The present manuscript specializes that exclusion to the theorem-scope comparison class fixed later. $\square$

Remark III.5 (Fixed local base object versus endpoint packet data). In the present manuscript, the symbol $\mathfrak{B}(G)$ denotes the fixed local theorem object strictly speaking: the bounded-region local observable net, its quasi-local algebra, the distinguished vacuum representation, and the vacuum Hamiltonian. The predecessor Yang–Mills theorem package also furnishes reconstruction-side endpoint data over that same fixed object. Those endpoint data are recalled separately when needed below; they do not redefine the fixed base object, but provide additional theorem-level structure over it.

Remark III.6 (Why only this package is recalled). The present paper needs only the fixed local observable net, the distinguished vacuum representation, the predecessor standard-form reconstruction and vacuum theorem, and the locality boundary of that theorem object, together with the theorem-level realization package recalled in Theorem III.3. Stating the predecessor input in this reduced form keeps the present manuscript from drifting into an implicit reproof of the earlier construction.

C. The fixed base object

Definition III.7 (Fixed base object). For each compact connected simple gauge group G , define

$$\mathfrak{B}(G) := \left(\mathcal{A}_{\text{loc}}(G), \pi_{\text{vac}}, \mathcal{H}_{\text{vac}}, \Omega_{\text{vac}}, H_{\text{vac}} \right).$$

Thus $\mathfrak{B}(G)$ is the bounded-region local gauge-invariant observable net together with the distinguished vacuum representation and vacuum Hamiltonian supplied by the earlier local theorem package.

All later constructions are relative to $\mathfrak{B}(G)$. In particular, the present paper does not redefine the local net, replace the vacuum representation, or enlarge the theorem-level local object by stipulation.

Corollary III.8 (Fixed-base reading rule). *Every construction in the present paper is relative to the fixed base object $\mathfrak{B}(G)$. Consequently:*

- (a) *a structure belonging to some bounded-region local algebra $\mathcal{A}_{\text{loc}}(O; G)$ is part of the predecessor local theorem object;*
- (b) *a structure organized through sectors or representations over $\mathfrak{B}(G)$ belongs to the new layer studied here; and*
- (c) *any construction that changes the bounded-region local algebras, the vacuum representation, or the locality boundary fixed by the earlier local theorem package lies outside the theorem scope of the present paper.*

Proof. Part (a) is immediate from Theorems III.1 and III.2. Part (b) is the meaning of working *over* the fixed base object rather than inside its bounded-region algebras. Part (c) follows because altering the local net, the vacuum representation, or the locality boundary changes the base object itself. $\square$

Remark III.9 (Boundary of recall). The predecessor theorem package functions here as fixed input. The present paper does not reopen that local construction, and the later sector layer is not allowed to back-propagate into the statement of the fixed local theorem object.

Dependency summary.

Inputs. the predecessor Yang–Mills local theorem package recalled at theorem level in Theorem III.2

Imported standard background. none beyond standard observable-net and vacuum-representation language

Conclusion at this node. the present paper works over one fixed local observable net and one fixed vacuum representation, packaged as $\mathfrak{B}(G)$

Used later in. Sections IV–VIII

IV. EXTENDED-SUPPORT OBSERVABLE CLASS

The main theorem is deliberately conditional on a fixed class of extended-support objects. This section fixes that class, its support labels, and the equivalence notions used later in the paper.

A. Underlying spacetime and support objects

Throughout, spacetime means the Minkowski spacetime on which the fixed local net of Section III is realized.

Definition IV.1 (Line support). A *line support* is the image of a piecewise smooth oriented closed curve

$$\gamma : S^1 \rightarrow \mathbb{R}^{1,3}$$

with compact image. Two parametrized curves define the same line support if they have the same image and differ only by an orientation-preserving piecewise smooth reparametrization.

Definition IV.2 (Surface support). A *surface support* is the image of a compact piecewise smooth oriented closed two-manifold

$$\Sigma : M^2 \rightarrow \mathbb{R}^{1,3}$$

with compact image and empty boundary. Two parametrized surfaces define the same surface support if they have the same image and differ only by an orientation-preserving piecewise smooth reparametrization.

Remark IV.3 (Closed supports only). The present paper treats only closed line and closed surface supports. Open curves with endpoints, surfaces with boundary, junction networks, and more elaborate defect webs are outside theorem scope.

B. Label data

Definition IV.4 (Admissible line and surface labels). For each compact connected simple gauge group G , fix two label sets:

- (i) a set $\Lambda_{\text{line}}(G)$ of admissible line labels; and

(ii) a set $\Lambda_{\text{surf}}(G)$ of admissible surface labels.

These sets are part of the theorem input for the present paper. They are not assumed to coincide with bounded-region local observable content, and they are not assumed to exhaust every nonlocal charge class.

Definition IV.5 (Formal extended-support objects). A *formal line-type object* is a pair (γ, λ) consisting of a line support γ and a label $\lambda \in \Lambda_{\text{line}}(G)$. A *formal surface-type object* is a pair (Σ, μ) consisting of a surface support Σ and a label $\mu \in \Lambda_{\text{surf}}(G)$.

The theorem-scope extended-support class is

$$\mathfrak{E}(G) := \mathfrak{E}_{\text{line}}(G) \sqcup \mathfrak{E}_{\text{surf}}(G).$$

Remark IV.6 (Support first, realization later). At this stage the objects in $\mathfrak{E}(G)$ are support-and-label data only. They are not yet assumed to be operators in any algebra or sectors over the net. Their realization is the subject of later sections.

Definition IV.7 (Forced theorem-scope sector role). A *forced theorem-scope sector role* is a role in the fixed extended-support comparison target whose occupancy is required by the support type and label data of Theorem IV.5. Thus a line or surface label in $\Lambda_{\text{line}}(G)$ or $\Lambda_{\text{surf}}(G)$ is not a free adornment of a bounded-region local observable. It names the role that the predecessor reconstruction-side realization package must occupy when that support-and-label datum is represented over the fixed local theorem object.

Proposition IV.8 (Role-cardinality discipline for the theorem-scope class). *Let $\mathcal{R}_{\text{adm}}(G)$ denote the admissible residue of the theorem-scope extended-support target after quotienting by support-equivalence and admissible deformation, and let $n_{\text{role}}(G)$ be the number of forced theorem-scope roles selected by the line and surface label data in Theorem IV.4. A realization of $\mathfrak{E}(G)$ is licensed as the specified theorem-scope sector layer only when*

$$|\mathcal{R}_{\text{adm}}(G)/\sim_{\text{role}}| = n_{\text{role}}(G).$$

In particular, singleton closure is only the special case $n_{\text{role}}(G) = 1$; the present sector-extension target may require several realized roles without thereby altering the bounded-region local net.

Proof. The local theorem object is closed only with respect to bounded-region gauge-invariant observables. By Theorem III.4, that closure does not populate extended-support sector

roles. The roles fixed here are therefore target roles for the reconstruction-side sector layer, not additional local generators. A realization that identifies two forced roles with different support-and-label residue fails to realize the declared target, while a realization that adds roles not fixed by the support-and-label data changes the theorem scope. The displayed equality is exactly the finite role-occupancy condition expressing this fixed-target discipline. $\square$

Remark IV.9 (Sector roles are not a failure of local closure). The need for a sector role is not evidence that the predecessor local theorem was incomplete. Local closure closes the bounded-region observable carrier. Extended-support sector roles are different theorem-scope roles organized over that carrier, and their realization is assessed by role occupancy rather than by membership in a bounded local algebra²².

C. Equivalence and deformation

Definition IV.10 (Support equivalence). Two formal line-type objects (γ, λ) and (γ', λ') are *support-equivalent* if $\lambda = \lambda'$ and the underlying line supports coincide as subsets of spacetime. Two formal surface-type objects (Σ, μ) and (Σ', μ') are *support-equivalent* if $\mu = \mu'$ and the underlying surface supports coincide.

Definition IV.11 (Admissible deformation). An *admissible deformation* of a formal line-type object (γ, λ) is a compactly supported ambient isotopy through oriented closed curves preserving the label λ . An *admissible deformation* of a formal surface-type object (Σ, μ) is a compactly supported ambient isotopy through compact oriented closed surfaces preserving the label μ .

Definition IV.12 (Deformation equivalence). Two formal line-type objects are *deformation-equivalent* if they are related by a finite chain of admissible deformations and support-equivalences. The same definition applies to formal surface-type objects.

Proposition IV.13 (Well-posedness of the comparison class). *Support-equivalence and deformation-equivalence are equivalence relations on $\mathfrak{E}_{\text{line}}(G)$ and on $\mathfrak{E}_{\text{surf}}(G)$.*

Proof. This is immediate from the definitions. Support-equivalence is equality of support together with equality of label. Deformation-equivalence is generated from support-equivalence and admissible ambient isotopy. $\square$

D. Frozen theorem scope

Proposition IV.14 (Frozen extended-support class). *For each compact connected simple gauge group G , the phrase extended-support observable class means exactly the class $\mathfrak{E}(G)$ of Theorem IV.5, considered modulo the support-equivalence and deformation notions above. In particular, the theorem scope of the paper includes only*

- (a) *closed oriented line supports with labels in $\Lambda_{\text{line}}(G)$, and*
- (b) *closed oriented surface supports with labels in $\Lambda_{\text{surf}}(G)$,*

and excludes open lines, endpoint fields, surfaces with boundary, junction networks, and all other extended-support structures not explicitly covered by Theorem IV.5.

Proof. The inclusion statement is the content of Theorem IV.5. The exclusion statement follows from Theorem IV.3. $\square$

Corollary IV.15 (No theorem-object drift at the support stage). *The present manuscript studies a fixed support-and-label class. Enlarging that class requires a new theorem statement and is not absorbed implicitly into the present one.*

Proof. The theorem scope is fixed by Theorem IV.14. Anything outside that class lies outside the present theorem statement. $\square$

E. Realization compatibility with the predecessor Yang–Mills endpoint package

Proposition IV.16 (Theorem-scope realization compatibility). *Let G be a compact connected simple gauge group, let $\mathfrak{B}(G)$ be the fixed local Yang–Mills theorem object, and let $\mathfrak{E}(G)$ be the theorem-scope extended-support class fixed in the present section. Then the predecessor Yang–Mills reconstruction-side endpoint package is compatible with $\mathfrak{E}(G)$ in the following sense: each theorem-scope extended-support object determines a corresponding reconstruction-side realization datum over the fixed local theorem object.*

Proof. The predecessor Yang–Mills theorem package already furnishes reconstruction-side sector and realization data over the fixed local theorem object by Theorem III.3. The present section fixes the theorem-scope extended-support comparison class in support-and-label form and, by Theorems IV.7 and IV.8, fixes the forced sector roles whose occupancy realizes

that class. The compatibility claim is that the support-and-label objects considered here are realized within that predecessor reconstruction-side package rather than introduced independently of it. The present manuscript uses that compatibility to pass from theorem-scope extended-support data to the realized sector layer over the fixed local net. $\square$

Proposition IV.17 (Realization respects support and deformation equivalence). *Let $\xi, \eta \in \mathfrak{E}(G)$.*

- (i) *If ξ and η are support-equivalent, then they determine the same reconstruction-side realization datum over the fixed local theorem object.*
- (ii) *If ξ and η are deformation-equivalent, then they determine transport-equivalent realized sector data over the fixed local net.*

Proof. Support-equivalence preserves the support-and-label data at theorem scope and therefore preserves the corresponding reconstruction-side realization datum. Deformation-equivalence is generated by support-equivalence together with admissible ambient deformation preserving the theorem-scope label. Since the predecessor realization package is used only up to the lawful transport and equivalence notions appropriate to the fixed local theorem object, deformation-equivalent theorem-scope objects determine transport-equivalent realized sector data. $\square$

Remark IV.18 (Status of the theorem-scope class). The theorem-scope class $\mathfrak{E}(G)$ is fixed here as the comparison class for the present manuscript. It is not introduced independently of the predecessor Yang–Mills endpoint package. Rather, it isolates, at paper scope, the extended-support class whose realization is carried by the reconstruction-side sector machinery already available at that endpoint.

Dependency summary.

Inputs. the fixed base object of Section III, together with the frozen choice of support and label classes introduced here

Imported standard background. only the ordinary smooth and topological language needed to define closed curves, closed surfaces, and ambient isotopy in Minkowski spacetime

Conclusion at this node. the paper now has a fixed theorem-scope class $\mathfrak{E}(G)$ of extended-support objects, together with the exact equivalence and deformation notions used later

Used later in. Sections V–VIII

V. LOCAL-NET BOUNDARY

This section proves the boundary theorem that separates the predecessor local net from the extended-support sector layer studied here. The two key points are that the fixed local net is insensitive to the theorem-scope global-form data carried by $\mathfrak{E}(G)$, and that the chosen extended-support objects are therefore not elements of bounded-region local algebras.

Imported exact-local-endpoint exclusion principle.

Proposition V.1. *The predecessor Yang–Mills endpoint theorem establishes that the exact bounded-region local observable net excludes extended-support sector data from its bounded local generators. Equivalently, the adjoint-local bounded generators of the local endpoint are insensitive to global-form distinctions whose residue is carried only by line, surface, or sector realization data.*

Consequently, any distinction carried purely at the level of reconstruction-side or sectoral realization cannot already belong to a bounded-region local algebra $\mathcal{A}_{\text{loc}}(O; G)$ of the fixed theorem object.

This is the local-boundary restatement of the recalled predecessor endpoint exclusion principle recorded in Theorem III.4.

A. Strictly local insensitivity

Definition V.2 (Local-net equivalence). Let G and G' be compact connected simple gauge groups whose predecessor theorem packages determine canonically identified bounded-region local observable nets. We say that G and G' are *locally equivalent at theorem level* if, for every bounded region O , there is a canonical isomorphism

$$\iota_{G \rightarrow G'}(O) : \mathcal{A}_{\text{loc}}(O; G) \xrightarrow{\sim} \mathcal{A}_{\text{loc}}(O; G')$$

compatible with isotony and with the distinguished vacuum representations recalled in Section III.

Proposition V.3 (Transport of strictly local observables). *Assume G and G' are locally equivalent at theorem level. Then every bounded-region local observable*

$$A \in \mathcal{A}_{\text{loc}}(O; G)$$

has a canonically corresponding observable

$$\iota_{G \rightarrow G'}(O)(A) \in \mathcal{A}_{\text{loc}}(O; G').$$

In particular, bounded-region local algebras cannot distinguish data that are invisible under the canonical local-net identification.

Proof. This is immediate from Theorem V.2. □

Proposition V.4 (Bounded local observables depend only on the bounded local transport class). *Let G and G' be compact connected simple gauge groups whose predecessor theorem packages determine canonically identified bounded-region local observable nets. Then every bounded-region local observable is determined entirely by the bounded local transport class of the fixed theorem object and cannot detect distinctions carried only by the realized sector layer.*

Proof. This is immediate from Theorem V.3: bounded-region local observables are preserved under the canonical local-net identifications of the fixed theorem object, so they depend only on the bounded local transport class and not on distinctions carried purely by the realized sector layer. □

Proposition V.5 (Imported exact-local-endpoint blindness principle). *Let G be a compact connected simple gauge group, and let $\mathfrak{B}(G)$ be the fixed local Yang–Mills theorem object. The predecessor exact-local-endpoint theorem implies that the bounded-region local endpoint is insensitive to distinctions carried only by the realized sector layer and not by the bounded local transport class itself. In particular, any extended-support distinction whose content survives only at sector level is invisible to the bounded-region local observable algebras of the fixed theorem object.*

Proof. By Theorem III.4, the predecessor Yang–Mills endpoint theorem already excludes extended-support sector data from the bounded local generators of the exact local endpoint. Therefore any distinction whose content is carried only by the realized sector layer and not by the bounded local transport class cannot be detected by an observable in a bounded-region local algebra of the fixed theorem object. □

Theorem V.6 (Strictly local global-form insensitivity). *Let G be a compact connected simple gauge group, and let $\mathfrak{B}(G)$ be the fixed base object of Section III. Then the bounded-region local observable net*

$$O \longmapsto \mathcal{A}_{\text{loc}}(O; G)$$

and its quasi-local algebra $\mathcal{A}_{\text{loc}}(G)$ are insensitive to the theorem-scope global-form data represented only through the extended-support class $\mathfrak{E}(G)$. Equivalently, any two realizations canonically identified at the level of the fixed local net cannot be separated by an observable belonging to a bounded-region local algebra.

Remark V.7 (Proof-home posture). This statement is a theorem-scope specialization of the imported exact-local-endpoint exclusion principle. The role of the present argument is to identify, for the class $\mathfrak{E}(G)$ fixed in Section IV, the corresponding global-form data carried only at sector level and therefore invisible to the strictly local observable net.

Proof. By Theorems V.1 and V.3, every bounded-region local observable is preserved under the canonical local-net identifications built into the fixed-base theorem package, while extended-support sector data remain excluded from the bounded local generators of the exact local endpoint. The theorem-scope global-form data treated in the present paper are, by construction, precisely those data carried only at sector level and not encoded in the bounded-region local observable net itself. Hence no bounded-region local observable can distinguish them. $\square$

Corollary V.8 (Strictly local blindness in fixed-base notation). *For every bounded region O , every observable*

$$A \in \mathcal{A}_{\text{loc}}(O; G)$$

is determined entirely at the level of the fixed base object $\mathfrak{B}(G)$. In particular, the bounded-region local net does not already carry the theorem-scope global-form data assigned in the present paper to $\mathfrak{E}(G)$.

Proof. This is just Theorem V.6 in the notation of Section III. $\square$

B. Exclusion from bounded-region local algebras

Definition V.9 (Bounded-region local realization). Let $\xi \in \mathfrak{E}(G)$. We say that ξ admits a *bounded-region local realization* if there exists a bounded region O and an observable

$$A_\xi \in \mathcal{A}_{\text{loc}}(O; G)$$

such that:

- (i) A_ξ is sufficient to represent the observable content of ξ within the fixed base object; and
- (ii) under every canonical local-net identification, the transport of A_ξ preserves all theorem-scope data assigned to ξ .

Remark V.10 (Local realization is algebraic, not geometric). A loop or surface may lie inside a bounded spacetime region as a subset, yet still fail to define a bounded-region local observable of the fixed net. Local realization means realization inside a bounded-region local algebra, not mere geometric containment.

Lemma V.11 (Local realization collapses global-form sensitivity). *If $\xi \in \mathfrak{E}(G)$ admits a bounded-region local realization, then its observable content is insensitive to the theorem-scope global-form data. In particular, any genuinely global-form-sensitive object of $\mathfrak{E}(G)$ fails to admit such a realization.*

Proof. Assume that ξ admits a bounded-region local realization $A_\xi \in \mathcal{A}_{\text{loc}}(O; G)$. By Theorems V.5 and V.6, every bounded-region local observable is insensitive to theorem-scope global-form data invisible to the local net. Hence A_ξ cannot preserve that information under the transport required in Theorem V.9. This contradicts clause (ii) of the definition. $\square$

Theorem V.12 (Extended-support objects are not elements of bounded-region local algebras). *Let $\mathfrak{E}(G)$ be the frozen extended-support class of Section IV. Then no nontrivial element of $\mathfrak{E}(G)$ belongs to a bounded-region local algebra $\mathcal{A}_{\text{loc}}(O; G)$ with O bounded. Equivalently, the chosen extended-support objects are not bounded-region local observables of the fixed net.*

Proof. By Theorem III.4, the predecessor Yang–Mills endpoint theorem already excludes extended-support sector data from the bounded local generators of the exact local endpoint. By Theorem IV.16, each theorem-scope extended-support object in $\mathfrak{E}(G)$ is realized through that extended-support sector layer over the fixed local theorem object. Therefore no nontrivial

theorem-scope extended-support object belongs to a bounded-region local algebra of the fixed exact local endpoint. $\square$

Corollary V.13 (The local theorem object remains strictly local). *The theorem-level local object $O \mapsto \mathcal{A}_{\text{loc}}(O; G)$ remains unchanged by the passage to the extended-support class $\mathfrak{E}(G)$. Later sections construct only an additional layer over the fixed base object $\mathfrak{B}(G)$.*

Proof. By Theorem V.12, the chosen extended-support objects are not elements of the bounded-region local algebras. Hence later constructions involving them cannot enlarge the local observable net itself. $\square$

Remark V.14 (Role of the boundary theorem). Theorems V.6 and V.12 provide the seam between the predecessor local theorem and the new sector layer. They show that the missing nonlocal data are neither already in the bounded-region local net nor recoverable there by relabeling.

Dependency summary.

Inputs. the fixed base object $\mathfrak{B}(G)$ from Section III and the frozen extended-support class $\mathfrak{E}(G)$ from Section IV

Imported standard background. none beyond standard observable-net language

Conclusion at this node. the local theorem object is strictly local, while the chosen extended-support objects are not bounded-region local observables

Used later in. Sections VI–VIII

VI. SECTOR CONSTRUCTION OVER THE LOCAL NET

Section V established that the theorem-scope extended-support objects are not elements of the bounded-region local algebras of the fixed net. By Section IV, each $\xi \in \mathfrak{E}(G)$ is realized through the reconstruction-side sector and realization package furnished by the predecessor Yang–Mills theorem, with explicit proof-home anchor in Companion III. The problem of the present section is therefore not to construct sector objects *ex nihilo*, but to organize these realized objects into a theorem-scope sector class over the fixed quasi-local algebra.

A. Sector objects over the fixed base

Recall the fixed quasi-local algebra $\mathcal{A}_{\text{loc}}(G)$ and the vacuum representation

$$\pi_{\text{vac}} : \mathcal{A}_{\text{loc}}(G) \rightarrow B(\mathcal{H}_{\text{vac}})$$

of Section III.

Definition VI.1 (Extended-support localized representation). Let $\xi \in \mathfrak{E}(G)$, and let $\text{supp}(\xi) \subset \mathbb{R}^{1,3}$ denote its underlying support. A representation

$$\pi_{\xi} : \mathcal{A}_{\text{loc}}(G) \rightarrow B(\mathcal{H}_{\xi})$$

is called an *extended-support localized representation over ξ* if for every bounded region O spacelike separated from $\text{supp}(\xi)$, the restricted representation

$$\pi_{\xi} \upharpoonright \mathcal{A}_{\text{loc}}(O; G)$$

is unitarily equivalent to the corresponding vacuum restriction

$$\pi_{\text{vac}} \upharpoonright \mathcal{A}_{\text{loc}}(O; G).$$

Definition VI.2 (Extended-support sector datum). An *extended-support sector datum* over $\mathfrak{B}(G)$ is a pair

$$(\xi, \pi_{\xi})$$

consisting of an object $\xi \in \mathfrak{E}(G)$ and an extended-support localized representation π_{ξ} in the sense of Theorem VI.1, arising from the reconstruction-side sector package of Companion III as recalled in Theorem VI.10.

Definition VI.3 (Admissible sector datum). An extended-support sector datum (ξ, π_{ξ}) is *admissible* if:

- (i) it is a representation of the fixed quasi-local algebra $\mathcal{A}_{\text{loc}}(G)$ and not a redefinition of that algebra;
- (ii) its local vacuum agreement away from $\text{supp}(\xi)$ is exactly that of Theorem VI.1; and
- (iii) its representation-theoretic content depends only on the support-and-label data fixed in Section IV, up to the equivalence and transport notions introduced below.

Remark VI.4 (Why the construction is representation-theoretic). The definitions are deliberately conservative. They do not attempt to realize ξ as an element of a bounded-region local algebra. Instead, they measure the effect of ξ through a representation of the fixed quasi-local algebra whose local behavior away from the support agrees with the vacuum sector.

B. Equivalence and transport

Definition VI.5 (Sector equivalence). Two admissible sector data

$$(\xi, \pi_\xi) \quad \text{and} \quad (\eta, \pi_\eta)$$

are *sector-equivalent* if:

- (i) ξ and η are deformation-equivalent in the sense of Theorem IV.12; and
- (ii) π_ξ and π_η are unitarily equivalent as representations of $\mathcal{A}_{\text{loc}}(G)$.

Proposition VI.6 (Sector equivalence is an equivalence relation). *The relation of Theorem VI.5 is an equivalence relation on the class of admissible sector data over $\mathfrak{B}(G)$.*

Proof. Reflexivity is immediate from reflexivity of deformation-equivalence and unitary equivalence. Symmetry follows by inverting the relevant admissible deformation chain and intertwining unitary. Transitivity follows by concatenating deformation chains and composing the corresponding unitaries. $\square$

Definition VI.7 (Deformation transport). Let (ξ, π_ξ) and (η, π_η) be admissible sector data with ξ deformation-equivalent to η . A *deformation transport* from (ξ, π_ξ) to (η, π_η) is the unitary intertwiner induced by the corresponding realization transport in the reconstruction-side sector package of Companion III under deformation-equivalence of ξ and η , concretely given by a map

$$T_{\xi \rightarrow \eta} : \mathcal{H}_\xi \rightarrow \mathcal{H}_\eta$$

implementing the equivalence of π_ξ and π_η as representations of $\mathcal{A}_{\text{loc}}(G)$.

Proposition VI.8 (Transport composition). *If*

$$T_{\xi \rightarrow \eta} : \mathcal{H}_\xi \rightarrow \mathcal{H}_\eta \quad \text{and} \quad T_{\eta \rightarrow \zeta} : \mathcal{H}_\eta \rightarrow \mathcal{H}_\zeta$$

are admissible deformation transports, then

$$T_{\eta \rightarrow \zeta} \circ T_{\xi \rightarrow \eta} : \mathcal{H}_\xi \rightarrow \mathcal{H}_\zeta$$

is again an admissible deformation transport.

Proof. The composite of two unitary intertwiners is again a unitary intertwiner. Compatibility with support data follows from transitivity of deformation-equivalence. $\square$

C. Existence theorem for the theorem-scope sector class

Definition VI.9 (Extended-support sector class). The *extended-support sector class* over the fixed base object $\mathfrak{B}(G)$ is the class

$$\mathfrak{S}_{\text{ext}}(G)$$

of sector-equivalence classes of admissible sector data (ξ, π_ξ) attached to the theorem-scope extended-support class $\mathfrak{E}(G)$.

Theorem VI.10 (Theorem-scope sector realization theorem). *Let $\mathfrak{B}(G)$ be the fixed base object of Section III, and let $\mathfrak{E}(G)$ be the frozen extended-support class of Section IV. Then for every $\xi \in \mathfrak{E}(G)$, the predecessor Yang–Mills reconstruction-side endpoint package furnishes an admissible realized sector datum*

$$(\xi, \pi_\xi)$$

over $\mathfrak{B}(G)$.

Proof. By Theorem III.3, the predecessor Yang–Mills theorem package already furnishes reconstruction-side sector and realization data over the fixed local theorem object. By Theorem IV.16, each theorem-scope extended-support object $\xi \in \mathfrak{E}(G)$ determines a corresponding reconstruction-side realization datum. By construction of the fixed-base representation language used in the present section, that realized datum determines a representation of the fixed quasi-local algebra whose local behavior away from the support agrees with the vacuum restriction, and whose sectoral content depends only on the theorem-scope support-and-label data up to the equivalence and transport notions fixed here. Hence it determines an admissible sector datum (ξ, π_ξ) over $\mathfrak{B}(G)$. $\square$

Proposition VI.11 (Theorem-scope realized sector data are admissible). *The realized sector data furnished by Theorem VI.10 satisfy the admissibility conditions of Theorem VI.3.*

Proof. The predecessor reconstruction-side realization package is organized over the fixed local theorem object and does not redefine the bounded-region local observable net. By Theorem IV.17, the realized sector data depend only on theorem-scope support-and-label data up to the permitted equivalence notions. By the very role of realized sector data over the fixed local net, the corresponding local restrictions away from support agree with the vacuum sector. Therefore the realized data satisfy clauses (i)–(iii) of Theorem VI.3. $\square$

Theorem VI.12 (Existence of the theorem-scope sector class over the fixed local net). *Let $\mathfrak{B}(G)$ be the fixed base object of Section III, and let $\mathfrak{E}(G)$ be the frozen extended-support class of Section IV. Then there exists a canonically defined extended-support sector class*

$$\mathfrak{S}_{\text{ext}}(G)$$

over $\mathfrak{B}(G)$ with the following properties.

- (i) *For every $\xi \in \mathfrak{E}(G)$ there exists at least one admissible sector datum (ξ, π_ξ) .*
- (ii) *Support-equivalent objects determine the same sector class.*
- (iii) *Deformation-equivalent objects determine canonically transport-equivalent sector classes.*
- (iv) *The class $\mathfrak{S}_{\text{ext}}(G)$ is built over the fixed quasi-local algebra $\mathcal{A}_{\text{loc}}(G)$ and does not enlarge the bounded-region local observable algebras themselves.*

Proof. Clause (i) is exactly Theorems VI.10 and VI.11. Clause (ii) follows from Theorem IV.17 together with the definition of sector-equivalence. Clause (iii) follows from the same proposition together with Theorems VI.7 and VI.8. Clause (iv) follows because all admissible sector data are representations over the fixed quasi-local algebra and not redefinitions of the bounded-region local net. $\square$

Remark VI.13 (Status of the existence theorem). Theorem VI.12 is not an abstract quotient construction detached from the Yang–Mills proof home. It is the theorem-scope realization and organization statement obtained by combining the predecessor reconstruction-side sector package with the support-and-label class fixed in Section IV.

Corollary VI.14 (Well-defined sector assignment on deformation classes). *There is a well-defined assignment*

$$\Xi_G : \mathfrak{E}(G)/\sim_{\text{def}} \longrightarrow \mathfrak{S}_{\text{ext}}(G)$$

from deformation-equivalence classes of theorem-scope extended-support objects to the extended-support sector class over the fixed local net.

Proof. By Theorem VI.12, every deformation-equivalence class admits an admissible sector representative, and by Theorem IV.17, support-equivalent and deformation-equivalent representatives descend to the same sector-equivalence class. Hence the assignment is well defined. $\square$

Corollary VI.15 (No local-net enlargement at the sector stage). *The passage from $\mathfrak{E}(G)$ to $\mathfrak{S}_{\text{ext}}(G)$ is not a passage to a larger bounded-region local observable net. It is a passage from the fixed local observable net to a representation-theoretic layer over that net.*

Proof. This is exactly clause (iv) of Theorem VI.12. $\square$

Remark VI.16 (Boundary of the present section). This section fixes the sector objects, their equivalence relation, and the existence theorem for the resulting sector class. It does not yet state which global-form distinctions are separated by that class; that is the content of Section VII. Nor does it state the spectral relation between the sector layer and the predecessor vacuum theorem; that is postponed to Section VIII.

Dependency summary.

Inputs. the fixed base object $\mathfrak{B}(G)$ from Section III, the frozen extended-support class $\mathfrak{E}(G)$ from Section IV, and the local-net boundary theorem of Section V

Imported standard background. only the ordinary representation-theoretic language of sector organization over a fixed observable net

Conclusion at this node. the nonlocal support-and-label data are organized as a sector class $\mathfrak{S}_{\text{ext}}(G)$ over the fixed local net rather than as additional bounded-region local observables

Used later in. Sections VII and VIII

VII. GLOBAL-FORM RECOVERY

The local-net boundary theorem shows that the fixed local net is insensitive to the theorem-scope global-form data carried by $\mathfrak{E}(G)$. The purpose of the present section is to state that insensitivity in fixed-base notation and then prove that the sector class constructed in Section VI distinguishes exactly those same data.

A. What is invisible locally

Definition VII.1 (Local shadow). Let $\xi \in \mathfrak{E}(G)$. Its *local shadow* is the support datum obtained from ξ after forgetting the theorem-scope label data and retaining only the underlying support type:

$$\text{Sh}_{\text{loc}}(\xi) := \begin{cases} \gamma, & \xi = (\gamma, \lambda) \in \mathfrak{E}_{\text{line}}(G), \\ \Sigma, & \xi = (\Sigma, \mu) \in \mathfrak{E}_{\text{surf}}(G). \end{cases}$$

Definition VII.2 (Global-form datum in theorem scope). For $\xi \in \mathfrak{E}(G)$, define the *theorem-scope global-form datum*

$$\text{GF}(\xi) := \begin{cases} \lambda, & \xi = (\gamma, \lambda) \in \mathfrak{E}_{\text{line}}(G), \\ \mu, & \xi = (\Sigma, \mu) \in \mathfrak{E}_{\text{surf}}(G). \end{cases}$$

Thus the theorem-scope global-form data are exactly the admissible line and surface labels fixed in Theorem IV.4, read as forced sector roles in the sense of Theorem IV.7.

Proposition VII.3 (Local-shadow dependence of strictly local observables). *Let $\xi, \eta \in \mathfrak{E}(G)$ satisfy*

$$\text{Sh}_{\text{loc}}(\xi) = \text{Sh}_{\text{loc}}(\eta).$$

Then no bounded-region local observable distinguishes ξ from η by the label data alone. Equivalently, any distinction between ξ and η that survives only in $\text{GF}(\xi)$ and $\text{GF}(\eta)$ is invisible at the level of the bounded-region local algebras $\mathcal{A}_{\text{loc}}(O; G)$.

Proof. If $\text{Sh}_{\text{loc}}(\xi) = \text{Sh}_{\text{loc}}(\eta)$, then the two objects differ, if at all, only by the label data frozen in Section IV. By Theorems V.6 and V.8, bounded-region local observables do not detect that theorem-scope global-form information. Hence the local observable algebras cannot separate ξ from η by label alone. $\square$

Corollary VII.4 (Strictly local invisibility of theorem-scope global-form data). *Let $\xi, \eta \in \mathfrak{E}(G)$ satisfy*

$$\text{Sh}_{\text{loc}}(\xi) = \text{Sh}_{\text{loc}}(\eta) \quad \text{and} \quad \text{GF}(\xi) \neq \text{GF}(\eta).$$

Then no observable belonging to a bounded-region local algebra $\mathcal{A}_{\text{loc}}(O; G)$ distinguishes the pair (ξ, η) by that difference.

Proof. This is the special case of Theorem VII.3 in which the two objects share the same support data and differ only by theorem-scope global-form datum. $\square$

B. Realized-sector inequivalence of the same data

Theorem VII.5 (Realized-sector inequivalence for distinct theorem-scope global-form data).

Let $\xi, \eta \in \mathfrak{E}(G)$ satisfy

$$\text{Sh}_{\text{loc}}(\xi) = \text{Sh}_{\text{loc}}(\eta) \quad \text{and} \quad \text{GF}(\xi) \neq \text{GF}(\eta).$$

Then the admissible realized sector data corresponding to ξ and η over the fixed local Yang–Mills theorem object are not unitarily equivalent.

Proof. By Theorem VI.10, each theorem-scope extended-support object determines an admissible realized sector datum over the fixed local theorem object. Since ξ and η have the same local shadow, any distinction between them that survives is carried entirely by the theorem-scope global-form datum. By Theorems IV.7 and IV.8, that datum is a role-bearing residue of the declared sector-extension target, not an arbitrary label attached after the fact. By Theorem V.6, the distinction is invisible to the bounded local observable net itself, so any surviving role distinction must be borne by the realized sector layer. If the corresponding realized sector data were unitarily equivalent, the sector assignment would identify two distinct forced roles while leaving the local shadow fixed. That would violate the role-cardinality discipline of Theorem IV.8 and collapse the theorem-scope target to a different comparison class. Hence the realized sector data are not unitarily equivalent. $\square$

Remark VII.6 (Theorem-scope exactness of sector-label inequivalence). Theorem VII.5 is exact at the declared theorem scope. It does not assert that arbitrary externally named line or surface labels are physically inequivalent. It states that labels admitted into $\Lambda_{\text{line}}(G)$ and $\Lambda_{\text{surf}}(G)$ as forced theorem-scope roles must be distinguished by any realization of the

declared target. If two such labels collapse under realization, then the declared target role structure has not been realized; the comparison class has changed rather than producing a faithful same-target realization.

Corollary VII.7 (Sector-class distinction for theorem-scope global-form data). *Let $\xi, \eta \in \mathfrak{E}(G)$ satisfy*

$$\text{Sh}_{\text{loc}}(\xi) = \text{Sh}_{\text{loc}}(\eta) \quad \text{and} \quad \text{GF}(\xi) \neq \text{GF}(\eta).$$

Then

$$\Xi_G([\xi]_{\text{def}}) \neq \Xi_G([\eta]_{\text{def}}).$$

In particular, the sector layer distinguishes theorem-scope global-form data that are invisible to the strictly local net.

Proof. By Theorem VII.5, the corresponding realized sector data are not unitarily equivalent. Therefore they cannot define the same sector-equivalence class in $\mathfrak{S}_{\text{ext}}(G)$. $\square$

C. Recovery theorem

Theorem VII.8 (Global-form recovery over the fixed local net). *Let G be a compact connected simple gauge group, let $\mathfrak{B}(G)$ be the fixed base object of Section III, and let Ξ_G be the sector assignment of Theorem VI.14. Then the following hold.*

- (i) *The strictly local observable net depends only on the local shadow $\text{Sh}_{\text{loc}}(\xi)$ of a theorem-scope extended-support object $\xi \in \mathfrak{E}(G)$ and is insensitive to the corresponding global-form datum $\text{GF}(\xi)$.*
- (ii) *The sector assignment Ξ_G distinguishes objects with the same local shadow but different global-form datum.*
- (iii) *Consequently, the sector structure $\mathfrak{S}_{\text{ext}}(G)$ distinguishes, at sector level, the theorem-scope global-form data that are invisible to the bounded-region local net.*

Proof. Part (i) is Theorem VII.3. Part (ii) is Theorem VII.7. Part (iii) is the conjunction of (i) and (ii). $\square$

Corollary VII.9 (Alignment with part (D) of the Main Theorem). *Part (D) of Theorem II.1 is exactly Theorem VII.8. In particular, “recovery of global-form data” means recovery of the theorem-scope labels through the sector class $\mathfrak{S}_{\text{ext}}(G)$ over the fixed local net $\mathfrak{B}(G)$.*

Proof. Immediate from Theorems VII.2 and VII.8. □

Remark VII.10 (What is and is not recovered). The present theorem is exact at the level of the theorem-scope labels fixed in Section IV. It does not claim that the sector layer reconstructs every possible nonlocal invariant of the theory. It claims only that the role-bearing theorem-scope label data fixed in Section IV are invisible to the bounded-region local net and distinguished at the sector level by the realized sector layer.

Dependency summary.

Inputs. the fixed base object $\mathfrak{B}(G)$ from Section III, the frozen extended-support class $\mathfrak{E}(G)$ from Section IV, the local-net boundary theorems of Section V, and the sector assignment of Section VI

Imported standard background. none beyond standard observable-net and representation-theoretic language

Conclusion at this node. the theorem-scope global-form data are invisible to the bounded-region local net but are distinguished, and therefore recovered, at the level of $\mathfrak{S}_{\text{ext}}(G)$

Used later in. Section VIII

VIII. COMPATIBILITY WITH THE VACUUM-SECTOR THEOREM AND KERNEL-FAITHFUL GNS SPECTRAL BRIDGE

The preceding sections construct a sectoral layer over the fixed base object $\mathfrak{B}(G)$ and show that this layer carries the global-form data invisible at the strictly local level. This section states how that layer relates to the predecessor vacuum-sector theorem. The answer has two parts. First, the sector construction leaves the fixed local theorem object unchanged. Second, the correct complete-theory bridge is not a claim about a naively enlarged “full Hamiltonian,” but a kernel-faithful spectral-invariance statement on scope-faithful extensions of that same fixed local theorem object.

Endpoint identity under admissibility. Identity of the endpoint object is fixed by admissibility-relevant invariants under lawful transformation. Any construction that alters these invariants does not produce the same object, but constitutes a change of domain rather than an alternative realization on the fixed domain.

A. Compatibility with the fixed local theorem object

Proposition VIII.1 (The fixed base object is preserved). *The constructions of Sections IV–VII do not alter:*

- (i) *the bounded-region local net $O \mapsto \mathcal{A}_{\text{loc}}(O; G)$;*
- (ii) *the quasi-local algebra $\mathcal{A}_{\text{loc}}(G)$;*
- (iii) *the distinguished vacuum representation $(\pi_{\text{vac}}, \mathcal{H}_{\text{vac}}, \Omega_{\text{vac}})$; or*
- (iv) *the vacuum Hamiltonian H_{vac} .*

In particular, the fixed base object $\mathfrak{B}(G)$ is unchanged.

Proof. Sections IV and VII fix the extended-support class, the local-net boundary, the sector class, and the associated recovery theorem. None of these constructions redefines the bounded-region local algebras, replaces the quasi-local algebra, or alters the distinguished vacuum representation recalled in Section III. They are all carried out over the fixed base object $\mathfrak{B}(G)$. $\square$

Theorem VIII.2 (Compatibility with the fixed local theorem object). *Let G be a compact connected simple gauge group, let $\mathfrak{B}(G)$ be the fixed base object of Section III, and let $\mathfrak{S}_{\text{ext}}(G)$ be the sector class of Section VI. Then the sector construction is compatible with the predecessor local theorem package in the following exact sense:*

- (i) *the bounded-region local net $O \mapsto \mathcal{A}_{\text{loc}}(O; G)$ is unchanged;*
- (ii) *the distinguished vacuum representation $(\pi_{\text{vac}}, \mathcal{H}_{\text{vac}}, \Omega_{\text{vac}}, H_{\text{vac}})$ is unchanged;*
- (iii) *the previously established vacuum-sector reconstruction and vacuum mass-gap theorem remain theorems about that same unchanged local net and that same unchanged vacuum representation; and*
- (iv) *the sector layer studied in the present paper is organized entirely over that fixed local theorem object and does not alter the theorem content already attached to it.*

Proof. Parts (i) and (ii) are Theorem VIII.1. Part (iii) follows because the recalled vacuum-sector theorem package is a theorem about the fixed local net and its distinguished vacuum representation. Since neither of those has changed, the theorem remains a theorem about the

same fixed local theorem object. Part (iv) is immediate from the construction of Sections VI and VII: the present manuscript adds a sectoral layer over $\mathfrak{B}(G)$ but does not feed that layer back into the theorem-level statement already attached to that fixed object. $\square$

Corollary VIII.3 (Alignment with part (E) of the Main Theorem). *Part (E) of Theorem II.1 is exactly Theorem VIII.2.*

Proof. Immediate from the statement of Theorem VIII.2. $\square$

B. Kernel necessity, UEAP claim legitimacy, and downstream fidelity

The spectral bridge below uses the admissibility kernel only in a fixed-domain, source-neutral way. The kernel is not optional structure placed on top of Yang–Mills theory. It is instantiated by necessity in every non-degenerate regime in which constructions are treated as determinate, validity-assessable, identity-preserving, and irreversible under lawful redescription. Once the fixed Yang–Mills theorem object is shown to be such a regime, every downstream sectoral, ultraviolet, or completion claim is legitimate only insofar as it remains faithful to that kernel.

The kernel results recorded in this subsection are included in the present manuscript in the restricted form required for the Yang–Mills endpoint. References to the upstream AASC corpus provide expanded provenance, examples, and formal development, but the endpoint argument uses only the stated kernel consequences established here and applied within the verified fixed-domain scope.

Definition VIII.4 (Non-degenerate construction regime). A regime R is *non-degenerate* when it has the following functional features, stated without using the kernel vocabulary:

- (i) objects or contents remain identifiable across construction stages;
- (ii) construction transitions can succeed or fail relative to the target task;
- (iii) there is a lawful equivalence relation under which presentation changes do not change the object being constructed;
- (iv) endpoint commitments are not freely reversible by later relabeling, reindexing, or post-hoc repair; and

(v) the regime is not a one-point or purely syntactic coding in which all distinctions collapse. Thus non-degeneracy is a functional condition on construction, comparison, and endpoint commitment. It is not defined by naming admissibility, standing, reference, or irreversibility in advance.

Theorem VIII.5 (Kernel necessity in non-degenerate regimes). *Let R be a non-degenerate construction regime. Then R necessarily instantiates the kernel*

$$K = \{\text{Adm}, \text{St}, \text{Ref}, \text{Irr}\}.$$

Thus admissibility, standing, determinate reference, and irreversibility are not optional axioms of a chosen formalism; they are the minimal conditions under which construction in R is determinate, validity-assessable, identity-preserving, and commitment-bearing.

Proof. The five functional features in Theorem VIII.4 force the four roles in K . Feature (i) requires determinate reference: without a fixed object or content that can be reidentified across stages, there is no object of construction. Feature (ii) requires an admissibility boundary: without a distinction between successful and failed transitions, the regime has no validity-assessable construction steps. Feature (iii) requires standing for lawful continuation: a transition that survives lawful equivalence must be licensed to participate in later construction rather than merely recorded as syntax. Feature (iv) requires irreversibility: if endpoint commitments can be undone by later relabeling or post-hoc repair, the regime has no stable commitment to a constructed object. Feature (v) excludes the degenerate cases in which those roles disappear because no substantive distinction remains to be preserved. Hence every non-degenerate construction regime instantiates K by necessity, not by optional stipulation. $\square$

Lemma VIII.6 (No lower same-regime generator of the kernel). *No same-regime condition L faithfully generates K from below unless L already presupposes the work of K or something lawfully equivalent to it.*

Proof. To function as a lower generator, L must itself have a determinate target, licensed inferential standing, a validity condition for its transitions, and stable commitment to its conclusion. Those are exactly the reference, standing, admissibility, and irreversibility roles named by K . If L lacks them, it is not a validity-assessable lower generator. If it has them,

it is not lower with respect to K ; it imports or restates the kernel work. Thus a faithful lower same-regime generator does not exist. $\square$

Lemma VIII.7 (No faithful same-regime kernel-violating counterexample). *There is no non-degenerate regime R' that preserves the same-regime identity of R , violates K , and remains a faithful validity-assessable same-regime construction.*

Proof. A violation of determinate reference loses endpoint identity; a violation of standing loses licensed theorem-step status; a violation of admissibility loses the valid/invalid construction boundary; and a violation of irreversibility loses non-repairable commitment. In each case the purported counterexample fails to be a faithful same-regime construction. It may be a different theory object, a degenerate coding, or external bookkeeping, but it is not a non-degenerate same-regime counterexample. $\square$

Corollary VIII.8 (Downstream fidelity requirement). *Every downstream construction over a non-degenerate theorem object is legitimate as a same-domain continuation only if it remains faithful to the instantiated kernel on that same domain. In the Yang–Mills setting this means preservation, under lawful redescription, of the fixed local observable carrier, vacuum restriction, implemented local dynamics, endpoint identity, and absence of any new same-domain authorizer, standing-transfer mechanism, or subgap carrier not licensed by the fixed theorem object.*

Proof. Apply Theorems VIII.5 and VIII.7. A downstream move that is not kernel-faithful is not a same-domain continuation of the original theorem object. It changes the carrier, collapses into bookkeeping, or fails as a licensed construction step. $\square$

Lemma VIII.9 (The fixed local Yang–Mills theorem object is non-degenerate). *Let*

$$Y(G) := (\mathcal{A}_{\text{loc}}(G), \omega_{\text{vac}}, U_{\text{vac}})$$

be the fixed local Yang–Mills theorem object supplied by the predecessor package. Then $Y(G)$ is a non-degenerate construction regime.

Proof. The predecessor package supplies nontrivial bounded local gauge-invariant observable content, a fixed vacuum state, implemented positive-energy local time-translation dynamics, and the vacuum-sector mass-gap commitment attached to that fixed object. These data

satisfy the functional features of Theorem VIII.4. The local observable net and vacuum state make the relevant objects identifiable across construction stages. The theorem package distinguishes successful theorem-level construction from failed or scope-changing moves. Lawful AQFT redescription is unitary equivalence preserving the represented local algebra, the vacuum vector, and the implemented local dynamics. The vacuum-sector mass-gap endpoint is a standing commitment, not freely reversible by later relabeling or indexing. Finally, the local observable content is nontrivial, so the regime is not a one-point coding. Thus $Y(G)$ is non-degenerate without first assuming the kernel terminology. $\square$

Corollary VIII.10 (Kernel instantiation in the Yang–Mills endpoint). *The kernel $K = \{\text{Adm}, \text{St}, \text{Ref}, \text{Irr}\}$ is instantiated in the fixed local Yang–Mills theorem object by necessity. It is not an optional interpretive overlay.*

Proof. Immediate from Theorems VIII.5 and VIII.9. $\square$

Proposition VIII.11 (UEAP claim-legitimacy map for the endpoint). *For the present manuscript, UEAP functions as a claim-legitimacy map²⁶: it records the carrier, scope, standing condition, and failure consequence for each theorem-bearing claim. The relevant map is:*

<i>Claim</i>	<i>Carrier</i>	<i>Scope</i>	<i>Standing condition</i>	<i>Failure consequence</i>
<i>Fixed local theorem object</i>	$(\mathcal{A}_{\text{loc}}(G), \omega_{\text{vac}}, U_{\text{vac}})$	<i>bounded local net</i>	<i>preservation of local observable content, vacuum state, and dynamics</i>	<i>different theorem object</i>
<i>Extended-support sector layer</i>	$\mathfrak{E}(G) \rightarrow \mathfrak{S}_{\text{ext}}(G)$	<i>theorem-scope line/surface roles</i>	<i>fixed local net preserved and forced role labels realized</i>	<i>target not realized</i>
<i>Same-theory completion</i>	$(\mathcal{A}, \omega, U)$ over $Y(G)$	<i>sectoral or ultraviolet completion</i>	<i>kernel-faithful domain continuation</i>	<i>same-theory inadmissible move or different object</i>
<i>Complete-theory gap</i>	$E_H([0, m^*))\mathcal{H} = \mathbb{C}\Omega$	<i>theorem-scope admissible completions</i>	<i>no new subgap local carrier or vacuum multiplicity</i>	<i>not a same-theory completion</i>

Thus UEAP does not add a physical axiom. It fixes which carrier a claim is about and what counts as standing-preserving continuation of that carrier.

Proof. The four rows are exactly the theorem-object boundaries established in Sections III–VII and in Theorems VIII.8 and VIII.10. A proposed completion that violates the fourth

row does not expose an omitted same-theory case; it fails the legitimacy condition for the same-theory completion claim. $\square$

C. Local Yang–Mills completion kernel packet and weakening resistance

The preceding kernel and UEAP map give the general fixed-domain constraints. We record their Yang–Mills completion form explicitly because the endpoint dispute is about completion status, not about sector notation alone.

Definition VIII.12 (Local Yang–Mills completion kernel packet). The local Yang–Mills completion packet $K_{\text{YM}}^{\text{comp}}(G)$ is the fixed-domain consequence layer for theorem-bearing same-domain completion use over $Y(G) = (\mathcal{A}_{\text{loc}}(G), \omega_{\text{vac}}, U_{\text{vac}})$.

Node	General role	Yang–Mills completion form
K1	Fixed-domain kernel	A theorem-bearing same-domain completion fixes the local observable carrier, vacuum restriction, local dynamics, endpoint identity, admissibility boundary, and act-time irreversibility.
K2	Reference fixation	A completion report cannot silently replace $\mathcal{A}_{\text{loc}}(G)$, ω_{vac} , U_{vac} , the vacuum sector, the group-level endpoint, or the complete-theory Hamiltonian claim.
K3	Standing/readout separation	Sector standing, local vacuum standing, completion standing, and complete-theory spectral readout are distinct roles. Sector classification is not mass-gap readout.
K4	Fail-closed bivalence	A completion is admitted as a same-theory completion or not admitted. Role-ambiguous, unsupported, deferred, and merely sectoral status are not third completion truth values.
K5	Unique admissible completion interior	The fixed local theorem object cannot carry plural independent same-domain completion interiors assigning different subgap status.
K6	Standing-admissibility identity	Completion standing cannot float outside same-domain admissibility. A completion claim with standing must be admissibly typed.
K7	No generators	A sector label, Hilbert-space summand, ultraviolet presentation, Wilson regularization, or sharp-field notation cannot generate completion standing by being written.
K8	No carriers	A nonlocal sector, charged representation, defect representation, direct-sum Hilbert space, auxiliary ultraviolet carrier, or regularization object cannot carry mass-gap status back to the fixed local theory without the GNS bridge.
K9	No post-selection repair	Later packaging of an unfiltered sector representation does not retroactively make it a same-theory completion.
K10	AMetric no-selector boundary	No preferred regularization, representation, sector label, Hilbert-space decomposition, cutoff, or UV scale decides same-theory completion status before admissibility is fixed.
K11	UEAP report preservation	A report may not exceed its carrier, scope, standing, and bridge support. Sector classification reports sector status only; completion status requires the filter.
K12	ATS skin discipline	Notation, labels, presentation choices, formal support objects, representative sector names, or regularization choices do not become tensor-level completion authority.
K13	Completion exhaustion	Once local carrier, vacuum restriction, dynamics, endpoint role, and admissibility boundary are fixed, a completion candidate is same-theory admissible completion, sector/support only, bookkeeping, carrier/domain shift, theorem-object change, or forbidden second same-domain completion authority.

Theorem VIII.13 (Local Yang–Mills completion packet). *Every theorem-bearing same-domain completion use over the fixed local Yang–Mills theorem object instantiates $K_{\text{YM}}^{\text{comp}}(G)$.*

Proof. By Theorems VIII.9 and VIII.10, the fixed local Yang–Mills theorem object already instantiates the kernel. By Theorems VIII.8 and VIII.11, every downstream same-domain completion claim must preserve the local observable carrier, vacuum restriction, local dynamics, endpoint identity, and report status. The listed K-nodes are exactly the local completion forms of reference, standing, admissibility, irreversibility, report preservation, no carrier transfer, no repair, no selector authority, ATS role discipline, and endpoint exhaustion. Hence the packet is in force for theorem-bearing same-domain completion use. $\square$

Definition VIII.14 (Core completion adequacy packet). The core completion adequacy packet $C_{\text{YM-comp}}^0(G)$ consists of the fixed local observable carrier $\mathcal{A}_{\text{loc}}(G)$, the fixed vacuum restriction ω_{vac} , the fixed local dynamics U_{vac} , the group-level endpoint, scope-faithful extension status, minimal report evaluability, act-time finality, and same-domain fidelity. This is the minimal packet already forced before the full K1–K13 consequence layer is invoked.

Definition VIII.15 (Completion-governance-permissive weakening). A completion-governance-permissive weakening W is a purported weaker same-domain completion regime that preserves $C_{\text{YM-comp}}^0(G)$ while permitting a completion candidate to introduce locally detectable subgap content or additional subgap vacuum multiplicity and still claim same-theory completion standing. Such a W is required, by definition, not to classify the candidate as sector/support only, bookkeeping, carrier shift, theorem-object change, or explicit inadmissibility.

Lemma VIII.16 (Local necessity of K5). *Under $C_{\text{YM-comp}}^0(G)$, two distinct endpoint-governing completion interiors for the same local theorem object either agree extensionally and are bookkeeping, split the theorem object, or introduce a second same-domain completion classifier.*

Proof. If two alleged interiors assign the same completion verdicts, the second contributes no endpoint work. If they disagree while preserving the same local carrier, vacuum restriction, dynamics, and endpoint role, the same theorem object receives two incompatible completion classifications. Resolving that disagreement requires a higher selector, a domain split, or an independent completion classifier. Those are precisely the alternatives K5 excludes. $\square$

Lemma VIII.17 (Local necessity of K6). *Under $C_{\text{YM-comp}}^0(G)$, completion standing outside admissibility is sector/support only, carrier shift, or a second completion gate.*

Proof. If the alleged standing changes no completion verdict, it is support or bookkeeping. If it changes the verdict while preserving the same local theorem object, it performs same-domain completion gate work outside the admissibility boundary. If it avoids that conclusion by changing carrier, vacuum restriction, dynamics, or endpoint role, it is carrier or theorem-object shift. $\square$

Lemma VIII.18 (Local necessity of K11). *Under $C_{\text{YM-comp}}^0(G)$, a theorem-bearing completion report that exceeds its carrier, scope, standing condition, and GNS-bridge support is status laundering, unsupported completion authority, or no-carrier transfer.*

Proof. Sector classification has sector scope; ultraviolet presentation has auxiliary-carrier scope; a same-theory completion claim has fixed-local-theorem-object scope. If a report moves from the weaker status to completion status without discharging the carrier, scope, standing, admissibility, and bridge conditions, it reports more than its support licenses. That is UEAP overreporting: status laundering, unsupported authority, or no-carrier transfer. $\square$

Lemma VIII.19 (Local necessity of K13). *Role-ambiguous, unfiltered, merely sectoral, direct-sum, deferred, or unsupported statuses are report or support statuses; they are not third same-theory completion truth values on the fixed local theorem object.*

Proof. Target fixation specifies what a same-theory completion must preserve. It does not create intermediate endpoint truth values. A candidate is either admitted as a same-theory completion, remains sector/support/bookkeeping, changes the theorem object, or imports forbidden second authority. Unknown, deferred, or unfiltered status may describe a construction attempt; it does not occupy completion truth. $\square$

Theorem VIII.20 (No strict same-domain weakening permits subgap completion governance). *No completion-governance-permissive weakening W preserving $C_{\text{YM-comp}}^0(G)$ yields a non-degenerate same-theory completion with new locally detectable subgap content or additional subgap vacuum multiplicity.*

Proof. Assume such a W and let its completion candidate carry new subgap content or extra subgap vacuum multiplicity. If the candidate has no completion effect, it is bookkeeping. If it changes $\mathcal{A}_{\text{loc}}(G)$, the vacuum restriction, the local dynamics, or endpoint identity, it changes theorem object. If it remains merely sectoral or auxiliary, it is support only. If it

claims same-theory completion force while preserving the same object, it performs completion governance. If that governance shares the unique admissible completion interior, it collapses to the admitted GNS bridge. If it differs, Theorem VIII.16 gives second classifier or domain split. If it claims standing outside admissibility, Theorem VIII.17 applies. If its report exceeds bridge support, Theorem VIII.18 applies. If it occupies neither admitted nor rejected completion status, Theorem VIII.19 applies. Thus every branch is endpoint-equivalent, support/bookkeeping, theorem-object shift, or forbidden second same-domain completion authority. No strict same-domain weakening survives. $\square$

Corollary VIII.21 (Weakening dichotomy for completion use). *Every weaker same-domain completion regime either preserves the existing completion endpoint behavior extensionally, lowers the datum to sector/support or bookkeeping, shifts the theorem object, or reintroduces forbidden second same-domain completion authority.*

Proof. This is the exhaustive conclusion of Theorem VIII.20. $\square$

D. Kernel-faithful GNS spectral bridge under scope-faithful extension

The correct endpoint theorem is a kernel-faithful GNS/spectral bridge stated over the same fixed-domain theorem object. The mass gap furnished by the earlier local theorem package is a spectral statement about the Hamiltonian in the vacuum representation generated by the fixed local observable algebra. Enlarging the observable structure therefore raises a precise question: when does such an enlargement leave the subgap spectrum unchanged? The answer is formulated below entirely in GNS, local-state, multiplicity, and spectral-subspace language.

Definition VIII.22 (Scope-faithful extension of the fixed local vacuum theory). A *scope-faithful extension* of the fixed local vacuum theory is a triple $(\mathcal{A}, \omega, U)$ such that:

- (i) $\mathcal{A}_{\text{loc}}(G) \subset \mathcal{A}$ as a unital C^* -subalgebra;
- (ii) $\omega|_{\mathcal{A}_{\text{loc}}(G)} = \omega_{\text{vac}}$;
- (iii) $U(t) = e^{-itH}$ is a strongly continuous positive-energy GNS implementation of an ω -preserving automorphism group of $\mathcal{A}$, so $H \geq 0$, $\omega \circ \alpha_t = \omega$, and $U(t)\Omega = \Omega$;
- (iv) $U(t)$ leaves $\mathcal{A}_{\text{loc}}(G)$ invariant and implements the same local time-translation dynamics

τ_t on $\mathcal{A}_{\text{loc}}(G)$ as the predecessor vacuum theory, that is,

$$U(t)\pi(A)U(t)^* = \pi(\tau_t(A)), \quad U_{\text{vac}}(t)\pi_{\text{vac}}(A)U_{\text{vac}}(t)^* = \pi_{\text{vac}}(\tau_t(A)) \quad (A \in \mathcal{A}_{\text{loc}}(G));$$

and

(v) in the GNS triple $(\pi, \mathcal{H}, \Omega)$ of ω , the closed cyclic subspace

$$\mathcal{H}_{\text{loc}} := \overline{\pi(\mathcal{A}_{\text{loc}}(G))\Omega}$$

is $U(t)$ -invariant.

Definition VIII.23 (GNS completion bridge admissibility object). The bridge admissibility object

$$B_{\text{YM}}^{\text{GNS}}(G; \mathcal{A}, \omega, U)$$

records the lawful same-domain route from the fixed local Yang–Mills theorem object into the GNS/spectral bridge. It is an admissibility and identity-preservation object. It does not contain the no-subgap consequences, and it does not contain the complete-theory gap conclusion, as primitive clauses. It consists of the following clauses.

- (B1) Same local observable carrier: $\mathcal{A}_{\text{loc}}(G) \subset \mathcal{A}$ as the same fixed bounded-region local gauge-invariant observable carrier, not a replacement net.
- (B2) Same vacuum restriction: $\omega|_{\mathcal{A}_{\text{loc}}(G)} = \omega_{\text{vac}}$.
- (B3) Same implemented positive-energy local dynamics: $U(t) = e^{-itH}$ is the scope-faithful GNS implementation, preserves $\mathcal{A}_{\text{loc}}(G)$, fixes Ω , and implements the predecessor local time translations there.
- (B4) Canonical local cyclic sector: $\mathcal{H}_{\text{loc}} = \overline{\pi(\mathcal{A}_{\text{loc}}(G))\Omega}$ is invariant for the implemented dynamics.
- (B5) Dynamical GNS comparison: the canonical local cyclic sector is the GNS representation of the restricted vacuum state and is therefore eligible for the dynamical GNS uniqueness bridge to the predecessor vacuum representation.
- (B6) Sector-to-completion filter passed: the datum is not merely a sector representation, but has passed the ATS/UEAP, scope-faithful, kernel-faithful completion filter.

(B7) Kernel-faithful same-domain continuation: reference, standing, admissibility boundary, irreversibility, local carrier, vacuum restriction, local dynamics, and endpoint identity are preserved.

The exclusions of locally detectable subgap states and additional subgap vacuum multiplicity are derived bridge consequences below, not defining clauses of the bridge object. The spectral projection $E_H([0, m^*))$ is the projection of the scope-faithful GNS Hamiltonian supplied by the completion datum, not an independent bridge-object assumption.

Proposition VIII.24 (The GNS bridge object is non-circular). *The bridge object $B_{\text{YM}}^{\text{GNS}}(G; \mathcal{A}, \omega, U)$ does not contain the complete-theory gap conclusion, the no-locally-detectable-subgap conclusion, or the no-extra-vacuum-multiplicity conclusion by definition. It fixes only the local carrier, vacuum restriction, implemented dynamics, local cyclic sector, GNS comparison posture, completion-filter status, and kernel-fidelity status required for the later GNS/spectral argument.*

Proof. Clauses B1–B5 identify the fixed local theorem object and the canonical local GNS comparison. Clause B6 records that the sector-to-completion filter has been discharged, and B7 records kernel-faithful same-domain continuation. None of these clauses asserts $E_H([0, m^*))\mathcal{H} = \mathbb{C}\Omega$. None asserts as primitive data that locally detectable subgap states or extra subgap vacuum multiplicity are absent. Those no-subgap conclusions are proved from scope-faithfulness, kernel fidelity, the sector-to-completion filter, subgap local-state conservation, subgap multiplicity exclusion, and standard GNS/spectral theory. Thus the bridge object is a route and admissibility object rather than a conclusion-bearing shortcut. $\square$

Theorem VIII.25 (GNS bridge object yields derived subgap exclusions). *If $B_{\text{YM}}^{\text{GNS}}(G; \mathcal{A}, \omega, U)$ holds, then the corresponding theorem-scope admissible completion has no locally detectable subgap local-state content and no additional subgap copy of the fixed local vacuum representation.*

Proof. Clauses B1–B5 provide the scope-faithful local carrier, restricted vacuum state, local dynamics, canonical local cyclic sector, and GNS comparison needed for the operator-algebraic bridge. Clauses B6 and B7 say that the sector-to-completion filter has been passed and that the extension is a kernel-faithful same-domain continuation. Therefore the completion is a theorem-scope admissible completion in the sense used below. The no-new-subgap-content

theorem, Theorem VIII.48, excludes both locally detectable subgap local-state content and an additional subgap vacuum copy for such completions. Hence the stated no-subgap exclusions are derived consequences of the admissible bridge object, not assumptions inside it. $\square$

Theorem VIII.26 (GNS bridge object yields the complete-theory gap). *If $B_{\text{YM}}^{\text{GNS}}(G; \mathcal{A}, \omega, U)$ holds, then*

$$E_H([0, m^*))\mathcal{H} = \mathbb{C}\Omega.$$

Proof. By Theorem VIII.25, the admissible bridge object yields the derived exclusions of locally detectable subgap states and additional subgap vacuum multiplicity. Clauses B1–B5 provide the canonical local GNS comparison with the predecessor vacuum representation, while B6 and B7 supply completion eligibility and kernel fidelity. The spectral projection is the projection of the scope-faithful GNS Hamiltonian in the completion datum, so the kernel-faithful GNS spectral bridge, Theorem VIII.57, applies and yields the displayed equality. $\square$

Lemma VIII.27 (Dynamical uniqueness of the canonical local vacuum sector). *Let $(\mathcal{A}, \omega, U)$ be a scope-faithful extension of the fixed local vacuum theory, with GNS triple $(\pi, \mathcal{H}, \Omega)$, and let*

$$\mathcal{H}_{\text{loc}} := \overline{\pi(\mathcal{A}_{\text{loc}}(G))\Omega}.$$

Then there exists a unitary

$$W_{\text{loc}} : \mathcal{H}_{\text{loc}} \rightarrow \mathcal{H}_{\text{vac}}$$

such that

$$W_{\text{loc}}\Omega = \Omega_{\text{vac}}, \quad W_{\text{loc}}\pi(A)|_{\mathcal{H}_{\text{loc}}}W_{\text{loc}}^* = \pi_{\text{vac}}(A) \quad (A \in \mathcal{A}_{\text{loc}}(G)),$$

and

$$W_{\text{loc}}U(t)|_{\mathcal{H}_{\text{loc}}}W_{\text{loc}}^* = U_{\text{vac}}(t) \quad (t \in \mathbb{R}).$$

Consequently, for the self-adjoint generators,

$$W_{\text{loc}}H|_{\mathcal{H}_{\text{loc}}}W_{\text{loc}}^* = H_{\text{vac}},$$

and for every Borel set $\Delta \subset \mathbb{R}$,

$$W_{\text{loc}}E_{H|_{\mathcal{H}_{\text{loc}}}}(\Delta)W_{\text{loc}}^* = E_{H_{\text{vac}}}(\Delta).$$

Proof. By standard GNS restriction, already recorded abstractly in the ultraviolet-scope note, $(\pi|_{\mathcal{A}_{\text{loc}}(G)}, \mathcal{H}_{\text{loc}}, \Omega)$ is the GNS representation of the restricted state ω_{vac} on $\mathcal{A}_{\text{loc}}(G)$. Since $(\pi_{\text{vac}}, \mathcal{H}_{\text{vac}}, \Omega_{\text{vac}})$ is also the GNS representation of ω_{vac} , uniqueness of the GNS construction yields a unique unitary

$$W_{\text{loc}} : \mathcal{H}_{\text{loc}} \rightarrow \mathcal{H}_{\text{vac}}$$

satisfying

$$W_{\text{loc}}\Omega = \Omega_{\text{vac}}, \quad W_{\text{loc}}\pi(A)|_{\mathcal{H}_{\text{loc}}}W_{\text{loc}}^* = \pi_{\text{vac}}(A) \quad (A \in \mathcal{A}_{\text{loc}}(G)).$$

It remains to show that W_{loc} also intertwines the implemented time-translation groups. Fix $t \in \mathbb{R}$ and define

$$V(t) := W_{\text{loc}}U(t)|_{\mathcal{H}_{\text{loc}}}W_{\text{loc}}^* \in \mathcal{B}(\mathcal{H}_{\text{vac}}).$$

For every $A \in \mathcal{A}_{\text{loc}}(G)$ one has, using that $U(t)$ implements the same automorphism group on the fixed local algebra and that $\mathcal{H}_{\text{loc}}$ is $U(t)$ -invariant,

$$\begin{aligned} V(t)\pi_{\text{vac}}(A)V(t)^* &= W_{\text{loc}}U(t)\pi(A)|_{\mathcal{H}_{\text{loc}}}U(t)^*W_{\text{loc}}^* \\ &= W_{\text{loc}}\pi(\tau_t(A))|_{\mathcal{H}_{\text{loc}}}W_{\text{loc}}^* \\ &= \pi_{\text{vac}}(\tau_t(A)). \end{aligned}$$

On the other hand, $U_{\text{vac}}(t)$ also implements τ_t in the vacuum representation, so

$$U_{\text{vac}}(t)\pi_{\text{vac}}(A)U_{\text{vac}}(t)^* = \pi_{\text{vac}}(\tau_t(A)).$$

Thus the unitary

$$X(t) := U_{\text{vac}}(t)^*V(t)$$

commutes with $\pi_{\text{vac}}(\mathcal{A}_{\text{loc}}(G))$. Moreover, the vacuum invariance used here is part of the definition of a scope-faithful extension. Hence

$$X(t)\Omega_{\text{vac}} = U_{\text{vac}}(t)^*V(t)\Omega_{\text{vac}} = U_{\text{vac}}(t)^*W_{\text{loc}}U(t)\Omega = \Omega_{\text{vac}},$$

since both $U(t)$ and $U_{\text{vac}}(t)$ fix their vacuum vectors. Because Ω_{vac} is cyclic for $\pi_{\text{vac}}(\mathcal{A}_{\text{loc}}(G))$, the relation

$$X(t)\pi_{\text{vac}}(A)\Omega_{\text{vac}} = \pi_{\text{vac}}(A)X(t)\Omega_{\text{vac}} = \pi_{\text{vac}}(A)\Omega_{\text{vac}}$$

for all $A \in \mathcal{A}_{\text{loc}}(G)$ implies that $X(t)$ is the identity on a dense subspace of $\mathcal{H}_{\text{vac}}$, hence on all of $\mathcal{H}_{\text{vac}}$. Therefore

$$W_{\text{loc}}U(t)|_{\mathcal{H}_{\text{loc}}}W_{\text{loc}}^* = U_{\text{vac}}(t) \quad (t \in \mathbb{R}).$$

By Stone's theorem, the self-adjoint generators are unitarily intertwined:

$$W_{\text{loc}} H|_{\mathcal{H}_{\text{loc}}} W_{\text{loc}}^* = H_{\text{vac}}.$$

The spectral-projection identity follows from the functional calculus for self-adjoint operators under unitary equivalence. $\square$

Remark VIII.28 (Canonical local vacuum sector inside an extension). Lemma VIII.27 identifies the cyclic local subrepresentation of any scope-faithful extension with the predecessor vacuum representation as a full covariant dynamical system, not merely as a representation of the fixed local algebra.

Lemma VIII.29 (Subgap rigidity of the canonical local vacuum sector). *Let $(\mathcal{A}, \omega, U)$ be a scope-faithful extension of the fixed local vacuum theory, with GNS triple $(\pi, \mathcal{H}, \Omega)$ and Hamiltonian H . Then*

$$E_H([0, m^*))\mathcal{H}_{\text{loc}} = \mathbb{C}\Omega.$$

Equivalently, the canonical local vacuum sector contains no non-vacuum vector in the subgap spectral window.

Proof. Set

$$H_{\text{loc}} := H|_{\mathcal{H}_{\text{loc}}}.$$

By Theorem VIII.22, the closed cyclic subspace

$$\mathcal{H}_{\text{loc}} := \overline{\pi(\mathcal{A}_{\text{loc}}(G))\Omega}$$

is invariant under $U(t) = e^{-itH}$ for every $t \in \mathbb{R}$; hence it is reducing for H , and H_{loc} is self-adjoint on $\mathcal{H}_{\text{loc}}$. By Theorem VIII.27, there exists a unitary

$$W_{\text{loc}} : \mathcal{H}_{\text{loc}} \rightarrow \mathcal{H}_{\text{vac}}$$

with

$$W_{\text{loc}} H_{\text{loc}} W_{\text{loc}}^* = H_{\text{vac}}, \quad W_{\text{loc}} \Omega = \Omega_{\text{vac}}.$$

By the spectral theorem for self-adjoint operators, unitary equivalence intertwines spectral projections; therefore, for every Borel set $\Delta \subset \mathbb{R}$,

$$W_{\text{loc}} E_{H_{\text{loc}}}(\Delta) W_{\text{loc}}^* = E_{H_{\text{vac}}}(\Delta).$$

Apply this with $\Delta = [0, m^*)$. The recalled predecessor theorem package states

$$\text{Spec}(H_{\text{vac}}) \subset \{0\} \cup [m^*, \infty), \quad \ker H_{\text{vac}} = \mathbb{C}\Omega_{\text{vac}}.$$

Hence

$$E_{H_{\text{vac}}}([0, m^*)) = P_{\Omega_{\text{vac}}},$$

where $P_{\Omega_{\text{vac}}}$ is the rank-one orthogonal projection onto $\mathbb{C}\Omega_{\text{vac}}$. Conjugating back by W_{loc} gives

$$E_{H_{\text{loc}}}([0, m^*)) = W_{\text{loc}}^* P_{\Omega_{\text{vac}}} W_{\text{loc}} = P_{\Omega},$$

so

$$E_{H_{\text{loc}}}([0, m^*))\mathcal{H}_{\text{loc}} = \mathbb{C}\Omega.$$

Because $\mathcal{H}_{\text{loc}}$ is reducing for H , the spectral projection of H restricted to $\mathcal{H}_{\text{loc}}$ agrees with the spectral projection of H_{loc} ; that is,

$$E_H([0, m^*))\mathcal{H}_{\text{loc}} = E_{H_{\text{loc}}}([0, m^*))\mathcal{H}_{\text{loc}} = \mathbb{C}\Omega.$$

This is the claimed subgap rigidity. □

Definition VIII.30 (Local vector state and local detectability). Let $(\pi, \mathcal{H}, \Omega)$ be the GNS triple of a scope-faithful extension and let H be the self-adjoint generator of the implemented time translations. For a unit vector $\psi \in \mathcal{H}$, define the associated *local vector state*

$$\omega_{\psi}^{\text{loc}}(A) := \langle \psi, \pi(A)\psi \rangle, \quad A \in \mathcal{A}_{\text{loc}}(G).$$

For $m > 0$, write

$$\mathcal{H}_{<m} := E_H([0, m))\mathcal{H}$$

for the spectral subspace of H in the half-open interval $[0, m)$. A *locally detectable subgap sector* over $\mathcal{A}_{\text{loc}}(G)$ is the GNS representation of a state $\omega_{\psi}^{\text{loc}}$ with $\psi \in \mathcal{H}_{<m^*}$, $\|\psi\| = 1$, and

$$\omega_{\psi}^{\text{loc}} \neq \omega_{\text{vac}}.$$

Definition VIII.31 (Theorem-scope admissible completion). A *theorem-scope admissible completion* of the fixed local vacuum theory is any scope-faithful extension $(\mathcal{A}, \omega, U)$ of Theorem VIII.22 that is also a kernel-faithful same-domain continuation of the fixed local Yang–Mills theorem object. Equivalently, it preserves the standing-bearing local observable

carrier, the vacuum restriction, the implemented local dynamics, the endpoint identity, and the UEAP same-theory completion row of Theorem VIII.11.

The present manuscript analyzes two concrete completion branches:

- (i) the extended-support sector layer classified in Sections IV–VII; and
- (ii) ultraviolet sharp-field enlargements that agree with the fixed local algebra on the macroscopic net in the sense described in the ultraviolet-scope note.

The definition is therefore not restricted by listing two examples first: same-theory status is controlled by scope-faithfulness and kernel fidelity, while the sectoral and ultraviolet branches are the completion branches treated in this paper.

Theorem VIII.32 (Operator-algebraic form of kernel-fidelity failure). *Let $(\mathcal{A}, \omega, U)$ be an extension over the fixed local Yang–Mills theorem object with GNS data $(\pi, \mathcal{H}, \Omega, H)$. Suppose that either:*

- (i) *there exists a unit vector $\psi \in E_H([0, m^*))\mathcal{H}$ with*

$$\omega_\psi^{\text{loc}} \neq \omega_{\text{vac}};$$

or

- (ii) *there exists a unit vector $\psi \in E_H([0, m^*))\mathcal{H}$ with $\psi \perp \Omega$ and*

$$\omega_\psi^{\text{loc}} = \omega_{\text{vac}}.$$

Then the extension is not a kernel-faithful same-domain completion of the fixed local Yang–Mills theorem object.

Proof. In case (i), the vector ψ implements a local vector state on $\mathcal{A}_{\text{loc}}(G)$ different from the predecessor vacuum state while lying below the predecessor gap. Equivalently, there exists some $A \in \mathcal{A}_{\text{loc}}(G)$ such that

$$\langle \psi, \pi(A)\psi \rangle \neq \omega_{\text{vac}}(A).$$

This is locally detectable new subgap content over the original local carrier, not a presentation change of the same carrier. It alters the standing-bearing local-state content recorded by the UEAP map of Theorem VIII.11.

In case (ii), the local vector state agrees with ω_{vac} , but the implementing vector is orthogonal to the vacuum ray and still lies in the subgap window. Although local expectation

values agree with the vacuum on $\mathcal{A}_{\text{loc}}(G)$, the Hilbert-space spectral representation contains an additional subgap vector implementing the vacuum local state. The extension therefore contains an additional subgap copy or multiplicity of the fixed local vacuum representation. That changes the endpoint's unique subgap-ray structure while claiming same-domain identity.

Both alternatives violate the downstream fidelity requirement of Theorem VIII.8: they create or import standing-positive subgap content, introduce a second same-domain carrier or authorizer, and alter the admissibility-relevant invariant bundle of the fixed theorem object. By Theorem VIII.10, the kernel is already in force for the Yang–Mills endpoint. Hence either alternative is a different theory object or an inadmissible same-domain move, not a kernel-faithful same-domain completion. $\square$

E. Endpoint-use ATS/UEAP filter for sector-to-completion promotion

The restricted ATS/UEAP packet in this subsection is used only to block unlicensed promotion from sector representation to same-theory completion. No broader ATS or UEAP machinery is required for the endpoint argument below.

Definition VIII.33 (Endpoint-use ATS role). For the present manuscript, ATS is used only in the restricted sector-to-completion sense needed at the endpoint^{24,25}. Relative to the fixed local Yang–Mills theorem object $Y(G)$, a datum x has one of the following endpoint-use roles:

- (i) *anchor*: a fixed precondition of the theorem object, not introduced by a completion and not alterable by a completion;
- (ii) *tensor*: load-bearing same-domain structure that may participate in continuation only if it preserves the fixed carrier, scope, and endpoint identity;
- (iii) *skin*: expressive, classificatory, or representational structure that may organize description but has no authority to alter the endpoint.

A datum is *role-typed* when its endpoint-use role is fixed as anchor, tensor, or skin. It is *role-ambiguous* when its endpoint-use role is not fixed relative to $Y(G)$.

Definition VIII.34 (Claim promotion and status laundering). A *claim promotion* is a move

from a weaker claim status to a stronger claim status. The promotion relevant here is

$$\text{sector representation} \longrightarrow \text{same-theory completion.}$$

Status laundering is claim promotion without discharging the carrier, scope, standing, admissibility, and failure-consequence conditions required by the stronger claim.

Lemma VIII.35 (ATS role typing is non-generative). *Endpoint-use ATS role typing does not create endpoint authority. It only classifies how a datum may relate to the already fixed theorem object $Y(G)$.*

Proof. Suppose role typing created endpoint authority. Then the act of classifying a datum as anchor, tensor, or skin would introduce a new same-domain authorizer for the endpoint. But by Theorems VIII.8 and VIII.10, same-domain continuation is legitimate only when it preserves the kernel already instantiated in $Y(G)$. A new authorizer would create or transfer standing rather than preserve it, contrary to the no-independent-carrier and conservation consequences recorded in the kernel packet. Therefore role typing cannot generate endpoint authority. It can only identify whether a datum functions as fixed precondition, load-bearing continuation structure, or classificatory skin relative to the fixed theorem object. $\square$

Lemma VIII.36 (Role ambiguity is fail-closed for endpoint use). *A role-ambiguous datum cannot enter a same-domain completion claim for $Y(G)$.*

Proof. A same-domain completion claim must preserve the fixed theorem object. To know whether a datum preserves that object one must know whether the datum is a fixed precondition, load-bearing continuation structure, or merely classificatory skin. If this role is ambiguous, the datum's endpoint authority is ambiguous. Ambiguous endpoint authority cannot license a same-domain theorem step because the step would not have a fixed carrier, fixed standing condition, or fixed failure consequence. Hence role-ambiguous data are rejected fail-closed for endpoint use. This does not deny that such data may exist as notation, representations, or sector-classification material; it denies only their eligibility for the complete-theory spectral bridge. $\square$

Definition VIII.37 (UEAP endpoint promotion map). For a theorem-bearing claim C , the restricted UEAP endpoint map records

$$\left(\text{Carrier}(C), \text{Scope}(C), \text{Standing}(C), \text{Adm}(C), \text{Fail}(C) \right),$$

where the five entries are respectively the object that bears the claim, the domain on which the claim is asserted, the condition licensing the claim to do theorem work, the same-domain condition that must be preserved, and the consequence of failure. This is the claim-level form of the UEAP map in Theorem VIII.11.

Lemma VIII.38 (Carrier adequacy for claim promotion). *A claim promotion is legitimate only if the promoted claim has an adequate carrier.*

Proof. A promoted claim performs theorem work not performed by the weaker claim. That work must attach to a determinate object. If no adequate carrier is fixed, there is no theorem object that bears the promoted status. Promotion without carrier adequacy is therefore an unlicensed claim promotion in the sense of Theorem VIII.34, not a licensed theorem step. $\square$

Lemma VIII.39 (Scope and standing adequacy for claim promotion). *A sector-classification claim cannot be promoted to a same-theory completion claim unless the promoted claim has adequate scope and standing for completion status.*

Proof. A sector representation has sector-classification scope: it organizes representation-theoretic information over the fixed observable algebra. A same-theory completion has fixed-domain completion scope: it claims to preserve the fixed local vacuum carrier, vacuum restriction, local dynamics, and endpoint identity while adding admissible structure. These are different claim statuses. Standing for the first therefore does not automatically provide standing for the second. If the scope and standing of the promoted claim are not verified, the promotion changes what the datum is being allowed to prove without licensing the stronger use. By Theorem VIII.37, such a promotion fails UEAP adequacy and is rejected as an unlicensed claim promotion. $\square$

Definition VIII.40 (Sector-to-completion admissibility filter). A theorem-scope realized sector datum $z = (\xi, \pi_\xi)$ passes the sector-to-completion admissibility filter when all of the following hold:

- (i) z is endpoint-use ATS role-typed and not role-ambiguous;
- (ii) the promoted same-theory completion claim for z has UEAP-adequate carrier, scope, standing condition, admissibility condition, and failure consequence;
- (iii) z is packaged by a triple $(\mathcal{A}_z, \omega_z, U_z)$ satisfying the scope-faithful extension conditions of Theorem VIII.22;

(iv) that package preserves the fixed local vacuum carrier and is kernel-faithful in the sense of Theorem VIII.31.

Theorem VIII.41 (Sector representation is not completion eligibility). *Let $z = (\xi, \pi_\xi)$ be a realized sector datum over the fixed local algebra $\mathcal{A}_{\text{loc}}(G)$. Then*

z is a sector representation $\not\Rightarrow z$ determines a scope-faithful same-theory completion.

Proof. A sector representation is representation-theoretic data over the fixed observable algebra. A scope-faithful same-theory completion is stronger. It must provide a state extension with the correct vacuum restriction, an ω -preserving positive-energy time-translation implementation, preservation of the fixed local dynamics, an invariant canonical local cyclic sector, and kernel fidelity on the fixed theorem object. None of these completion requirements follows from representation-theoretic sector status alone. Therefore sector representation does not imply completion eligibility. The theorem does not deny the sector representation; it denies only automatic promotion of that representation to a vacuum-preserving same-domain completion. $\square$

Theorem VIII.42 (No one-step sector-to-completion promotion). *For a theorem-scope realized sector datum $z = (\xi, \pi_\xi)$, membership in $\mathfrak{S}_{\text{ext}}(G)$ does not imply membership in the class of theorem-scope admissible same-theory completions. A sector datum enters completion status only after passing the sector-to-completion admissibility filter.*

Proof. This is the endpoint-use form of Theorem VIII.41. Sector status is representation-theoretic data over the fixed observable algebra. Completion status additionally requires ATS role typing, UEAP carrier/scope/standing adequacy, a scope-faithful extension package, and kernel fidelity over the fixed theorem object; these are exactly the filter requirements of Theorems VIII.40 and VIII.43. $\square$

Theorem VIII.43 (Sector-to-completion admissibility filter). *A theorem-scope realized sector datum is eligible to enter the complete-theory spectral bridge only after passing the sector-to-completion admissibility filter of Theorem VIII.40. Conversely, once the datum is packaged as a scope-faithful, kernel-faithful same-domain completion datum satisfying that filter, it may be used as an input to the bridge.*

Proof. If a sector datum contributes to the spectral bridge, it is being used as part of a same-theory completion claim, not merely as sector-classification data. By Theorems VIII.35 and VIII.36, it must therefore have a fixed endpoint-use role and cannot be role-ambiguous. By Theorems VIII.38 and VIII.39, the promoted completion claim must have adequate carrier, scope, standing, admissibility condition, and failure consequence. Finally, by Theorem VIII.31, the datum must be packaged as a scope-faithful, kernel-faithful same-domain continuation. Hence it passes the filter.

Conversely, once the datum is packaged as a same-domain completion datum satisfying the filter, it is no longer being used merely as a sector representation. The filter supplies role typing, UEAP adequacy for the promoted completion claim, a scope-faithful extension package, preservation of the fixed local vacuum carrier, and kernel fidelity. Therefore the datum is eligible to enter the kernel-faithful GNS/spectral bridge as part of the sectoral completion branch. $\square$

Proposition VIII.44 (Filtered sector realizations yield scope-faithful extensions). *Every theorem-scope realized sector datum furnished by Theorem VI.10 that passes the sector-to-completion admissibility filter determines a completion-eligible scope-faithful extension of the fixed local vacuum theory in the sense of Theorem VIII.22.*

Proof. Passing the filter includes a package $(\mathcal{A}_z, \omega_z, U_z)$ satisfying Theorem VIII.22. The ATS part of the filter prevents the sector datum from being used outside its endpoint role. The UEAP part verifies that the promoted completion claim has the correct carrier, scope, standing condition, admissibility condition, and failure consequence. Kernel fidelity then ensures that the package preserves the fixed local vacuum carrier rather than introducing a new same-domain authorizer. Hence the filtered datum determines a completion-eligible scope-faithful extension of the fixed local vacuum theory. $\square$

Proposition VIII.45 (Filtered sector realizations are theorem-scope admissible completions). *The sectoral branch of theorem-scope admissible completions consists precisely of theorem-scope realized sector data that pass the sector-to-completion admissibility filter. Sector data that do not pass the filter remain part of the sector-classification layer only and do not enter the complete-theory spectral bridge.*

Proof. By Theorem VIII.41, sector representation alone does not imply same-theory completion eligibility. By Theorem VIII.43, a sector datum may enter the spectral bridge only

after passing the ATS/UEAP, scope-faithful, kernel-faithful filter. A datum that passes the filter is, by Theorem VIII.44, a scope-faithful extension and, by the final clause of the filter, a kernel-faithful same-domain continuation. It is therefore a theorem-scope admissible completion in the sense of Theorem VIII.31. Conversely, any sectoral theorem-scope admissible completion is by definition a sector datum used as a same-domain completion claim; hence it must pass the filter by Theorem VIII.43. $\square$

Theorem VIII.46 (Yang–Mills endpoint package satisfies the kernel-side conditions). *Let*

$$\left(\mathcal{A}_{\text{loc}}(G), \omega_{\text{vac}}, U_{\text{vac}}\right)$$

be the fixed local Yang–Mills theorem object together with the predecessor reconstruction-side endpoint package recalled in Sections III and II. Then this fixed Yang–Mills theorem object satisfies the exact structural side conditions required for application of the theorem-derived admissibility, uniqueness, and conservation constraints recorded in the compressed kernel packet above and developed in the upstream provenance corpus.

Proof. By the predecessor Yang–Mills theorem package, the fixed local object is a nontrivial bounded local observable theory with distinguished vacuum representation, exact local endpoint, and reconstruction-side realization structure. By Theorem V.1, extended-support sector data are excluded from the bounded local generators of that exact local endpoint. By Theorems VIII.43 to VIII.45, exactly the filtered theorem-scope realized sector data that enter the completion branch are organized over the same fixed local theorem object. Thus the fixed Yang–Mills theorem object together with its realized endpoint layer satisfies exactly the side conditions needed for the kernel-side admissibility, uniqueness, and conservation constraints recorded above to apply in the fixed-domain comparison carried out below. $\square$

Remark VIII.47 (Framework-language translation for theorem scope). In the language of the foundational pair, a theorem-scope admissible completion is a same-domain, scope-faithful enlargement of a fixed theorem object. The admissibility role here is therefore not to generate a new local theorem object from below, but to show that the enlarged realization preserves the same fixed local theorem object without introducing new subgap local-state content, new subgap multiplicity, or any other governance-non-equivalent same-scope alternative. A non-vacuum sector representation may remain a valid sector-classification object without

thereby entering the complete-theory spectral bridge; completion eligibility begins only after the sector-to-completion filter is passed.

Corollary VIII.48 (No new subgap same-domain content in theorem-scope completions). *Let $(\mathcal{A}, \omega, U)$ be a theorem-scope admissible completion of the fixed local vacuum theory. Then the completion has no additional standing-positive subgap content over $\mathcal{A}_{\text{loc}}(G)$ in either of the following forms:*

- (i) *no locally detectable subgap sector whose local vector state differs from ω_{vac} ; and*
- (ii) *no orthogonal locally generated subgap copy of the fixed local vacuum representation.*

Proof. A theorem-scope admissible completion is, by Theorem VIII.31, a same-domain completion claim over the fixed local theorem object. By Theorems VIII.8 and VIII.10, such a claim is legitimate only when it is kernel-faithful. If item (i) or item (ii) occurred, Theorem VIII.32 would show that the extension is not a kernel-faithful same-domain completion. Hence neither form of subgap content can occur in a theorem-scope admissible completion. $\square$

Lemma VIII.49 (Exclusion of orthogonal subgap local content under same-scope admissibility). *Let $(\mathcal{A}, \omega, U)$ be a theorem-scope admissible completion of the fixed local theory, with GNS triple $(\pi, \mathcal{H}, \Omega)$ and Hamiltonian H . If*

$$\psi \in E_H([0, m^*))\mathcal{H}, \quad \|\psi\| = 1,$$

then

$$\psi \in \mathcal{H}_{\text{loc}}.$$

Equivalently, no theorem-scope admissible completion contains a subgap vector in the orthogonal complement of the canonical local vacuum sector.

Proof. Because $\mathcal{H}_{\text{loc}}$ is reducing for H , so is $\mathcal{H}_{\text{loc}}^\perp$. Decompose

$$\psi = \psi_{\text{loc}} + \psi_\perp, \quad \psi_{\text{loc}} \in \mathcal{H}_{\text{loc}}, \quad \psi_\perp \in \mathcal{H}_{\text{loc}}^\perp.$$

Since $\psi \in E_H([0, m^*))\mathcal{H}$ and both summands belong to reducing subspaces, one has

$$\psi_{\text{loc}}, \psi_\perp \in E_H([0, m^*))\mathcal{H}.$$

Assume toward contradiction that $\psi_\perp \neq 0$, and set

$$\eta := \frac{\psi_\perp}{\|\psi_\perp\|} \in E_H([0, m^*))\mathcal{H}_{\text{loc}}^\perp.$$

If $\omega_\eta^{\text{loc}} \neq \omega_{\text{vac}}$, then η realizes a locally detectable subgap sector in the sense of Theorem VIII.30, contradicting Theorem VIII.48. If $\omega_\eta^{\text{loc}} = \omega_{\text{vac}}$, then the cyclic representation generated by η is a vacuum local representation carried in $\mathcal{H}_{\text{loc}}^\perp$; this is an orthogonal locally generated subgap copy of the fixed local vacuum representation, again excluded by Theorem VIII.48. Hence $\psi_\perp = 0$ and $\psi \in \mathcal{H}_{\text{loc}}$. $\square$

Lemma VIII.50 (Subgap vacuum-multiplicity-one for the fixed local theory). *Let*

$$K \subset \mathcal{H}_{\text{vac}}$$

be a closed $\pi_{\text{vac}}(\mathcal{A}_{\text{loc}}(G))$ -invariant subspace satisfying

$$K \subset E_{H_{\text{vac}}}([0, m^*))\mathcal{H}_{\text{vac}}.$$

Then

$$K = \mathbb{C}\Omega_{\text{vac}}.$$

Equivalently, the fixed local vacuum representation occurs with multiplicity one in the subgap spectral window.

Proof. This is immediate from Theorem VIII.29 applied to the scope-faithful extension given by the fixed local vacuum theory itself. Equivalently, it is exactly the predecessor gap statement on the canonical local vacuum sector. $\square$

Lemma VIII.51 (Local detectability of non-vacuum subgap vectors). *Let $(\mathcal{A}, \omega, U)$ be a theorem-scope admissible completion of the fixed local theory, with GNS triple $(\pi, \mathcal{H}, \Omega)$ and Hamiltonian H . If*

$$\psi \in E_H([0, m^*))\mathcal{H}, \quad \|\psi\| = 1,$$

and

$$\omega_\psi^{\text{loc}} \neq \omega_{\text{vac}},$$

then the completion contains an additional locally detectable subgap sector over $\mathcal{A}_{\text{loc}}(G)$.

Proof. By definition, $\omega_\psi^{\text{loc}}(A) = \langle \psi, \pi(A)\psi \rangle$ for $A \in \mathcal{A}_{\text{loc}}(G)$. Since $\omega_\psi^{\text{loc}} \neq \omega_{\text{vac}}$, its GNS triple is a representation of $\mathcal{A}_{\text{loc}}(G)$ distinct from the predecessor vacuum representation. Because $\psi \in E_H([0, m^*))\mathcal{H}$, this representation is realized by a vector in the subgap spectral window of the completion. This is exactly a locally detectable subgap sector in the sense of Definition VIII.30. $\square$

Lemma VIII.52 (Exclusion of locally detectable subgap sectors in theorem-scope admissible completions). *No theorem-scope admissible completion of the fixed local theory contains an additional locally detectable subgap sector over $\mathcal{A}_{\text{loc}}(G)$.*

Proof. This is item (i) of Theorem VIII.48. Equivalently, if such a sector existed, Theorem VIII.51 would exhibit it as new subgap local-state content over the original fixed local algebra, which theorem-scope admissibility forbids. $\square$

Corollary VIII.53 (Subgap local-state conservation). *Let $(\mathcal{A}, \omega, U)$ be a theorem-scope admissible completion, with Hamiltonian H . Then every unit vector*

$$\psi \in E_H([0, m^*))\mathcal{H}$$

satisfies

$$\omega_\psi^{\text{loc}} = \omega_{\text{vac}}.$$

Proof. If not, Theorem VIII.51 would produce an additional locally detectable subgap sector. This is impossible by Theorem VIII.52. $\square$

Lemma VIII.54 (Multiplicity decomposition for equal local vector states). *Let $\psi, \varphi \in \mathcal{H}$ be unit vectors such that*

$$\omega_\psi^{\text{loc}} = \omega_\varphi^{\text{loc}} =: \omega_0$$

on $\mathcal{A}_{\text{loc}}(G)$. Let

$$\mathcal{K}_\psi := \overline{\pi(\mathcal{A}_{\text{loc}}(G))\psi}, \quad \mathcal{K}_\varphi := \overline{\pi(\mathcal{A}_{\text{loc}}(G))\varphi}.$$

Then:

- (i) $(\pi|_{\mathcal{K}_\psi}, \mathcal{K}_\psi, \psi)$ and $(\pi|_{\mathcal{K}_\varphi}, \mathcal{K}_\varphi, \varphi)$ are both GNS triples for the same state ω_0 ;
- (ii) there exists a unitary

$$W : \mathcal{K}_\psi \rightarrow \mathcal{K}_\varphi$$

such that

$$W\pi(A)|_{\mathcal{K}_\psi}W^* = \pi(A)|_{\mathcal{K}_\varphi}, \quad W\psi = \varphi;$$

(iii) on the closed subspace

$$\mathcal{K} := \overline{\mathcal{K}_\psi + \mathcal{K}_\varphi},$$

the representation of $\mathcal{A}_{\text{loc}}(G)$ admits a multiplicity decomposition

$$\mathcal{K} \cong \mathcal{H}_{\omega_0} \otimes M, \quad \pi|_{\mathcal{K}} \cong \pi_{\omega_0} \otimes \mathbf{1}_M,$$

where $(\pi_{\omega_0}, \mathcal{H}_{\omega_0}, \Omega_{\omega_0})$ is the GNS triple of ω_0 , and M is a multiplicity space;

(iv) under this identification,

$$\psi = \Omega_{\omega_0} \otimes m_\psi, \quad \varphi = \Omega_{\omega_0} \otimes m_\varphi,$$

for unit vectors $m_\psi, m_\varphi \in M$.

Proof. The first two statements are standard uniqueness of the GNS construction for a fixed state. Since $\omega_\psi^{\text{loc}} = \omega_\varphi^{\text{loc}} = \omega_0$, the cyclic representations generated by ψ and φ are both GNS realizations of the same state ω_0 , hence are unitarily equivalent by a unique intertwiner sending the cyclic vector of one realization to the cyclic vector of the other.

The third and fourth statements are the standard multiplicity decomposition for two copies of the same cyclic representation inside a common Hilbert space: the algebra acts through the same representation on each copy, and any difference between ψ and φ lies entirely in the multiplicity factor M , on which the local algebra acts as the identity. $\square$

Lemma VIII.55 (Subgap vacuum-multiplicity exclusion). *Let $(\mathcal{A}, \omega, U)$ be a theorem-scope admissible completion, with Hamiltonian H . Suppose*

$$\psi, \varphi \in E_H([0, m^*))\mathcal{H}$$

are unit vectors satisfying

$$\omega_\psi^{\text{loc}} = \omega_\varphi^{\text{loc}}.$$

Then the multiplicity space M in Theorem VIII.54 is one-dimensional.

Proof. By Theorem VIII.53, both restricted local states equal the vacuum state:

$$\omega_\psi^{\text{loc}} = \omega_\varphi^{\text{loc}} = \omega_{\text{vac}}.$$

Apply Theorem VIII.54 with $\omega_0 = \omega_{\text{vac}}$. On the subspace generated by ψ and φ , the local algebra therefore acts as

$$\pi_{\text{vac}} \otimes \mathbf{1}_M.$$

If $\dim M > 1$, the completion contains two independent subgap carriers of the same fixed local vacuum standing. That is precisely an additional orthogonal subgap copy of the vacuum representation, excluded by item (ii) of Theorem VIII.48. Hence M is one-dimensional. $\square$

Corollary VIII.56 (Subgap local-state rigidity). *Let $(\mathcal{A}, \omega, U)$ be a theorem-scope admissible completion of the fixed local theory, with GNS triple $(\pi, \mathcal{H}, \Omega)$ and Hamiltonian H . If*

$$\psi, \varphi \in E_H([0, m^*))\mathcal{H}$$

are unit vectors and

$$\omega_\psi^{\text{loc}} = \omega_\varphi^{\text{loc}},$$

then

$$\varphi = e^{i\theta}\psi$$

for some $\theta \in \mathbb{R}$.

Proof. Apply Theorem VIII.54 and then Theorem VIII.55. Since the multiplicity space is one-dimensional, the corresponding unit vectors differ only by phase. $\square$

Theorem VIII.57 (Kernel-faithful GNS spectral bridge for theorem-scope admissible completions). *Let $(\mathcal{A}, \omega, U)$ be a theorem-scope admissible completion of the fixed local vacuum theory in the sense of Theorem VIII.31, and let $(\pi, \mathcal{H}, \Omega, H)$ be its GNS and Hamiltonian data. Then*

$$E_H([0, m^*))\mathcal{H} = \mathbb{C}\Omega.$$

Equivalently,

$$\text{Spec}(H) \cap (0, m^*) = \emptyset, \quad \ker H = \mathbb{C}\Omega.$$

In particular, the predecessor vacuum gap m^ remains the gap of every theorem-scope admissible completion.*

Proof. Let $\psi \in E_H([0, m^*))\mathcal{H}$ be a unit vector. By Theorem VIII.53,

$$\omega_\psi^{\text{loc}} = \omega_{\text{vac}}.$$

Applying Theorem VIII.56 with $\varphi = \Omega$, we obtain

$$\psi = e^{i\theta}\Omega.$$

Hence every subgap vector is proportional to Ω , so

$$E_H([0, m^*))\mathcal{H} = \mathbb{C}\Omega.$$

The stated spectral consequences follow immediately. $\square$

Corollary VIII.58 (Complete-theory endpoint for theorem-scope admissible completions). *Let $(\mathcal{A}, \omega, U)$ be a theorem-scope admissible completion of the fixed local vacuum theory. Then the predecessor vacuum-sector gap*

$$\text{Spec}(H_{\text{vac}}) \subset \{0\} \cup [m^*, \infty), \quad \ker H_{\text{vac}} = \mathbb{C}\Omega_{\text{vac}}$$

is the mass-gap statement of the complete Yang–Mills theory in the theorem scope fixed by the present manuscript.

Proof. Apply Theorem VIII.57. By Theorem VIII.2, the fixed local vacuum theory is unchanged, and by the bridge theorem no new subgap spectral content appears in a theorem-scope admissible completion. Therefore the predecessor vacuum gap is the gap of the complete theory in the theorem scope under consideration. $\square$

Theorem VIII.59 (Falsification burden for the complete-theory endpoint). *Within the fixed dependency ledger of the present manuscript, and assuming the standard GNS/spectral lemmas are correctly applied, once the fixed local Yang–Mills theorem object has been verified as non-degenerate and the kernel has been instantiated there by necessity, the complete-theory endpoint can be defeated only by at least one of the following:*

- (i) *a faithful lower derivation of the kernel from strictly weaker same-regime conditions that do not already presuppose the work of admissibility, standing, reference, or irreversibility;*
- (ii) *a faithful same-regime counterexample preserving kernel fidelity while violating the endpoint, i.e. a completion $Y'(G)$ such that*

$$Y'(G) \sim_{\text{same regime}} Y(G), \quad Y'(G) \text{ is kernel-faithful}, \quad E_{H'}([0, m^*))\mathcal{H}' \neq \mathbb{C}\Omega';$$

- (iii) *a proof that the fixed local Yang–Mills theorem object is not a non-degenerate construction regime.*

Absent one of these, a proposed objection does not reach the theorem-level endpoint.

Proof. By Theorems VIII.9 and VIII.10, the fixed local Yang–Mills theorem object is a non-degenerate regime and the kernel is instantiated in it. By Theorem VIII.6, a lower derivation of the kernel is impossible unless it already imports the kernel work; this is exactly alternative (i). By Theorems VIII.7, VIII.8 and VIII.32, a same-regime counterexample must preserve the kernel while still violating the endpoint; this is alternative (ii). If the fixed Yang–Mills object were not non-degenerate, the kernel-instantiation premise would fail; this is alternative (iii). The three alternatives exhaust the ways to attack the theorem-content within the stated dependency ledger. $\square$

Remark VIII.60 (Source-neutral challenge boundary). Validity of the endpoint depends on theorem-content, not on authorial provenance of the corpus from which that theorem-content was developed. Methodological preference, citation provenance, or preference for a different proof route does not by itself identify a false premise, invalid inference, failed hypothesis match, or same-regime counterexample.

Non-claim. Theorem VIII.57 is *not* a theorem about an arbitrarily assembled direct-sum or direct-integral Hilbert space. It is a kernel-faithful spectral-invariance theorem for theorem-scope admissible completions of the fixed local vacuum theory.

Non-claim. The present section does *not* assert a charged-sector or defect-sector Hamiltonian theorem independent of the fixed local vacuum representation. Its conclusion is that theorem-scope admissible sectoral and ultraviolet completions introduce no new subgap local-state content over the fixed local algebra.

Corollary VIII.61 (Alignment with part (F) of the Main Theorem). *Part (F) of Theorem II.1 is exactly the kernel-faithful GNS spectral bridge of Theorem VIII.57 together with the complete-theory endpoint of Theorem VIII.58.*

Proof. This is precisely the role assigned to part (F) in Theorem II.1: theorem-scope admissible sectoral or ultraviolet completions satisfy the kernel-faithful local-state conservation and rigidity results proved above, and therefore inherit the predecessor vacuum gap as the complete-theory gap in the theorem scope fixed by the paper. $\square$

Remark VIII.62 (Why the section now ends at the complete-theory endpoint). The paper has now reached the theorem-level spectral endpoint stated in Theorem II.1. The predecessor local theorem remains unchanged, the admissible extension problem has been recast in

kernel-faithful GNS/spectral language using standard operator-algebraic machinery, and the requisite subgap conservation and rigidity statements have been proved for the theorem-scope completions treated here. No further direct-sum Hamiltonian theorem is needed.

Dependency summary.

Inputs. the fixed base object $\mathfrak{B}(G)$ from Section III, the compatibility theorem above, the local-net boundary and sector-construction results of Sections V–VII, and standard uniqueness/multiplicity results for GNS representations of the fixed local algebra

Imported standard background. standard C^* -algebra, GNS uniqueness, representation multiplicity, and spectral-projection language; additional theorem-content used here consists of kernel necessity, downstream fidelity, ATS role discipline, UEAP carrier discipline, and the sector-to-completion admissibility filter established earlier in this section

Conclusion at this node. the correct complete-theory spectral endpoint is obtained by a kernel-faithful GNS/spectral-invariance theorem for admissible sectoral and ultraviolet completions of the fixed local vacuum theory

Used later in. the final theorem-level reading of part (F) of Theorem II.1

Remark VIII.63 (Framework reading of completion classification). The classification theorem below is the Yang–Mills translation of the fixed-domain distinction used in the foundational pair: either an enlargement preserves the same theorem object on the same scope, or it changes that object and therefore exits the same fixed-domain meaningful regime. There is no third admissible same-scope option.

Theorem VIII.64 (Completion classification over the fixed local Yang–Mills theory). *Let*

$$(\mathcal{A}_{\text{loc}}(G), \omega_{\text{vac}}, U_{\text{vac}})$$

be the fixed local Yang–Mills theory furnished by the earlier local theorem package, with vacuum GNS triple

$$(\pi_{\text{vac}}, \mathcal{H}_{\text{vac}}, \Omega_{\text{vac}})$$

and Hamiltonian H_{vac} satisfying

$$\text{Spec}(H_{\text{vac}}) \subset \{0\} \cup [m^*, \infty), \quad \ker H_{\text{vac}} = \mathbb{C}\Omega_{\text{vac}}.$$

Let $(\mathcal{A}, \omega, U)$ be an extension built over $\mathcal{A}_{\text{loc}}(G)$ that is scope-faithful in the sense of Theorem VIII.22. Thus it contains $\mathcal{A}_{\text{loc}}(G)$ as a unital C^ -subalgebra, restricts to ω_{vac} on $\mathcal{A}_{\text{loc}}(G)$,*

carries a positive-energy ω -preserving GNS implementation $U(t) = e^{-itH}$, implements the same local time-translation dynamics on $\mathcal{A}_{\text{loc}}(G)$, and has invariant canonical local cyclic subspace. Assume moreover that the added structure is one of the completion branches treated in this manuscript:

1. an extended-support sector layer over the fixed local net, as constructed in the sector part of this manuscript and passed through the sector-to-completion admissibility filter; or
2. an ultraviolet enlargement agreeing with the fixed macroscopic local net on $\mathcal{A}_{\text{loc}}(G)$, in the sense isolated in the sharp-local-fields scope note.

Then exactly one of the following holds.

(I) **Same-theory completion.** The extension is kernel-faithful and preserves the same fixed local theorem object, equivalently the same fixed-domain meaningful regime at the local level. In this case it is a theorem-scope admissible completion, and:

(a) the cyclic subspace

$$\mathcal{H}_{\text{loc}} := \overline{\pi(\mathcal{A}_{\text{loc}}(G))\Omega}$$

is canonically unitarily equivalent to the predecessor vacuum representation;

(b) the subgap space satisfies

$$E_H([0, m^*))\mathcal{H} = \mathbb{C}\Omega;$$

(c) and therefore the mass gap of the extension is exactly the predecessor mass gap m^* .

(II) **Non-same-theory extension.** The extension fails kernel fidelity by introducing additional locally detectable subgap content or additional subgap multiplicity for the fixed local vacuum representation. In that case it does not preserve the same fixed local theorem object, and hence does not remain in the same fixed-domain meaningful regime at the local level; it is a different theory object.

Equivalently: within the scope-faithful completion branches treated in this manuscript, an extension counts as the same Yang–Mills theory over the fixed local theorem object only when it is kernel-faithful, hence introduces no new locally detectable subgap sector and no additional subgap copy of the fixed local vacuum representation. In the language of the closure papers, this is exactly the condition that the extension remain a theorem-scope admissible completion of the same fixed domain rather than introducing a governance-non-equivalent alternative.

Proof. Let $(\pi, \mathcal{H}, \Omega)$ be the GNS triple of $(\mathcal{A}, \omega)$, and define

$$\mathcal{H}_{\text{loc}} := \overline{\pi(\mathcal{A}_{\text{loc}}(G))\Omega}.$$

By restriction of the state and invariance of the local dynamics,

$$(\pi|_{\mathcal{A}_{\text{loc}}(G)}, \mathcal{H}_{\text{loc}}, \Omega)$$

is the GNS representation of ω_{vac} as a C^* -dynamical system on the fixed local algebra. By the dynamical uniqueness lemma proved above, there is a unitary

$$W_{\text{loc}} : \mathcal{H}_{\text{loc}} \rightarrow \mathcal{H}_{\text{vac}}$$

intertwining both the local algebra representation and the time-translation group. Hence

$$W_{\text{loc}} H|_{\mathcal{H}_{\text{loc}}} W_{\text{loc}}^* = H_{\text{vac}}$$

and therefore

$$W_{\text{loc}} E_{H|_{\mathcal{H}_{\text{loc}}}}(\Delta) W_{\text{loc}}^* = E_{H_{\text{vac}}}(\Delta)$$

for every Borel set $\Delta \subset \mathbb{R}$. Since the predecessor vacuum theory has gap m^* ,

$$E_{H_{\text{vac}}}([0, m^*)) = P_{\Omega_{\text{vac}}},$$

so

$$E_{H|_{\mathcal{H}_{\text{loc}}}}([0, m^*)) = P_{\Omega}.$$

Thus the fixed local vacuum sector contributes exactly one subgap ray.

Now consider the full extension space $\mathcal{H}$. If

$$E_H([0, m^*))\mathcal{H} = \mathbb{C}\Omega,$$

then the extension introduces neither a locally detectable subgap sector over $\mathcal{A}_{\text{loc}}(G)$ nor an additional subgap copy of the vacuum representation. If, in addition, the extension is kernel-faithful as a same-domain continuation, this is precisely case (I), and the spectral bridge theorem already proved above gives the preserved mass gap m^* .

Conversely, if

$$E_H([0, m^*))\mathcal{H} \neq \mathbb{C}\Omega,$$

then one of two things must occur:

1. there exists a subgap vector whose restricted local vector state differs from ω_{vac} , yielding a new locally detectable subgap sector over the fixed local net; or
2. every subgap vector restricts to ω_{vac} , but the subgap space has dimension > 1 , yielding additional subgap multiplicity of the vacuum representation.

The bridge lemmas proved earlier exclude both possibilities for same-theory completions built over the fixed local theorem object. Hence if either occurs, the extension cannot be regarded as a completion of the same Yang–Mills theory over $\mathcal{A}_{\text{loc}}(G)$. It is instead a different theory object. The two cases are exhaustive and mutually exclusive because, on the fixed domain, the closure results leave no third admissible same-scope alternative between preservation and theory change. $\square$

Corollary VIII.65 (Exhaustion of admissible completions over the fixed local theorem object). *Within the completion class treated in this manuscript, every admissible extension over the fixed local Yang–Mills theory is either:*

1. *a same-theory completion preserving the predecessor vacuum mass gap m^* , or*
2. *a genuine theory change, because it introduces new locally detectable subgap content or new subgap vacuum multiplicity.*

There is no third case.

F. Route-locus exhaustion for same-domain completion alternatives

Definition VIII.66 (Native completion counterexample package). A native completion counterexample package

$$\text{CEX}_{\text{YM}}^{\text{comp}}(G; \mathcal{A}, \omega, U, \psi)$$

consists of the same fixed local carrier $\mathcal{A}_{\text{loc}}(G)$, the same vacuum restriction ω_{vac} , the same local dynamics U_{vac} , the same predecessor gap m^* , a scope-faithful extension $(\mathcal{A}, \omega, U)$, and a unit vector $\psi \in E_H([0, m^*))\mathcal{H}$ such that either $\omega_{\psi}^{\text{loc}} \neq \omega_{\text{vac}}$ or $\psi \perp \Omega$ while $\omega_{\psi}^{\text{loc}} = \omega_{\text{vac}}$, together with claimed same-theory completion standing.

Theorem VIII.67 (Native completion counterexample collapse). *A native completion counterexample package has exactly the following same-domain dispositions: sector/support only, bookkeeping, carrier/vacuum/dynamics shift, non-same-theory extension, unfiltered*

sector-to-completion promotion, or forbidden second same-domain completion authority. There is no autonomous same-theory completion-counterexample role outside this list.

Proof. If the package is not used as a same-theory completion claim, it is sector/support or bookkeeping. If it changes the local carrier, vacuum restriction, or local dynamics, it is a theorem-object shift. If it preserves those data but lacks the sector-to-completion filter, it is unfiltered promotion. If it passes the filter and remains kernel-faithful, Theorem VIII.57 excludes the asserted subgap vector. If it claims same-domain authority outside the GNS bridge, it introduces a second completion classifier, excluded by Theorem VIII.20. The listed cases are exhaustive. $\square$

Definition VIII.68 (Completion route loci). A same-domain endpoint-resolving completion alternative for the Yang–Mills endpoint enters at least one of the following loci:

- L1. local observable carrier drift;
- L2. vacuum restriction drift;
- L3. local dynamics drift;
- L4. extended-support-to-local promotion;
- L5. sector-to-completion promotion;
- L6. charged or defect subgap import;
- L7. additional vacuum multiplicity;
- L8. ultraviolet or sharp-field carrier shift;
- L9. direct-sum or direct-integral enlargement;
- L10. global-form label overreporting;
- L11. Wilson or regularization endpoint drift;
- L12. repair or reclassification of an unfiltered datum;
- L13. endpoint classifier or second completion authority.

Theorem VIII.69 (Completion route-locus exhaustion). *Every same-domain endpoint-resolving completion alternative over the fixed local Yang–Mills theorem object enters at least one of L1–L13. If it enters none, it does no completion work.*

Proof. A completion alternative must either change local carrier, vacuum restriction, dynamics, local membership, sector/completion status, subgap content, vacuum multiplicity, ultraviolet carrier, Hilbert-space carrier, global-form report level, regularization endpoint, act-time status, or endpoint classifier. Those are L1–L13. If none changes, the alternative preserves every admissibility-relevant component of the fixed theorem object and changes no completion verdict; it is endpoint-inert. $\square$

Theorem VIII.70 (Route-locus collapse). *No route-locus alternative survives as a same-domain completion counterexample unless it satisfies the GNS bridge and is therefore a theorem-scope admissible completion.*

Proof. L1–L3 are theorem-object changes. L4 contradicts the local-net boundary. L5 is blocked by Theorem VIII.42. L6 is blocked by the sector-to-completion filter and local-state conservation. L7 is blocked by subgap multiplicity exclusion. L8 requires scope-faithfulness; otherwise it is carrier shift. L9 is outside the theorem if it is arbitrary Hilbert-space assembly; if it is scope-faithful and kernel-faithful, the GNS bridge applies. L10 is UEAP overreporting. L11 is blocked by the group-level endpoint guard. L12 violates no-repair. L13 is excluded by the local completion packet and Theorem VIII.20. Therefore the only surviving route is the admitted GNS bridge route. $\square$

Corollary VIII.71 (No surviving same-domain completion alternative). *Every completion alternative preserving the same local Yang–Mills theorem object is either a theorem-scope admissible completion satisfying the GNS bridge, sector/support or bookkeeping, a theorem-object shift, or forbidden second same-domain completion authority.*

Proof. Immediate from Theorems VIII.69 and VIII.70. $\square$

G. Forced constrained-Hamiltonian realization layer

The spectral bridge above isolates the correct complete-theory endpoint over the fixed local vacuum theory. The matured corpus sharpens the constructional reading of that endpoint by separating two logical roles.

Constraint forcing. The admissibility-derived rigidity results for gauge and constraint dynamics force the standard constrained-Hamiltonian architecture on the regular stratum; it is not inserted as an independent formal preference. With standing surface $C \subset S$, quotient

map $\pi : C \rightarrow S_{\text{phys}}$, pulled-back form $\omega_C := \iota^* \omega$, and local constraint generators $\{\phi_a\}$, the forced regular-stratum package is

$$D := \ker(d\pi) = \ker(\omega_C), \quad \{\phi_a, \phi_b\} \approx \sum_c C_{ab}^c \phi_c, \quad H_\alpha \approx H_{\text{phys}} + \sum_a u_\alpha^a \phi_a.$$

Thus coisotropic standing, characteristic redescription, first-class closure, and multiplier-type Hamiltonian dynamics are theorem-level consequences of admissibility. In Yang–Mills language, S is the local phase space of gauge fields and conjugate momenta, C is the Gauss-law standing surface, D is the gauge-redescription distribution, and $S_{\text{phys}} = C/\sim$ is the reduced classical phase space²¹.

Completion closure. Once the gauge-invariant observable carrier and fixed local theorem object are in place, the same-scope operator closure theorem controls alternatives. No same-scope operator can totalize a partial construction on the standing-bearing carrier; every totalization attempt is either bookkeeping-equivalent, inadmissible, a change of theorem object, or an extensional collapse back to the fixed gate/interior. No new admissible same-scope family emerges by composition, iteration, diagonalization, or foundation swap²⁰.

Non-generative boundary. The closure layer does not generate Yang–Mills gauge dynamics. Constraint forcing fixes the gauge/constraint architecture; closure then excludes same-domain alternatives once that architecture and the fixed local theorem object have already been obtained. This division prevents the completion theorem from being read as a substitute for the predecessor analytic construction.

H. Recalled Haag–Kastler/Wightman realization of the fixed local theorem object

This subsection recalls the standard-form local QFT object supplied by the predecessor Yang–Mills theorem package. It does not introduce a new theory class and it does not leave the local algebras, locality, covariance, positivity, or vacuum representation to be constructed inside the present sector manuscript.

Recalled endpoint object. For each bounded spacetime region O , the predecessor package supplies the bounded-region local gauge-invariant algebra

$$O \longmapsto \mathcal{A}_{\text{loc}}(O; G),$$

with quasi-local algebra $\mathcal{A}_{\text{loc}}(G)$, vacuum state ω_{vac} , GNS triple $(\pi_{\text{vac}}, \mathcal{H}_{\text{vac}}, \Omega_{\text{vac}})$, and implemented time translations $U_{\text{vac}}(t) = e^{-itH_{\text{vac}}}$. This is the fixed local theorem object used throughout the present paper.

Standard-form properties recalled from the proof home. The predecessor proof home supplies, for the fixed compact connected simple gauge group G :

- (i) the bounded local gauge-invariant net and its quasi-local algebra;
- (ii) isotony, locality, and covariance of the local net in the standard Haag–Kastler sense;
- (iii) the vacuum state, positive-energy GNS representation, and implemented Minkowski time-translation dynamics;
- (iv) the OS/Wightman/Haag–Kastler reconstruction route and the non-triviality witness;
and
- (v) the vacuum-sector spectral statement

$$\text{Spec}(H_{\text{vac}}) \subset \{0\} \cup [m^*, \infty), \quad \ker H_{\text{vac}} = \mathbb{C}\Omega_{\text{vac}}.$$

The present manuscript uses these items as theorem-level input. Its new work is the extended-support role layer, local-net boundary, sector realization, completion classification, and complete-theory spectral bridge over that fixed object.

Same-domain uniqueness. If two Haag–Kastler/Wightman presentations are sound and complete for the same bounded local gauge-invariant observable content, the same vacuum restriction, and the same local time-translation dynamics, then they are lawfully equivalent on the fixed local theorem object. Any nontrivial divergence either changes the theorem object or introduces same-domain operator content not fixed by the standing-bearing carrier, contrary to closure by exhaustion.

Group-level endpoint guard. The endpoint is attached to the compact connected simple gauge group G and the fixed bounded local gauge-invariant net. Faithful finite-dimensional Wilson or sharp-field choices belong to the proof route and realization machinery; they are not additional parameters of the theorem object.

I. Mass-gap theorem package in the fixed-domain vacuum representation

Lemma VIII.72 (Hilbert-space closure of admissible excitations). *Let $(\mathcal{A}, \omega, U)$ be a scope-faithful extension of the fixed local vacuum theory, with GNS triple $(\pi, \mathcal{H}, \Omega)$ and canonical local vacuum sector*

$$\mathcal{H}_{\text{loc}} := \overline{\pi(\mathcal{A}_{\text{loc}}(G))\Omega}.$$

Then every vector in $\mathcal{H}_{\text{loc}}$ is the norm-limit of vectors of the form $\pi(A)\Omega$ with $A \in \mathcal{A}_{\text{loc}}(G)$. The canonical local vacuum sector is invariant under the functional calculus of the Hamiltonian, so the spectral projections $E_H(B)$ preserve that local closure for every Borel set $B \subset \mathbb{R}$. Consequently, after Theorem VIII.49, every theorem-scope subgap vector is controlled by the canonical local vacuum sector.

Proof. The first statement is immediate from the definition of $\mathcal{H}_{\text{loc}}$ as the closed cyclic subspace generated by the fixed local algebra acting on the vacuum. Invariance under the implemented time-translation group gives reduction for the self-adjoint generator and hence invariance under its spectral projections. Finally, Theorem VIII.49 excludes orthogonal subgap local content in theorem-scope admissible completions, so the full subgap window is governed by the canonical local sector. $\square$

Mass-gap proof architecture. The predecessor theorem verifies the vacuum-sector gap on the fixed local theorem object. The present manuscript verifies that theorem-scope admissible sectoral and ultraviolet completions introduce neither new locally detectable subgap content nor new subgap vacuum multiplicity. The bridge is therefore:

- (MG1) the predecessor package supplies positive-energy vacuum dynamics and the vacuum-sector gap;
- (MG2) dynamical GNS uniqueness identifies the canonical local sector of a scope-faithful extension with that predecessor vacuum representation;
- (MG3) same-scope admissibility, standing conservation, and no-independent-carrier closure exclude new subgap local-state content and new subgap vacuum multiplicity;
- (MG4) local-state conservation and rigidity force every subgap vector to be proportional to the vacuum; and

(MG5) the spectral theorem yields

$$E_H([0, m^*))\mathcal{H} = \mathbb{C}\Omega, \quad \text{Spec}(H) \cap (0, m^*) = \emptyset.$$

Thus the mass-gap role in this manuscript is not a second construction of the local Yang–Mills theory. It is completion control over the already constructed fixed local theorem object.

Definition VIII.73 (Clay-scope Yang–Mills realization). A *Clay-scope Yang–Mills realization* for the fixed compact connected simple gauge group G is a translation-covariant quantum Yang–Mills realization on $\mathbb{R}^4$ whose bounded local gauge-invariant observables determine a net

$$O \longmapsto \tilde{\mathcal{A}}_{\text{loc}}(O; G)$$

on bounded regions, together with a vacuum state $\tilde{\omega}_{\text{vac}}$ and a positive-energy time-translation group $\tilde{U}(t)$. Any additional extended-support or ultraviolet structure is regarded as an enlargement built over that bounded local gauge-invariant net.

Remark VIII.74 (Clay-scope terminology). The term “Clay-scope” is not a replacement for the official Clay problem statement and does not self-certify admissibility by definition. It names the class of Yang–Mills realizations already satisfying the predecessor package’s standard local existence, reconstruction, non-triviality, and vacuum-gap assumptions in the scope relevant to the present fixed-domain completion theorem.

Remark VIII.75 (Clay/local-net alignment at the completion stage). The official local-field and existence-side requirements for Yang–Mills theory—local gauge-invariant curvature-polynomial field content, OS/Wightman-strength reconstruction, positive energy, non-triviality, and a vacuum-sector spectral gap—belong to the predecessor local theorem package. The present completion argument begins only after those requirements have produced the fixed local theorem object $Y(G) = (\mathcal{A}_{\text{loc}}(G), \omega_{\text{vac}}, U_{\text{vac}})$. Its endpoint is therefore not another local-existence theorem, but the statement that every scope-faithful, kernel-faithful same-domain completion of that Clay-scope local object preserves the predecessor gap as the complete-theory gap.

Definition VIII.76 (Local equivalence of Clay-scope realizations). Let

$$\mathfrak{T}_1(G) = (\mathcal{A}_{\text{loc}}^{(1)}(\cdot; G), \omega_{\text{vac}}^{(1)}, U^{(1)}), \quad \mathfrak{T}_2(G) = (\mathcal{A}_{\text{loc}}^{(2)}(\cdot; G), \omega_{\text{vac}}^{(2)}, U^{(2)})$$

be Clay-scope Yang–Mills realizations. We say that $\mathfrak{T}_1(G)$ and $\mathfrak{T}_2(G)$ are *locally equivalent* if there exists a family of $*$ -isomorphisms

$$\Phi_O : \mathcal{A}_{\text{loc}}^{(1)}(O; G) \xrightarrow{\cong} \mathcal{A}_{\text{loc}}^{(2)}(O; G) \quad (O \text{ bounded})$$

compatible with isotony and translations, intertwining the implemented local time-translation dynamics, and such that

$$\omega_{\text{vac}}^{(2)}(\Phi_O(A)) = \omega_{\text{vac}}^{(1)}(A)$$

for all $A \in \mathcal{A}_{\text{loc}}^{(1)}(O; G)$ and all bounded O .

Proposition VIII.77 (AQFT interface to the closure theorems). *Let*

$$\tilde{\mathfrak{T}}(G) = (\tilde{\mathcal{A}}_{\text{loc}}(\cdot; G), \tilde{\omega}_{\text{vac}}, \tilde{U})$$

be a Clay-scope Yang–Mills realization. Then its bounded local gauge-invariant observable net, vacuum state, lawful local equivalences, and positive-energy time-translation dynamics determine a fixed local meaningful regime to which the closure theorems from the foundational papers apply in the following precise sense.

- (i) *The bounded local net together with the vacuum state determines a stable family of determinate local expectation values and correlation functions on bounded regions.*
- (ii) *Lawful local equivalences in the sense of Theorem VIII.76 preserve that expectation-value content and the local time-translation dynamics.*
- (iii) *Consequently, the local observable/dynamical content of $\tilde{\mathfrak{T}}(G)$ is a fixed-domain realization in the sense needed to apply the closure theorems: no faithful lower generator, no faithful same-domain extension, unique gate classification and admissible interior, and conservation.*

Proof. Part (i) is immediate from the AQFT data: for each bounded region O , the pair

$$(\tilde{\mathcal{A}}_{\text{loc}}(O; G), \tilde{\omega}_{\text{vac}}|_{\tilde{\mathcal{A}}_{\text{loc}}(O; G)})$$

defines a determinate local expectation-value functional, and the time-translation group $\tilde{U}(t)$ generates the corresponding local time evolution. Thus the realization carries a stable family of determinate local observables and their vacuum correlations.

Part (ii) follows from the definition of local equivalence: the maps Φ_O preserve the bounded local observable content, the vacuum restriction, isotony, and translations. Hence they preserve exactly the local observable/dynamical data that individuate the realization on bounded regions.

For part (iii), the relevant closure inputs are now available with no further translation gap. Theorem 2.14 and Corollary 2.15 of *Minimal Conditions for Derivation* place admissibility, standing, reference, and irreversibility already in force for any meaningful regime supporting non-degenerate construction, while Theorems 4.6, 4.7, and 5.8 exclude a faithful lower generator and any governance-non-equivalent rival architecture in the same regime. Theorems 4.29 and 4.30 of *The Structure of Admissibility* exclude faithful lower generators and faithful same-domain extensions on a fixed domain. Theorems 6.17, 7.15, and 7.17 of that same paper give uniqueness of gate classification, uniqueness of the admissible interior, and conservation. Since the bounded local observable/dynamical content of $\tilde{\mathfrak{T}}(G)$ is exactly the fixed-domain content preserved by lawful local equivalence, these closure theorems apply to the local AQFT realization. $\square$

Proposition VIII.78 (AQFT consequences of the closure theorems). *For the fixed gauge group G , the closure theorems enter the present manuscript through the following operator-algebraic consequences for the bounded local gauge-invariant observable content of any Clay-scope Yang–Mills realization.*

- (i) **No faithful lower realization.** *There is no faithful lower generator of the fixed local observable/dynamical system and no faithful same-domain extension that produces a second same-theory realization of that system from strictly weaker or additional same-domain data.*
- (ii) **Uniqueness of the fixed local theory.** *Once the bounded local gauge-invariant observable content, the vacuum restriction, and the local time-translation dynamics are fixed, the resulting local theory is unique up to unique local equivalence in the sense of Theorem VIII.76.*
- (iii) **Conservation under same-theory enlargement.** *A same-theory enlargement over the fixed local observable content cannot create, amplify, transfer, or recover new locally detectable subgap content or additional subgap multiplicity for the fixed local vacuum representation.*

- (iv) **Exhaustion.** *Any putative alternative to the previous two cases is excluded: a realization is either locally equivalent to the fixed local theorem object, or it changes that theorem object and is therefore a different theory object.*

Proof. Part (i) is the AQFT consequence of Theorems 4.6, 4.7, and 5.8 of *Minimal Conditions for Derivation* together with Theorems 4.29 and 4.30 of *The Structure of Admissibility*, applied through Theorem VIII.77: once the bounded local observable/dynamical content is fixed, there is no faithful lower generator and no faithful same-domain extension producing a second same-theory realization from weaker or additional same-domain data.

Part (ii) is the operator-algebraic form of uniqueness of governance work, unique gate classification, and unique admissible interior: Theorem 5.8 of *Minimal Conditions for Derivation* rules out governance-non-equivalent rival architectures, and Theorems 6.17 and 7.15 of *The Structure of Admissibility* rule out distinct same-domain admissible interiors once the fixed content is specified. In AQFT language, this is exactly uniqueness up to local equivalence of the same fixed-domain theorem object.

Part (iii) is the AQFT consequence of conservation: Theorem 7.17 and Corollary 7.18 of *The Structure of Admissibility*, together with the standing-conservation papers, imply that same-domain enlargement cannot create or transport new standing-positive content. In the present bounded local AQFT setting, this means that a same-theory enlargement cannot introduce new locally detectable subgap content or additional subgap multiplicity for the fixed local vacuum representation.

Part (iv) is the exhaustion statement obtained by combining parts (i)–(iii): if a realization is not locally equivalent to the fixed local theorem object, then by definition it changes the bounded local observable/dynamical content. The closure theorems exclude any third admissible alternative between those two cases. $\square$

J. AQFT as a non-degenerate compositional regime

The final identification step requires one explicit interface statement connecting standard AQFT realizations of Yang–Mills theory to the closure theorems used in the manuscript. We state that interface here entirely in operator-algebraic language, but we keep the framework vocabulary explicit: fixed domain, lawful redescription, non-degenerate construction, no faithful lower generator, no faithful same-domain extension, uniqueness, and conservation.

Definition VIII.79 (Lawful AQFT equivalence on the fixed local theorem object). Let

$$(\mathcal{A}_{\text{loc}}(G), \omega_{\text{vac}}, U_{\text{vac}})$$

be the fixed local theorem object. A lawful AQFT equivalence of realizations over the fixed local theorem object is a unitary equivalence of C^* -dynamical systems

$$W : \mathcal{H}_1 \rightarrow \mathcal{H}_2$$

such that

$$W\pi_1(A)W^* = \pi_2(A) \quad (A \in \mathcal{A}_{\text{loc}}(G)), \quad WU_1(t)W^* = U_2(t), \quad W\Omega_1 = \Omega_2.$$

Definition VIII.80 (AQFT realization domain over the fixed local theorem object). An AQFT realization domain over the fixed local theorem object is the triple consisting of:

1. the fixed local gauge-invariant observable algebra $\mathcal{A}_{\text{loc}}(G)$;
2. the vacuum state ω_{vac} restricted to that algebra;
3. the implemented local time-translation dynamics $t \mapsto U_{\text{vac}}(t)$.

The admissibility-relevant content of a realization over the fixed local theorem object is the content invariant under lawful AQFT equivalence in the sense of Definition VIII.79; this is the operator-algebraic instance of lawful redescription on the same fixed domain.

Lemma VIII.81 (AQFT realizations determine non-degenerate compositional systems). *Let*

$$(\tilde{\mathcal{A}}, \tilde{\omega}, \tilde{U})$$

be a Yang–Mills AQFT realization on $\mathbb{R}^4$ satisfying the Clay-relevant assumptions used throughout this manuscript:

1. *a local gauge-invariant observable net with bounded-region local algebra*

$$\mathcal{A}_{\text{loc}}(G);$$

2. *a vacuum state whose restriction to $\mathcal{A}_{\text{loc}}(G)$ is ω_{vac} ;*
3. *translation covariance with positive energy, implemented by a strongly continuous unitary group whose restriction to the fixed local algebra agrees with U_{vac} ;*

4. *nontriviality of the local observable content in the sense that the local algebra contains observables with nonconstant vacuum correlations.*

Then the fixed local observable algebra together with its state and dynamics determines a non-degenerate compositional regime in the sense required by the closure theorems. More precisely:

1. *the fixed local observable algebra, its vacuum state, and its implemented dynamics determine a fixed domain of evaluation;*
2. *lawful AQFT equivalences preserve the local observable content, the vacuum state, and the local dynamics, and therefore serve as lawful redescriptions on that fixed domain;*
3. *the realization is non-degenerate: the observable content is not trivial, and the positive-energy vacuum dynamics give a stable distinction between vacuum-preserving continuation and representation-theoretically distinct alternatives;*
4. *consequently, the closure theorems apply to the fixed local theorem object extracted from the AQFT realization.*

Proof. The fixed local observable algebra together with the restricted vacuum state and implemented time-translation dynamics is, by construction, a bounded domain on which expectation values, correlation functions, and dynamical evolution are evaluated. This proves the first point.

For the second point, any lawful AQFT equivalence in the sense of Definition VIII.79 preserves the represented local algebra, the vacuum vector, and the implementing time-translation group. Therefore it preserves exactly the local observable content, state assignments, and dynamical structure that individuate the fixed local theorem object. Such equivalences are precisely the representation-theoretic redescriptions that leave the fixed local content unchanged.

For the third point, nontriviality of the local observable content excludes collapse to a trivial one-point realization. Positive energy and the existence of the distinguished vacuum state provide the fixed reference point for local state evaluation and distinguish the vacuum sector from any genuinely different represented local state. In particular, the local algebra, state, and dynamics support the same role played abstractly by a non-degenerate compositional regime: there is a fixed domain, a lawful notion of redescription, and a nontrivial class of observables

whose expectation values and time-evolved correlations are meaningfully evaluated.

The fourth point is now immediate. Since the AQFT realization determines a fixed local theorem object together with its lawful AQFT equivalences, and since this object is non-degenerate in the preceding sense, the closure theorems apply to it through the operator-algebraic realization just described. $\square$

Theorem VIII.82 (AQFT interface to the closure theorems). *Let*

$$(\tilde{\mathcal{A}}, \tilde{\omega}, \tilde{U})$$

be a Yang–Mills AQFT realization satisfying the hypotheses of Lemma VIII.81. Then:

- 1. no faithful lower realization of the same fixed local theorem object exists;*
- 2. no faithful same-domain extension can introduce new standing-positive content while preserving the fixed local observable content, vacuum restriction, and local dynamics;*
- 3. any two realizations with the same admissibility-relevant content are lawfully equivalent on the fixed local theorem object.*

Accordingly, Theorem 5.8 of Minimal Conditions for Derivation and Theorems 4.29, 4.30, 6.17, 7.15, and 7.17 of The Structure of Admissibility apply to AQFT Yang–Mills realizations through this interface.

Proof. By Lemma VIII.81, the AQFT realization determines a non-degenerate fixed-domain regime on the fixed local theorem object with lawful AQFT equivalence as its admissible redescription class. Theorem 5.8 of *Minimal Conditions for Derivation* therefore applies to the realization at the level of governance work: no rival same-regime architecture with different admissibility-relevant content can perform the same work on the fixed local theorem object.

Likewise, Theorems 4.29 and 4.30 of *The Structure of Admissibility* apply to the fixed local domain extracted from the AQFT realization. They exclude both faithful lower realizations and faithful same-domain extensions introducing new standing-positive content while preserving the same fixed-domain object. Theorem 6.17 gives uniqueness of gate classification on that fixed scope, Theorem 7.15 gives uniqueness of the admissible interior, and Theorem 7.17 gives conservation.

Therefore any AQFT Yang–Mills realization satisfying the hypotheses of the present theorem is governed by the same closure alternatives: either a purported additional structure

has no independent work, is a forbidden same-domain extension, violates conservation, or collapses extensionally back to the same gate/interior. This is exactly the interface needed in the sequel. $\square$

K. Structural theorem provenance for the closure layer

Before stating the fixed-domain closure consequences, we record the exact source posture used in this section. The structural primitives appearing below are not treated here as newly posited axioms. They are used here through theorem-content recorded in the restricted kernel packet below and developed in the upstream provenance corpus: the governance-level bivalence theorem, the AMetric boundary and non-parameterization package, uniqueness of the admissible interior, and standing–reuse-stable-admissibility equivalence^{13–16}.

The role of the present section is therefore twofold. First, it states what the fixed local Yang–Mills theorem object forces in operator-algebraic language once same-domain comparison is made coherent. Second, it separates genuine same-theory continuation from bookkeeping, forbidden same-domain extension, and change of theorem object. This is a theorem-provenance statement, not a re-axiomatization.

L. Constructed theorem objects lie on the necessity-constrained side

The final proof-posture issue is not whether the structural kernel exists upstream, but whether the Yang–Mills theorem object constructed here lies on the side of that kernel relevant to same-domain comparison. We record that bridge explicitly.

Definition VIII.83 (Standing-bearing fixed local theorem object). A fixed local theorem object

$$\left(\mathcal{A}_{\text{loc}}(G), \omega_{\text{vac}}, U_{\text{vac}}\right)$$

is standing-bearing if:

1. its bounded local gauge-invariant observable content is nontrivial;
2. lawful redescription is fixed to unitary equivalence preserving local observable content, vacuum vector, and implemented local dynamics;
3. same-domain continuation is evaluated only relative to that fixed content.

Proposition VIII.84 (Constructed Yang–Mills theorem objects are standing-bearing). *Let the predecessor Yang–Mills theorem package provide a non-trivial fixed local theorem object together with the vacuum-sector mass-gap theorem used in the present manuscript. Then the fixed local theorem object is standing-bearing in the sense of Definition VIII.83.*

Proof. Nontriviality is part of the predecessor construction: the bounded local gauge-invariant observable content is not vacuous and the vacuum representation is not a one-point collapse. Lawful redescription is fixed in the present paper to unitary equivalence preserving the represented local algebra, the vacuum vector, and the implemented local dynamics. Finally, every same-domain comparison carried out in the manuscript is explicitly relative to that fixed bounded local content. So all three conditions hold. $\square$

Theorem VIII.85 (Standing-bearing fixed theorem objects instantiate the endpoint necessity constraints). *Let*

$$\left(\mathcal{A}_{\text{loc}}(G), \omega_{\text{vac}}, U_{\text{vac}}\right)$$

be a standing-bearing fixed local theorem object satisfying the predecessor vacuum-sector mass-gap theorem. Then, for purposes of same-domain comparison at the endpoint, the object lies on the side of the theorem-derived necessity constraints established in Subsection VIII Q. More precisely:

1. *the governing admissibility interface is bivalent and fail-closed;*
2. *no admissibility-relevant parameterization or indexed plurality survives at the boundary;*
3. *the admissible interior is determined uniquely up to lawful extensional equivalence on that same fixed domain;*
4. *standing and admissibility cannot coherently come apart on that fixed domain.*

Proof. By Proposition VIII.84, the theorem object is nontrivial, has a fixed law of lawful redescription, and is compared only relative to a fixed bounded local domain. The contradiction chains written in Subsection VIII Q require exactly that much for the endpoint application.

For item (1), once same-domain evaluation is fixed relative to a nontrivial represented local object, any intermediate standing-bearing status would either be promotable without fresh gating (contradicting the fixed-domain irreversibility posture of the endpoint) or inert bookkeeping. Hence the interface is bivalent and fail-closed.

For item (2), any admissibility-relevant selector, ranking, or index at the boundary would introduce multiple positive-side alternatives over one and the same fixed local object. But such a differentiator is exactly what the in-paper AMetric contradiction chain excludes.

For item (3), if two distinct admissible interiors existed over the same fixed local theorem object, they would determine competing admissibility classifications for the same bounded local content. The in-paper uniqueness contradiction then applies directly.

For item (4), if standing and admissibility came apart on that same fixed object, one would obtain either standing-neutral admissibility or standing-bearing authorization distinct from the gate. Both branches are excluded by the in-paper standing–admissibility contradiction chain.

So the fixed Yang–Mills theorem object lies on the necessity-constrained side relevant to the endpoint. The explicit hypothesis verification is recorded separately in Theorem VIII.101. $\square$

Corollary VIII.86 (Endpoint use of theorem-derived necessity constraints is internal to the manuscript). *For the constructed fixed local Yang–Mills theorem object, the endpoint use of bivalence, AMetric boundary, uniqueness of admissible interior, and standing–reuse-stable-admissibility equivalence proceeds through Theorem VIII.85. The endpoint therefore does not depend merely on external labels or on an unformalized appeal to “coherent domains”.*

Proof. Immediate from Proposition VIII.84, Theorem VIII.85, and Theorem VIII.101. $\square$

Kernel-status reminder. The endpoint arguments in this section do not attempt to generate the structural kernel from within a downstream axiomatic regime. They apply previously established theorem-level necessity content to the fixed Yang–Mills theorem object. In this exact sense, admissibility is not an added axiom layer local to the present manuscript, but the governing closure architecture already proved for the relevant class of fixed-domain meaningful regimes.

M. Admissibility equals validity on the fixed Yang–Mills domain

We now make explicit the final identification claim required for the endpoint: on the fixed Yang–Mills theorem object treated in this manuscript, admissibility is not a merely external label attached to otherwise independent notions of mathematical validity. Rather, within

the same-domain comparison fixed here, admissibility coincides with mathematical validity for purported continuations of the original acts on the original fixed domain.

Definition VIII.87 (Mathematical validity on the fixed Yang–Mills domain). A purported continuation of the fixed local Yang–Mills theorem object is mathematically valid on the fixed domain if it governs the same original admissibility-relevant acts on that same fixed domain without illicit structure import, trivialization of the original acts, collapse of the base goods needed for fixed-domain evaluation, or silent extensional reidentification of a different gate as though it were an independent rival.

Remark VIII.88 (Why this is the correct validity notion here). The present paper is not classifying arbitrary theory objects on arbitrary scopes. It is classifying purported *same-domain continuations* of one fixed Yang–Mills theorem object. In that setting, the relevant notion of mathematical validity is faithful same-domain validity for the original acts on the original fixed domain. The closure theorems recorded in the manuscript and developed in the upstream provenance corpus are stated exactly for that notion of faithfulness.

Lemma VIII.89 (No faithful same-domain mathematical validity outside the gate). *Let*

$$\left(\mathcal{A}_{\text{loc}}(G), \omega_{\text{vac}}, U_{\text{vac}}\right)$$

be the fixed local Yang–Mills theorem object. There exists no faithful same-domain source of mathematical validity for the original acts on the original fixed domain outside the admissibility gate governing that theorem object.

Proof. The manuscript’s fixed-domain comparison is a comparison of purported continuations of the same original admissibility-relevant acts on the same fixed domain. So any candidate source of mathematical validity outside the gate would have to function as an independent same-domain authorizer for those original acts.

By the non-existence of a faithful lower generator, any purported same-domain lower source of authority for admissibility, standing, reference, or irreversibility ends in exactly four outcomes: illicit structure import, triviality, violation of the base assumptions, or extensional collapse back to the gate itself. This is the theorem-level closure result recorded in the necessity/structure papers. The same conclusion is stable under faithful same-domain extension. Therefore any purported independent same-domain source of mathematical

validity lies in one of the four excluded classes or collapses extensionally to the admissibility gate. There is no fifth option. $\square$

Lemma VIII.90 (Every mathematically valid same-domain continuation is admissible). *Every mathematically valid same-domain continuation of the fixed local Yang–Mills theorem object is admissible.*

Proof. Let $\mathfrak{C}$ be a mathematically valid same-domain continuation. If $\mathfrak{C}$ were not admissible, then either:

1. $\mathfrak{C}$ nevertheless functions as an independent same-domain authorizer for the original acts;
or
2. $\mathfrak{C}$ changes no same-domain verdict for the original acts and so has no independent same-domain authority.

Case (1) is impossible by Lemma VIII.89: there is no faithful same-domain source of mathematical validity outside the gate. Any such attempt either imports illicit structure, trivializes the original acts, violates the base goods, or collapses extensionally back to the gate.

In case (2), the purported continuation has no independent same-domain validity content distinct from the gate. If it changes nothing, then it is not a non-admissible mathematically valid rival at all; it is bookkeeping or extensional restatement. Thus no mathematically valid same-domain continuation can fail to be admissible. $\square$

Lemma VIII.91 (Every inadmissible same-domain continuation is not mathematically valid). *Every inadmissible purported same-domain continuation of the fixed local Yang–Mills theorem object is not mathematically valid on that same fixed domain.*

Proof. Let $\mathfrak{C}$ be a purported same-domain continuation that is inadmissible. If $\mathfrak{C}$ were mathematically valid on the same fixed domain, then by Lemma VIII.90 it would be admissible. This contradicts the hypothesis. So every inadmissible same-domain continuation is not mathematically valid on that same fixed domain. $\square$

Theorem VIII.92 (Admissibility equals validity on the fixed Yang–Mills domain). *Let*

$$\mathfrak{Y}(G) := (\mathcal{A}_{\text{loc}}(G), \omega_{\text{vac}}, U_{\text{vac}})$$

be the fixed local Yang–Mills theorem object satisfying the manuscript’s required existence standard and vacuum-sector mass-gap endpoint. Then, for purported continuations of the original acts on the original fixed domain, admissibility coincides with mathematical validity. More precisely:

1. every mathematically valid same-domain continuation is admissible;
2. every inadmissible same-domain continuation is not mathematically valid;
3. equivalently, there is no mathematically valid same-domain rival lying outside the admissible class.

Proof. Item (1) is Lemma VIII.90. Item (2) is Lemma VIII.91. Item (3) is the conjunction of items (1) and (2). The substantive content is Lemma VIII.89, which rules out any faithful same-domain source of mathematical validity outside the admissibility gate by the internal and extension-stable non-derivability closure theorems. $\square$

Corollary VIII.93 (No mathematically valid same-domain rival to the Yang–Mills endpoint). *There exists no mathematically valid same-domain rival to the Yang–Mills endpoint lying outside the admissible class.*

Proof. Immediate from Theorem VIII.92. $\square$

N. The Yang–Mills endpoint is a representational commitment under standing

We now close the definition-matching point by verifying that the Yang–Mills endpoint used throughout the present section is a representational commitment under standing in the exact technical sense needed by the irreversibility theorem.

Proposition VIII.94 (The Yang–Mills endpoint is a representational commitment under standing). *Let*

$$\left(\mathcal{A}_{\text{loc}}(G), \omega_{\text{vac}}, U_{\text{vac}}\right)$$

be the fixed local Yang–Mills theorem object together with the vacuum-sector mass-gap endpoint proved in the earlier local theorem package and carried through the present paper. Then this endpoint is a representational commitment under standing in the following precise sense:

1. *it selects, under standing, a specific admissible redescription of the Yang–Mills theory on the fixed domain;*

2. *the selected redescription serves as a standing-bearing reference anchor for subsequent same-domain comparison;*
3. *future admissible redescriptions are constrained by that selected endpoint.*

Proof. For item (1), the present manuscript does not work over an unstructured proposal space. It works over a fixed local theorem object: bounded local gauge-invariant observable content, a fixed vacuum state on that content, and fixed local time-translation dynamics, together with the proved vacuum-sector mass-gap endpoint. That is exactly the selection of a specific admissible redescription on a fixed domain.

For item (2), the selected endpoint is standing-bearing because the paper's same-domain comparison always treats it as the reference object relative to which later continuations are evaluated. In particular, lawful same-domain redescription is defined only relative to preservation of the fixed local observable content, the vacuum restriction, and the local dynamics. Thus the endpoint functions as the standing-bearing reference anchor in the precise technical sense relevant here.

For item (3), future admissible redescriptions are constrained by the selected endpoint because the later completion, individuation, and endpoint theorems all evaluate purported continuations by asking whether they preserve or alter that fixed theorem object. A continuation that changes the bounded local content, the vacuum restriction, or the local dynamics is not a same-domain continuation of the selected endpoint. A continuation that leaves them fixed remains subordinate to the spectral and structural consequences already attached to that endpoint. Hence future admissible redescriptions are constrained by the selected endpoint. □

Corollary VIII.95 (Applicability of admissibility-irreversibility to the Yang–Mills endpoint). *The Yang–Mills endpoint satisfies the definition-matching hypotheses needed for Theorem VIII.96. Therefore the irreversibility theorem applies to the endpoint without a definition-level gap.*

Proof. Immediate from Proposition VIII.94 together with the standing-bearing fixed-domain posture established earlier in this section. □

O. Admissibility-irreversibility of the Yang–Mills endpoint

Theorem VIII.96 (Admissibility-irreversibility of the Yang–Mills endpoint). *Once the Yang–Mills endpoint commitment is established, there exists no admissible sequence capable of globally retracting, undoing, or neutralizing that commitment without violating admissibility or closure.*

Proof. By Proposition VIII.94, the Yang–Mills endpoint is a representational commitment under standing in the exact technical sense relevant here. By Corollary VIII.95, the irreversibility theorem applies to that endpoint. Therefore no admissible global rollback exists. $\square$

P. Hypothesis verification for the endpoint necessity constraints

We now verify, one theorem at a time, that the constructed fixed local Yang–Mills theorem object satisfies the hypotheses required by the theorem-derived necessity constraints used at the endpoint. The goal of the present subsection is purely logical hygiene: not to restate the endpoint, but to record explicitly that the hypotheses of the theorem-derived constraints are matched by the constructed object in the precise theorem scope used here.

Proposition VIII.97 (Verification for the bivalence theorem). *The constructed fixed local Yang–Mills theorem object satisfies the hypotheses of Theorem VIII.108. More precisely:*

1. Stable reference: *the bounded local gauge-invariant observable content is nontrivial and remains fixed under lawful redescription.*
2. Identity persistence: *lawful redescription in the present paper is unitary equivalence preserving the represented local algebra, the vacuum vector, and the implemented local dynamics; hence identity is preserved across lawful update.*
3. Irreversible commitment: *by Theorem VIII.96, once the fixed local theorem object is coupled to the vacuum-sector mass-gap theorem already proved in the manuscript, there is no nontrivial same-domain continuation preserving local observable content, vacuum restriction, and local dynamics while undoing the vacuum-sector spectral separation already established.*

Proof. For item (1), nontriviality is part of the predecessor construction: the bounded local gauge-invariant observable content is not vacuous and the vacuum representation is not a one-point collapse. Since lawful redescription is fixed here to preserve that represented local content, reference remains stable under lawful redescription.

For item (2), the paper has fixed lawful redescription to unitary equivalence preserving the local algebra, vacuum vector, and local dynamics. This is exactly the persistence condition needed for the theorem: the theorem object is not silently reidentified across lawful update.

For item (3), Theorem VIII.96 supplies the needed irreversibility directly from the physical Yang–Mills theorem object: once the fixed local observable content, the vacuum restriction, and the local dynamics are held fixed, no same-domain continuation can undo the already-established vacuum-sector spectral separation. Hence the irreversibility hypothesis needed for Theorem VIII.108 is satisfied in the theorem scope relevant to the endpoint without circular appeal to the later closure layer. $\square$

Proposition VIII.98 (Verification for the AMetric boundary theorem). *The constructed fixed local Yang–Mills theorem object satisfies the hypothesis posture needed for Theorem VIII.109. In the theorem scope used here, same-domain admissibility is evaluated only relative to fixed local observable content, fixed vacuum restriction, and fixed local dynamics, so no admissibility-relevant selector, ranking, or indexed positive-side plurality may be introduced without changing the theorem object or introducing forbidden same-domain structure.*

Proof. The AMetric theorem excludes admissibility-relevant parameterization at the boundary. In the present manuscript, the only admissibility-relevant domain for endpoint comparison is the fixed local theorem object. Any selector, ranking, or index that purported to distinguish multiple positive-side same-domain continuations over that object would have to do one of two things.

Either it changes the represented local content, vacuum restriction, or local dynamics. Then it is not a same-domain differentiator but a change of theorem object. Or it leaves those fixed and yet still purports to distinguish admissibility-relevant positive alternatives. Then it introduces extra same-domain governance work not carried by the fixed theorem object itself, which is exactly the forbidden same-domain structure excluded by the completion and individuation theorems proved earlier in this section. Hence the theorem scope relevant to the endpoint satisfies the non-parameterization posture required by Theorem VIII.109. $\square$

Proposition VIII.99 (Verification for uniqueness of the admissible interior). *The constructed fixed local Yang–Mills theorem object satisfies the hypothesis posture needed for Theorem VIII.111. In particular, same-domain comparison is fixed on a single bounded local observable/dynamical object, and no competing admissibility classification may be introduced on that same object without violating the endpoint completion / individuation layer.*

Proof. Uniqueness of admissible interior requires that competing admissibility classifications for the same fixed domain be excluded. That exclusion is exactly what the endpoint section has already established for the theorem object constructed here. By the domain-coherence and individuation theorems proved earlier, same-theory same-domain comparison is coherent only when the bounded local observable content, vacuum restriction, and local dynamics are fixed up to lawful equivalence. Once that fixation is in place, any purported competing admissibility classification is either a disguised change of theorem object, a forbidden same-domain extension, or bookkeeping only. So the fixed local Yang–Mills theorem object satisfies the uniqueness hypothesis posture needed for Theorem VIII.111. $\square$

Proposition VIII.100 (Verification for standing–reuse-stable-admissibility equivalence). *The constructed fixed local Yang–Mills theorem object satisfies the hypothesis posture needed for Theorem VIII.110. In the theorem scope of the endpoint, every same-domain admissibility determination is already a standing-bearing determination about the fixed theorem object, and no second same-domain authorizer is available.*

Proof. At the endpoint, admissibility is not evaluated on an abstract free-floating proposal space but only on purported same-domain continuations of the fixed theorem object. A positive same-domain admissibility determination therefore already commits the paper to treating that continuation as lawful over the fixed theorem object; that is exactly the standing-bearing role relevant here.

Conversely, any standing-bearing authorization distinct from same-domain admissibility would introduce a second same-domain authorizer. By the earlier completion classification and individuation theorems, no such second authorizer is available: if it changes the fixed local content, it is a different theorem object; if it does not, it is bookkeeping only. Hence the fixed local Yang–Mills theorem object satisfies the no-dual-track posture required for Theorem VIII.110. $\square$

Theorem VIII.101 (Hypotheses of the endpoint necessity constraints are satisfied). *The constructed fixed local Yang–Mills theorem object satisfies the hypotheses, in the theorem scope relevant to the endpoint, of:*

1. *Theorem VIII.108;*
2. *Theorem VIII.109;*
3. *Theorem VIII.111;*
4. *Theorem VIII.110.*

Accordingly, those theorem-derived necessity constraints apply to the endpoint by explicit hypothesis verification in the manuscript itself.

Proof. Combine Theorem VIII.96 with Propositions VIII.97, VIII.98, VIII.99, and VIII.100. □

Q. Compressed kernel packet retained for proof-home completeness

This subsection retains the minimal endpoint kernel packet as proof-home material. It is not a second introduction of the kernel and it does not duplicate the operative argument of Section VIII B. The operative chain is already fixed there: non-degenerate construction functionally forces the kernel; the fixed local Yang–Mills theorem object is non-degenerate; downstream completion claims are legitimate only by kernel fidelity. The present packet records the corresponding theorem-content in compact hypothesis/conclusion form for later cross-reference and source-neutral inspection.

Definition VIII.102 (Kernel of non-degenerate construction). The endpoint kernel is

$$K := \{\text{Adm}, \text{St}, \text{Ref}, \text{Irr}\},$$

where Adm names the valid/invalid construction boundary, St standing for licensed continuation, Ref determinate reference, and Irr non-repairable endpoint commitment.

Theorem VIII.103 (Minimal assessability boundary). *Any regime whose outputs are evaluated as determinate mathematical construction steps requires identifiable objects, a success/failure boundary for transitions, lawful preservation under redescription, and non-free*

rollback of endpoint commitments. In the notation of Theorem VIII.102, those functional requirements instantiate Ref, Adm, St, and Irr.

Proof. This is the compact form of Theorem VIII.5: object identifiability yields Ref, success/failure assessment yields Adm, licensed continuation under lawful equivalence yields St, and non-free rollback yields Irr. □

Theorem VIII.104 (Non-degenerate construction forces the kernel). *Every non-degenerate construction regime instantiates the full kernel K .*

Proof. Immediate from Theorems VIII.4 and VIII.103. □

Theorem VIII.105 (No lower generator on the same meaningful regime). *No member of K is faithfully generated from a strictly lower same-regime condition unless that condition already presupposes the work of K or something lawfully equivalent to it.*

Proof. This is Theorem VIII.6 in endpoint-kernel notation. □

Theorem VIII.106 (Regime-invariant non-derivability). *Changing formal dress, meta-language, or coding does not derive the kernel from below. A purported derivation is either not validity-assessable or already relies on the kernel roles needed for derivation.*

Proof. Any asserted derivation requires a target, licensed steps, a validity boundary, and stable commitment to its conclusion. These are precisely the kernel roles. □

Theorem VIII.107 (No faithful lower generator and no faithful same-domain extension). *For the admissibility boundary on a fixed domain, there is no faithful lower generator and no faithful same-domain extension that converts admissibility, standing, reference, or irreversibility into lower theorems for the original acts on the original fixed domain.*

Proof. Combine Theorems VIII.7, VIII.105 and VIII.106. □

Theorem VIII.108 (Necessity and bivalence of admissibility). *On a fixed domain supporting licensed construction, every admissibility-relevant status collapses extensionally to a binary split: admitted for standing-bearing continuation or not admitted.*

Proof. If a status performs no gate work, it is bookkeeping; if it performs gate work, it is the gate. No third same-domain status performs independent admissibility-relevant work. □

Theorem VIII.109 (AMetric boundary and non-parameterization). *At the admissibility boundary, no selector, ranking, grading, metric, or discrete indexing carries admissibility-relevant authority beyond the binary admitted/not-admitted split.*

Proof. Any such parameter is either not invariant under lawful redescription, effectless bookkeeping, or a disguised restatement of the binary gate. $\square$

Theorem VIII.110 (Standing equals reuse-stable admissibility). *On a fixed scope, standing is admissibility preserved through licensed same-scope continuation. It is not an additional positive classifier.*

Proof. A positive split beyond reuse-stable admissibility would either perform forbidden extra gate work or do nothing. $\square$

Theorem VIII.111 (Extensional uniqueness of the admissible interior). *For a fixed system on a fixed domain, the admissible interior is unique up to lawful extensional equivalence.*

Proof. Two interiors sound and complete for the same standing acts cannot disagree without assigning two standing verdicts to the same act on the same scope. $\square$

Theorem VIII.112 (Conservation of standing). *Within a fixed domain, licensed construction can preserve or destroy standing, but cannot create, amplify, transfer, or recover it.*

Proof. Creation, amplification, transfer, and recovery each require either a second same-domain authorizer, positive-side grading, post-hoc repair, or an independent carrier. The preceding kernel results exclude all four. $\square$

Proposition VIII.113 (Kernel violation destroys licensed same-domain construction of the endpoint). *A purported same-domain continuation of the fixed local Yang–Mills theorem object that violates the kernel is not a licensed same-domain construction of that endpoint.*

Proof. By Theorems VIII.8 and VIII.10, the Yang–Mills endpoint already instantiates the kernel and downstream legitimacy requires kernel fidelity. A kernel-violating move changes the carrier, exits the same domain, or fails as a theorem step. $\square$

Corollary VIII.114 (Internal endpoint packet available for Yang–Mills). *For the fixed local Yang–Mills theorem object, the endpoint has the following internal packet available:*

kernel necessity, no lower generator, no faithful same-domain extension, bivalent admissibility, AMetric boundary, standing-admissibility equivalence, unique admissible interior, and standing conservation.

Proof. Immediate from the results in this subsection together with Theorem VIII.10. $\square$

R. Direct AQFT derivation of the admissibility structure

We now state, in standard operator-algebraic language, the four structural features that the closure framework later packages as bivalence, AMetric boundary, uniqueness, and conservation. The point of the present subsection is that these features are not introduced as additional hypotheses once the fixed local theorem object and lawful AQFT equivalence have been specified.

Definition VIII.115 (Admissibility-relevant classification on the fixed local theorem object). An admissibility-relevant classification on the fixed local theorem object is a classification of realizations or enlargements over

$$(\mathcal{A}_{\text{loc}}(G), \omega_{\text{vac}}, U_{\text{vac}})$$

that affects one of the following on the same local domain:

1. the represented bounded local observable content;
2. the restricted vacuum state on $\mathcal{A}_{\text{loc}}(G)$;
3. the implemented local time-translation dynamics;
4. the existence of locally detectable subgap content or subgap multiplicity over the fixed local vacuum sector.

Classifications that change none of these are bookkeeping only.

Proposition VIII.116 (AQFT bivalence on the fixed local theorem object). *Let*

$$(\tilde{\mathcal{A}}, \tilde{\omega}, \tilde{U})$$

be a Yang–Mills AQFT realization satisfying the hypotheses of Lemma VIII.81. Then any admissibility-relevant classification on the fixed local theorem object is binary: either the realization is admitted as a same-theory realization on that fixed domain, or it is not. No third same-domain status performs independent work.

Proof. Let a purported same-domain status distinction be given. If it affects the represented bounded local observable content, the restricted vacuum state, the local dynamics, or the presence of locally detectable subgap content, then it changes the fixed local theorem object or its same-domain completion class. In that case it is not an intermediate status of the same theory on the same domain, but a different theory object. If it affects none of those items, then by Definition VIII.115 it is bookkeeping only. Hence there are only two admissibility-relevant possibilities on the fixed domain: same-theory admission or non-admission. $\square$

Proposition VIII.117 (AQFT AMetric boundary of the fixed local theorem object). *Let*

$$(\tilde{\mathcal{A}}, \tilde{\omega}, \tilde{U})$$

be a Yang–Mills AQFT realization satisfying the hypotheses of Lemma VIII.81. Then no admissibility-relevant grading, selector, ranking, or nontrivial metric survives on the pre-admissible side of the fixed local theorem object. Any such purported structure is either:

- 1. not invariant under lawful AQFT equivalence and therefore not admissibility-relevant;*
- 2. invariant but effectless on the fixed local theorem object and therefore bookkeeping only; or*
- 3. extensional restatement of the already-fixed same-theory / different-theory dichotomy.*

Proof. A grading, selector, ranking, or metric is admissibility-relevant only if it changes the classification of same-domain realizations. If it is not invariant under lawful AQFT equivalence, then it depends on presentation rather than on the fixed local theorem object and is therefore not admissibility-relevant. If it is invariant but does not change any admissibility-relevant item of Definition VIII.115, then it is bookkeeping only. If it is invariant and does change the classification, then by Proposition VIII.116 the only admissibility-relevant classification available is the binary same-theory / different-theory split, so the purported metric or ranking contributes no additional independent structure beyond that split. Thus no nontrivial admissibility-relevant metric structure survives on the fixed local theorem object. $\square$

Proposition VIII.118 (AQFT uniqueness of the admissible interior). *Let*

$$(\tilde{\mathcal{A}}, \tilde{\omega}, \tilde{U})$$

be a Yang–Mills AQFT realization satisfying the hypotheses of Lemma VIII.81. Then the admissible interior of the fixed local theorem object is unique up to lawful AQFT equivalence. Equivalently, there is no plurality of same-domain, same-theory admissible interiors over

$$(\mathcal{A}_{\text{loc}}(G), \omega_{\text{vac}}, U_{\text{vac}}).$$

Proof. Suppose there were two distinct same-domain admissible interiors over the fixed local theorem object. If they differ in represented bounded local observable content, restricted vacuum state, local dynamics, or subgap realization class, then they are different theory objects rather than two interiors of the same one. If they differ in none of those items, then they are lawfully AQFT equivalent by Definition VIII.79 and differ only by bookkeeping. Hence there is no genuine plurality of same-domain admissible interiors. $\square$

Proposition VIII.119 (AQFT conservation on the fixed local theorem object). *Let*

$$(\tilde{\mathcal{A}}, \tilde{\omega}, \tilde{U})$$

be a Yang–Mills AQFT realization satisfying the hypotheses of Lemma VIII.81. Then within the same fixed local theorem object, standing-positive status can be preserved or destroyed, but not created, amplified, transferred, or repaired without fresh gating.

Proof. By the spectral bridge and completion classification theorems proved earlier in this section, every same-theory completion of the fixed local theorem object has subgap space equal to the vacuum ray. Therefore a same-theory enlargement cannot create new locally detectable subgap content or new subgap multiplicity. If a purported enlargement does create such content, then by Definition VIII.115 it changes the same-domain theory object and is not a same-theory continuation. Thus standing-positive status on the fixed local theorem object may be preserved or destroyed under passage to another realization, but it cannot be created, amplified, transferred, or repaired within the same theory on the same domain. $\square$

Theorem VIII.120 (AQFT forces the admissibility structure on the fixed local theorem object). *Let*

$$(\mathcal{A}_{\text{loc}}(G), \omega_{\text{vac}}, U_{\text{vac}})$$

be the fixed local Yang–Mills theorem object, and let

$$(\tilde{\mathcal{A}}, \tilde{\omega}, \tilde{U})$$

be a Yang–Mills AQFT realization on $\mathbb{R}^4$ satisfying the hypotheses of Lemma VIII.81. Then, relative to the fixed local theorem object, the AQFT realization forces the following admissibility structure on the same domain:

- (a) binary gate, by Proposition VIII.116;
- (b) AMetric boundary, by Proposition VIII.117;
- (c) unique gate classification, as the binary same-theory / different-theory split of Proposition VIII.116;
- (d) unique admissible interior, by Proposition VIII.118;
- (e) conservation, by Proposition VIII.119.

Accordingly, the closure theorems of Minimal Conditions for Derivation and The Structure of Admissibility apply to Yang–Mills AQFT realizations through the fixed local theorem object itself, rather than through an added external assumption.

Proof. The theorem is the conjunction of Propositions VIII.116, VIII.117, VIII.118, and VIII.119, together with the interface theorem VIII.82. Once the fixed local theorem object is specified and lawful AQFT equivalence is taken as the admissibility-relevant notion of redescription, the admissibility structure is forced on the same domain rather than imported from outside it. $\square$

Theorem VIII.121 (Domain coherence forces fixation of the local theorem object). *Let*

$$\mathfrak{T}_1 = (\mathcal{A}_1, \omega_1, U_1), \quad \mathfrak{T}_2 = (\mathcal{A}_2, \omega_2, U_2)$$

be Yang–Mills AQFT realizations on $\mathbb{R}^4$ for the same compact simple gauge group G . Assume that $\mathfrak{T}_1$ and $\mathfrak{T}_2$ are claimed to be realizations of the same Yang–Mills theory on the same domain. Then the bounded local gauge-invariant observable content, the vacuum restriction, and the local time-translation dynamics must coincide up to lawful AQFT equivalence. Equivalently, the two realizations determine the same fixed local theorem object

$$(\mathcal{A}_{\text{loc}}(G), \omega_{\text{vac}}, U_{\text{vac}})$$

up to local equivalence.

If these data do not coincide, then the claim of “same Yang–Mills theory on the same domain” is incoherent: one has either a different theory object, a forbidden same-domain

extension, or a purported difference with no independent work, which collapses extensionally back to the same gate/interior.

Proof. By Lemma VIII.81 and Theorem VIII.120, each Yang–Mills AQFT realization satisfying the Clay-relevant assumptions determines a fixed admissibility structure on its bounded local gauge-invariant observable content: the gate is binary, the gate classification is unique, the admissible interior is unique up to lawful AQFT equivalence, and conservation forbids creation, amplification, transfer, or repair of new same-domain standing-positive content.

Now suppose that $\mathfrak{T}_1$ and $\mathfrak{T}_2$ are claimed to be realizations of the same Yang–Mills theory on the same domain, but that their bounded local gauge-invariant observable content, vacuum restriction, or local time-translation dynamics do not coincide up to lawful AQFT equivalence. Then one of two things must happen.

First, the difference performs independent admissibility-relevant work on the same domain. But Theorem 5.8 of *Minimal Conditions for Derivation* excludes any same-regime architecture with different governance work once the domain is fixed, and Theorem VIII.82 applies that exclusion to the present AQFT setting. In particular, one cannot have two non-equivalent same-domain Yang–Mills realizations carrying different admissibility-relevant local work while still claiming to be realizations of the same theory.

Second, the difference performs no independent admissibility-relevant work. Then, by Theorem VIII.82 together with Theorems 6.17 and 7.15 of *The Structure of Admissibility*, it cannot define a distinct gate classification or a distinct admissible interior on the same domain. The purported difference therefore collapses extensionally back to the same gate/interior and is bookkeeping only.

The final possibility would be a same-domain extension introducing new same-domain content while preserving the fixed domain. But Theorems 4.29 and 4.30 of *The Structure of Admissibility*, applied through Theorem VIII.82, exclude precisely such faithful lower realizations and faithful same-domain extensions. Finally, conservation (Theorem 7.17 of *The Structure of Admissibility*) excludes any attempt to preserve same-theory identity by creating, transporting, or repairing new same-domain positive content.

Accordingly, coherence of same-theory same-domain comparison forces fixation of the bounded local gauge-invariant observable content, the vacuum restriction, and the local time-translation dynamics up to lawful AQFT equivalence. This is exactly the fixation of the local theorem object. $\square$

Theorem VIII.122 (AQFT individuation by the fixed local theorem object). *Let*

$$\tilde{\mathfrak{T}}(G) := \left(\tilde{\mathcal{A}}_{\text{loc}}(\cdot; G), \tilde{\omega}_{\text{vac}}, \tilde{U} \right)$$

be a Clay-scope Yang–Mills realization for the fixed gauge group G . Then the realization’s same-domain standing-relevant content is exhausted, up to lawful AQFT equivalence, by the predecessor local theorem object

$$\left(\mathcal{A}_{\text{loc}}(G), \omega_{\text{vac}}, U_{\text{vac}} \right).$$

More precisely, exactly one of the following holds.

- (i) **Same-theory realization.** $\tilde{\mathfrak{T}}(G)$ *is a realization of the same Yang–Mills theory on the same domain as the predecessor local theorem object. Then, by Theorem VIII.121, its bounded local gauge-invariant observable content, vacuum restriction, and local time-translation dynamics coincide with those of the predecessor theorem object up to lawful AQFT equivalence. In particular, its canonical local vacuum sector is dynamically unitarily equivalent to the predecessor vacuum representation, and every further same-domain admissible enlargement is a theorem-scope admissible completion in the sense of Theorem VIII.31.*
- (ii) **Different-theory realization.** $\tilde{\mathfrak{T}}(G)$ *is not a realization of the same Yang–Mills theory on the same domain as the predecessor local theorem object. Then it changes the bounded local gauge-invariant observable/dynamical content and is therefore a different theory object.*

Equivalently: any purported additional same-domain structure over the predecessor local theorem object either has no independent governance work, is a forbidden same-domain extension, violates conservation, or collapses extensionally back to the same gate/interior. There is no third admissible alternative.

Proof. Apply Theorems VIII.78, VIII.120 and VIII.121. If $\tilde{\mathfrak{T}}(G)$ is claimed to be a realization of the same Yang–Mills theory on the same domain as the predecessor local theorem object, then Theorem VIII.121 forces fixation of the bounded local gauge-invariant observable content, the vacuum restriction, and the local time-translation dynamics up to lawful AQFT equivalence. This is case (i). The dynamical uniqueness lemma proved earlier in this section then identifies the canonical local vacuum sector of $\tilde{\mathfrak{T}}(G)$ with the predecessor vacuum

representation, and Theorem VIII.78 shows that every further same-theory enlargement falls under the completion classification theorem already proved.

If, on the other hand, $\tilde{\mathfrak{T}}(G)$ is not a realization of the same theory on the same domain, then by Theorem VIII.121 the failure of local fixation is not a harmless same-domain difference: it changes the bounded local observable/dynamical content and therefore defines a different theory object. This is case (ii).

The final equivalence statement is exactly the operator-algebraic reading of Theorems VIII.78 and VIII.120: once the local theorem object is fixed, any purported further same-domain structure either has no independent work, is a forbidden same-domain extension, violates conservation, or collapses extensionally back to the same gate/interior. Hence there is no third admissible alternative. $\square$

Theorem VIII.123 (No same-theory theorem-scope completion introduces new subgap local multiplicity). *No theorem-scope admissible completion preserving the fixed local Yang–Mills theorem object and its realized sector layer introduces new locally detectable subgap multiplicity over the fixed local algebra.*

Proof. Let $(\mathcal{A}, \omega, U)$ be such a completion. By Theorem VIII.45, any such completion in the sectoral branch is a filtered sector realization and hence a scope-faithful extension of the fixed local vacuum theory. Hence the dynamical uniqueness and subgap rigidity results of this section apply.

In particular, the canonical local vacuum sector remains unitarily equivalent to the predecessor vacuum representation, and its subgap spectral window is exactly the vacuum ray. Any additional locally detectable subgap multiplicity would therefore define a second locally generated subgap sector over the fixed local algebra, contradicting the exclusion and rigidity results proved earlier in the section. Hence no such new subgap local multiplicity can arise. $\square$

Corollary VIII.124 (Clay endpoint under AQFT individuation and completion). *Let $\tilde{\mathfrak{T}}(G)$ be any Clay-scope Yang–Mills realization for the fixed gauge group G that is a realization of the same Yang–Mills theory on the same domain as the predecessor local theorem object. Then $\tilde{\mathfrak{T}}(G)$ is uniquely locally equivalent on its canonical local vacuum sector to the predecessor local theorem object. Every further same-domain admissible enlargement of that canonical*

local vacuum object is a theorem-scope admissible completion, and the Hamiltonian on each such canonical local vacuum realization or theorem-scope admissible completion satisfies

$$\text{Spec}(\widetilde{H}) \cap (0, m^*) = \emptyset, \quad \ker \widetilde{H} = \mathbb{C}\widetilde{\Omega}.$$

In particular, the predecessor gap m^* is the mass gap of the Yang–Mills theory in the same-domain, Clay-scope sense fixed by the local-net theorem package and the admissible-completion bridge.

Proof. By Theorems VIII.46, VIII.86, VIII.101 and VIII.122, any same-theory Clay-scope realization is uniquely locally equivalent to the predecessor local theorem object on its bounded local gauge-invariant observable content, and every further same-domain admissible enlargement is a theorem-scope admissible completion. The spectral bridge and completion classification theorems proved earlier in this section then apply directly to the canonical local vacuum realization and to those theorem-scope admissible completions, yielding

$$\text{Spec}(\widetilde{H}) \cap (0, m^*) = \emptyset, \quad \ker \widetilde{H} = \mathbb{C}\widetilde{\Omega}.$$

Thus the predecessor gap m^* is the mass gap of the Yang–Mills theory in the Clay-relevant same-domain sense. $\square$

Remark VIII.125 (Group-level endpoint, not regularization-level endpoint). The endpoint of Theorem VIII.124 is the Yang–Mills theory for the compact connected simple gauge group G . Auxiliary Wilson, sharp-field, or faithful finite-dimensional choices belong to the predecessor proof route and reconstruction machinery. They are not additional endpoint parameters and do not weaken the group-level mass-gap statement.

Proof-home note for the endpoint. The endpoint obtained in this section is governed by exact proof-home discipline. The vacuum-sector construction and vacuum mass-gap theorem remain the proof home for the fixed local Yang–Mills theorem object. The present section adds the sectoral and fixed-domain closure layer over that object, together with the corresponding completion and individuation consequences. It does not claim a fresh lower derivation of the theorem-derived kernel-side constraints recorded above, and it does not claim a classification of arbitrary theory objects outside the same-domain comparison fixed here.

Endpoint proof-home clarification. The endpoint theorem uses the constructed theorem object as its proof home. The theorem-derived necessity constraints are not floating meta-level decorations; they are applied only after the fixed local Yang–Mills theorem object has been constructed and recognized, in-paper, as lying on the necessity-constrained side relevant to same-domain comparison.

Proof-home note on validity. At the endpoint, mathematical validity is not an external rival notion standing alongside admissibility. On the fixed Yang–Mills domain treated in this manuscript, the internal and extension-stable closure theorems force admissibility to exhaust faithful same-domain mathematical validity for the original acts on the original fixed domain. That is the exact sense in which the endpoint excludes mathematically valid same-domain rivals.

IX. HOSTILE-REFEREE AUDIT AND SAME-MODE DENIAL BURDEN

This section adds no proof content. It records the denial burden for same-mode objections.

A. Wrong-mode objections

Objection	Status
This is not a new local Yang–Mills construction.	Granted. The predecessor package is the proof home for local existence and the vacuum-sector gap.
This does not prove a charged-sector mass gap.	Granted. Sector representations are not automatically same-theory completions.
Line and surface operators should be part of the theory.	They are analyzed as extended-support sector data over the fixed local net, not as bounded-region local observables.
Lean does not formalize all analytic Yang–Mills.	Granted. Lean audits theorem routing and endpoint hygiene; it does not replace the analytic proof home.
A direct-sum Hilbert space could add subgap states.	Wrong carrier unless the enlargement is a scope-faithful, kernel-faithful same-domain completion.
A different ultraviolet completion could have new subgap content.	Then it is not same-domain, not kernel-faithful, or a different theorem object unless it passes the completion filter and GNS bridge.
AASC is optional vocabulary.	Wrong-mode unless the objection breaks kernel necessity, fixed-domain instantiation, or the local completion packet.

B. Right-mode objections

A right-mode objection must target a named theorem link: fixed-base handoff; local/nonlocal boundary; sector realization over the fixed net; sector-to-completion admissibility filtering; the GNS bridge object; dynamical GNS uniqueness; local-state conservation; subgap vacuum-multiplicity exclusion; the local completion packet; weakening resistance; route-locus exhaustion; or Lean-to-manuscript correspondence where Lean is cited.

C. Referee burden worksheet

Target	Successful objection must produce
Fixed-base handoff	A reason that $\mathfrak{B}(G)$ is not the predecessor local theorem object or that this manuscript silently changes it.
Local-net boundary	A nontrivial object of $\mathfrak{E}(G)$ that belongs to a bounded-region algebra $\mathcal{A}_{\text{loc}}(O; G)$ without changing the local theorem object.
Sector-to-completion filter	A sector representation that is automatically a same-theory completion without carrier, scope, standing, admissibility, dynamics, and kernel-fidelity data.
GNS bridge object	A clause of $B_{\text{YM}}^{\text{GNS}}(G; \mathcal{A}, \omega, U)$ that smuggles the gap conclusion, assumes no-subgap consequences rather than deriving them, or fails to preserve the completion interface.
Subgap counterexample	A scope-faithful, kernel-faithful same-domain completion with $\psi \in E_H([0, m^*))\mathcal{H} \setminus \mathbb{C}\Omega$.
Direct-sum enlargement	A direct-sum or direct-integral construction that preserves fixed local carrier, vacuum restriction, local dynamics, and kernel fidelity while adding subgap content.
Weakening resistance	A weaker same-domain completion regime preserving endpoint determinacy while allowing subgap completion governance outside support, bookkeeping, carrier shift, theorem-object change, and forbidden second authority.
Lean boundary	A mismatch between the Lean audit surface and the manuscript theorem chain.

DATA AVAILABILITY STATEMENT

Data sharing is not applicable to this article as no new data were created or analyzed in this study.

Appendix A: Formalization boundary and proof-home posture

This appendix records the scope of the accompanying Lean 4 formal audit layer and the proof-home posture of the present manuscript. It adds no mathematics. Its role is to identify what has been formalized at theorem-bearing level, which theorem inputs remain outside that formal scope, and how the present paper sits relative to the predecessor Yang–Mills theorem package.

Formal audit archive

The manuscript-facing Yang–Mills local-net and Clay-extension audit layer is supported by a dedicated Lean 4 archive, *AASC Yang–Mills Lean Audit Endpoint*, Version 1.0.1, published June 3, 2026, archived at Zenodo under DOI 10.5281/zenodo.20523056 and mirrored at the public GitHub repository <https://github.com/somamaley-ux/AASC-YangMills-Lean-Audit>²⁷. The archive separates the Yang–Mills Lean work from the larger AASC working repository and packages the theorem-routing, endpoint, seam-checking, and audit-control files relevant to the present manuscript.

The release was checked with

```
powershell -ExecutionPolicy Bypass -File  
scripts/check-yang-mills-a-plus-audit.ps1
```

The check confirms that the Yang–Mills Lean package builds, the A+ audit files run successfully, the audited Yang–Mills files contain no live project-level `axiom`, `sorry`, `admit`, or `unsafe`, and the Lean/mathlib environment is recorded. The environment recorded for the release is Lean `leanprover/lean4:v4.28.0`, mathlib revision `8f9d9cff6bd728b17a24e163c9402775d9e6a365`, Version 1.0.1.

Current formalization scope

The accompanying formal layer is theorem-bearing and manuscript-facing rather than analytic in full depth. Concretely, the Lean 4 archive formalizes the theorem-level architecture and audit endpoint of the present paper, including:

- (i) the fixed-base handoff from the predecessor local-net endpoint to the base object used here;
- (ii) the extended-support class, its support-equivalence and deformation structure, and the fixed theorem-scope support discipline of the present manuscript;
- (iii) the local-net boundary layer, including strictly local insensitivity and exclusion of the chosen extended-support class from bounded-region local algebras;
- (iv) the representation-theoretic sector construction over the fixed quasi-local algebra together with its equivalence and deformation-transport structure;
- (v) the endpoint-use ATS/UEAP sector-to-completion filter, including the distinction between sector representation and same-domain completion eligibility;
- (vi) the kernel-fidelity and claim-carrier routing used to block new subgap same-domain content; and
- (vii) the recovery, compatibility, and spectral-boundary statements proved in Sections VII and VIII.

The formal companion is therefore an auxiliary verification layer for theorem-level organization, dependency locality, fixed-base discipline, sector-to-completion classification hygiene, spectral-endpoint routing, and claim boundaries.

What the formal companion does not claim

The formalization posture is restrained. It is *not* a claim that every analytic argument lying beneath the predecessor Yang–Mills construction has been reproved natively inside the present sector manuscript. In particular, it does not claim:

- (a) a fresh end-to-end Lean formalization, inside the present manuscript alone, of the full local Yang–Mills analytic construction carried by the predecessor paper and its proof homes;
- (b) native discharge, inside the present manuscript alone, of every analytic or constructive argument used to establish the predecessor vacuum-sector theorem package;

- (c) a reproof from first principles of standard Lean/classical foundations or standard imported mathematical background; or
- (d) any charged-sector, defect-sector, or other nonvacuum spectral theorem beyond the explicit boundary stated in Section VIII.

The formal boundary is therefore this: the theorem-bearing sector-extension, fixed-domain, completion-classification, spectral-boundary, and endpoint-hygiene layer is formalized as a layer over the fixed predecessor local theorem package, while the deeper analytic construction of that package remains carried by the earlier Yang–Mills corpus. Standard Lean foundations and imported mathematical background remain part of the explicit boundary. Dependency reports may therefore mention standard foundations such as `propext`, `Classical.choice`, and `Quot.sound`; these are not project-specific Yang–Mills axioms.

This is the no-overclaim rule for the formal layer: Lean corroborates theorem-routing where the objects and hypotheses have been supplied; it is not presented as a native reconstruction of the analytic Yang–Mills proof home^{23,27}.

Imported theorem-level inputs

The theorem-bearing input left outside the present manuscript is the predecessor local theorem package itself, together with standard background theorems used in the usual AQFT and operator-algebraic sense. More precisely, the present paper treats as fixed:

- (i) the existence of the bounded-region local gauge-invariant observable net;
- (ii) its distinguished vacuum representation; and
- (iii) the predecessor vacuum-sector reconstruction and vacuum mass-gap theorem attached to that fixed local object.

Beyond that fixed theorem package, the present paper uses standard observable-net, sector, GNS, spectral-projection, and OS/Wightman terminology in the usual sense^{9–12}. The formal audit archive is correspondingly an audit endpoint for the manuscript-facing proof route, not a claim to rederive every standard C^* -algebraic, Hilbert-space, OS/Wightman, or spectral-theory theorem from first principles.

Proof-home posture

The proof-home posture is one-directional.

- (i) The predecessor Yang–Mills project remains the proof home for the bounded-region local net, its vacuum reconstruction, and its vacuum-sector mass-gap theorem^{1–4}.
- (ii) The present manuscript is the proof home for the extended-support class, the local-net boundary theorem, the sector-construction layer, the ATS/UEAP sector-to-completion filter, the global-form recovery theorem, and the compatibility and spectral-boundary statements proved here.
- (iii) No theorem proved in the present paper is used to strengthen, reinterpret, or silently enlarge the predecessor local theorem package.

Accordingly, the formal layer supplied for the present paper is an audit and verification aid for the theorem-bearing sector-extension, completion-classification, spectral-boundary, and endpoint-hygiene structure over the fixed base object. It is neither a substitute for the mathematical exposition nor a hidden source of additional analytic theorem content.

Boundary note for hardening additions

The proof-class lock, no-one-step promotion theorem, local Yang–Mills completion packet, GNS bridge object, weakening-resistance theorem, route-locus audit, native completion counterexample collapse, theorem ladder, anti-circularity audit, and hostile-referee worksheet are manuscript-level defensive refinements of the theorem-routing surface already represented in the Lean audit: fixed-base handoff, local-net boundary, sector-to-completion filtering, kernel-fidelity routing, and spectral endpoint. They do not add a new first-principles analytic Yang–Mills formalization burden and do not alter the predecessor local theorem proof home.

Potential future Lean-facing anchors, if the support archive is extended, would cover the GNS bridge object, bridge non-circularity, no-one-step promotion, the local completion packet, weakening resistance, route-locus exhaustion, and native completion-counterexample collapse. These names identify audit-surface refinements only; they are not new analytic proof-home claims.

Appendix B: Theorem ladder and endpoint audit map

1. Mathematical-status, proof-class, and source-use locks.
2. Fixed-base handoff from the predecessor local theorem object.
3. Frozen extended-support class and local-net boundary.
4. Sector realization over the fixed net and global-form recovery.
5. Compatibility with the predecessor local theorem object.
6. No one-step sector-to-completion promotion.
7. Sector-to-completion admissibility filtering.
8. Local Yang–Mills completion K1–K13 packet.
9. Weakening resistance for same-domain completion use.
10. GNS bridge-admissibility object, bridge-object non-circularity, and derived no-subgap consequences.
11. Dynamical GNS uniqueness and local subgap rigidity.
12. Local-state conservation and subgap multiplicity exclusion as derived bridge consequences.
13. Kernel-faithful spectral bridge: $E_H([0, m^*))\mathcal{H} = \mathbb{C}\Omega$.
14. Completion classification and route-locus exhaustion.
15. Clay-scope endpoint and hostile-referee burden worksheet.

Appendix C: Anti-circularity and no-overclaim audit

- The GNS bridge object does not contain the mass-gap conclusion and does not assume the no-subgap consequences. It fixes bridge-admissibility and identity data; local-state conservation, subgap multiplicity exclusion, and the spectral endpoint are derived later.
- Sector representation does not already supply completion; completion eligibility begins only after the filter is passed.

- Geometric containment of an extended-support object is not membership in a bounded-region local algebra.
- The completion theorem does not redo the local Yang–Mills construction; the predecessor package remains the proof home.
- Completion closure does not generate gauge dynamics; it controls same-domain completions after the fixed local theorem object exists.
- Direct-sum Hilbert spaces matter only if they are scope-faithful, kernel-faithful, and preserve the fixed local carrier, vacuum restriction, dynamics, and endpoint identity.
- UV and sharp-field carriers enter only through macroscopic local-net agreement and kernel fidelity; otherwise they are carrier shifts.
- Lean audits theorem routing and proof boundaries. It is not offered as a first-principles analytic Yang–Mills proof.
- Non-vacuum sector representations are lawful sector data. They are not same-theory completions unless they pass the completion filter.
- The theorem is about theorem-scope admissible completions, not arbitrary completions.

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